

Tail Risk Targeting: Target VaR and CVaR Strategies

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Abstract

We present dynamic trading strategies that target a predefined level of risk measured by volatility, Value-at-Risk (VaR) or Conditional-Value-at-Risk (CVaR). Recent studies have shown that volatility targeting increases the risk-adjusted performance and heightens utility gains for mean-variance investors. We find that downside risk targeting outperforms volatility targeting in terms of higher Sharpe Ratios, better drawdown protection and higher utility gains for mean-variance, CRRA and loss-averse investors. In particular, a loss-averse investor is not willing to pay a positive fee to switch from a static portfolio to a volatility managed strategy, whereas the same investor would pay a fee of 18% per year to have access to the downside risk managed strategy. The performance of risk targeting can further be enhanced by switching between volatility and CVaR targeting based on estimates of whether the market will be in a bull or bear regime. This strategy successfully reduces the drawdowns during the global financial crisis and the recent corona crisis.

Keywords: Volatility; Value at Risk; Conditional Value at Risk; Risk targeting; Extreme Value Theory; Dynamic trading strategies

JEL classification: C53; G11; G17

1 Introduction

During extreme market crashes, due to an increase of correlations, diversification fails as a risk management tool. Especially when financial markets exhibit huge downturn periods, assets typically crash together, and thus lower the benefit of diversification just when it is most needed (Ang and Bekaert, 2002, Butler and Joaquin, 2002, Chabi-Yo et al., 2018, Guidolin and Timmermann, 2008, Karolyi and Stulz, 1996, Patton, 2004, Poon et al., 2004). Investors typically

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overestimate the benefits of diversification in bear markets and underestimate return potentials in bull markets. This leads to too high equity exposures in bear markets, whereas the equity allocation is too low in bull markets ([Ang and Chen, 2002](#), [Longin and Solnik, 2001](#)).

For that reason, more tactical tools, such as volatility targeting, have become popular in the financial industry and academic literature.¹ The aim of volatility targeting is to build a portfolio, consisting of a risky and a riskless asset, that has a (predetermined) constant level of portfolio volatility over time. In order to achieve this constant level of portfolio volatility, the target volatility strategy allocates money between the risky and the riskless asset, based on a forecast of the risky asset's volatility: if the risky asset's volatility is expected to be high, the weight of the risky asset is decreased and vice versa (see [Bollerslev et al. \(2018\)](#) for example). By choosing a portfolio's exposure inversely to the portfolio's volatility, left tail risk can be significantly reduced ([Dreyer and Hubrich, 2019](#), [Harvey et al., 2018](#)). Generally, the economic value of volatility timing, measured by significant performance and utility gains for investors who time volatility in multi-asset portfolios, has been extensively examined by [Fleming et al. \(2001\)](#), [Fleming et al. \(2003\)](#), [Han \(2005\)](#), [Kirby and Ostdiek \(2012\)](#) and [Taylor \(2014\)](#).² In a single asset scenario, using long-only equity portfolios, the economic value of volatility timing is examined by [Dreyer and Hubrich \(2019\)](#), [Marquering and Verbeek \(2004\)](#), [Bollerslev et al. \(2018\)](#) and [Moreira and Muir \(2017\)](#). The authors find vast utility gains of volatility timing and that volatility timing is superior to return timing. [Dreyer and Hubrich \(2019\)](#) find that the utility gains of volatility targeting mainly come from a reduction of left tail risk. Thus, volatility targeting works especially well for assets with non-normally distributed returns, like equities or dynamic trading strategies ([Perchet et al., 2016](#)). [Barroso and Santa-Clara \(2015\)](#), [Barroso and Maio \(2017\)](#), [Cederburg et al. \(2020\)](#) and [Moreira and Muir \(2017\)](#) examine volatility targeting applied to different long-short factor portfolios and find significant improvements in the risk-

¹For academic research on target volatility strategies see [Barroso and Santa-Clara \(2015\)](#), [Bollerslev et al. \(2018\)](#), [Moreira and Muir \(2017\)](#), [Barroso and Maio \(2017\)](#), [Moreira and Muir \(2019\)](#), [Cederburg et al. \(2020\)](#) among others. For research from practitioners see, e.g., [Banerjee et al. \(2016\)](#), [Dreyer and Hubrich \(2019\)](#), [Hocquard et al. \(2013\)](#), [Benson et al. \(2014\)](#) and [Perchet et al. \(2016\)](#).

²[Fleming et al. \(2001\)](#), [Fleming et al. \(2003\)](#), [Han \(2005\)](#), [Kirby and Ostdiek \(2012\)](#) and [Taylor \(2014\)](#) assess the economic value of volatility timing in a multivariate setting. This approach is slightly different to volatility targeting but demonstrates that investment decisions that rely on volatility (or more precisely covariances) solely work well in empirical applications.

adjusted performance, especially for strategies with a high left tail risk, like momentum or betting against beta.³ This also highlights a nice characteristic of volatility targeting. Volatility targeting can be used for any underlying investment strategy, i.e. volatility targeting can be separated from the fund manager's asset allocation choice, where the asset allocation is chosen first and this portfolio is then overlayed by a volatility targeting strategy as market timing tool (Harvey et al., 2018, Hocquard et al., 2013, Moreira and Muir, 2017, Zakamulin, 2015). Busse (1999) examines the impact of volatility timing for the institutional fund industry and concludes that “funds that reduce systematic risk when conditional market volatility is high earn higher risk-adjusted returns” and that funds who time volatility the most are associated with higher Sharpe Ratios (Busse, 1999, p. 1010 and 1027). Generally, volatility is frequently used as a market timing tool by practitioners (Christoffersen and Diebold, 2006, Copeland and Copeland, 1999). This does not only hold for investors with short investment horizons. Moreira and Muir (2019) assess the economic value of volatility timing for long-term investors and the authors find that even long-term investors should time short-term volatility. This supports the finding of Benartzi and Thaler (1995) that even long-term investors have short evaluation periods and are concerned about short-term risk.

So far, studies on risk targeting focus on volatility as a risk measure: the weight of the risky asset is a function of the risky asset's volatility. However, since asset returns are typically skewed, fat-tailed and non-normally distributed, the choice of volatility as a measure of market risk is not appropriate (see Szegö (2002), Poon and Granger (2003), Kuester et al. (2006, p. 56) and Bali et al. (2009)). Xiong and Idzorek (2011), Guidolin and Timmermann (2008), Jondeau and Rockinger (2006), Jondeau and Rockinger (2012) and Ghysels et al. (2016) examine the impact of skewness and fat-tails on the asset allocation and show that incorporating higher

³Barroso and Santa-Clara (2015) successfully use a target volatility strategy to manage the risk of the momentum portfolio and show that targeting a constant level of volatility extremely reduces the drawdowns of the momentum portfolio, the so called “momentum crashes”. This high crash risk reduction translates into a superior risk-adjusted performance compared to the non-managed momentum portfolio (see also Daniel and Moskowitz (2016) and Rickenberg (2020a)). Moreira and Muir (2017) use a volatility timing strategy for different factor portfolios and show that the risk-adjusted performance of the volatility managed portfolios is superior to the performance of the non-managed portfolios. This finding is most pronounced for the momentum strategy. Barroso and Maio (2017) use volatility targeting for several factor strategies and find huge improvements of the volatility targeting strategies for all strategies except for the size factor. The best results are found for the momentum strategy and the “betting against beta” strategy of Frazzini and Pedersen (2014). This finding is confirmed by Cederburg et al. (2020).

moments, as done by adequately measuring downside risk, is beneficial compared to mean-variance optimization. Furthermore, most investors have preferences for higher skewness and lower kurtosis (see [Kraus and Litzenberger \(1976\)](#), [Scott and Horvath \(1980\)](#), [Guidolin and Timmermann \(2008\)](#) among others). [Bali et al. \(2009\)](#) and [Kelly and Jiang \(2014\)](#) find that an increase of tail risk predicts higher kurtosis and lower (or more negative) skewness. Hence, investors who dislike a negative skewness and a high kurtosis should better manage a portfolio's downside risk. This also fits well to the observation that investors are more concerned about downside risks instead of return deviations ([Bollerslev et al., 2015](#), [Kelly and Jiang, 2014](#), [Lee and Rao, 1988](#)).⁴ Similarly, most investors weight losses higher than gains, which implies that avoiding huge losses is crucial in order to increase a loss-averse investor's utility ([Aït-Sahalia and Brandt, 2001](#), [Ang et al., 2005, 2006a](#), [Benartzi and Thaler, 1995](#)). Avoiding crashes is particularly important since most investors are crash-averse and have a demand for portfolio insurance methods, especially in times of extremely negative returns ([Bollerslev and Todorov, 2011](#), [Chabi-Yo et al., 2018](#)). Therefore, timing an asset's downside risk instead of volatility fits better to most investors' preferences. Furthermore, [Benson et al. \(2014\)](#) find that the superior performance of volatility targeting does result from mitigating drawdowns. Similarly, [Dreyer and Hubrich \(2019\)](#) and [Harvey et al. \(2018\)](#) find that volatility targeting reduces the likelihood of extremely negative returns, which is an important source of the outperformance of volatility targeting. Consequently, if drawdown protection is a main driver of the superior (risk-adjusted) performance of risk targeting, choosing the risky asset's weight based on a forecast of the risky asset's downside risk should be more successful in mitigating drawdowns and should eventually result in a superior risk-adjusted performance.

In this paper, we show how the idea of volatility targeting can be extended to tail risk targeting, where tail risk is measured by Value at Risk (VaR) or Conditional Value at Risk (CVaR).⁵ These strategies aim to keep the VaR or CVaR of the portfolio constant over time by shifting money between the risky and the riskless asset, based on a forecast of the risky asset's tail risk.

⁴See [Arzac and Bawa \(1977\)](#) who examine the portfolio allocation of safety-first investors who are concerned about financial tail events. See also [van Oordt and Zhou \(2016\)](#) for an asset pricing study based on this theory.

⁵These risk measures are frequently used by practitioners and are important for regulators and banks ([Berkowitz et al., 2011](#), [Berkowitz and O'Brien, 2002](#), [Du and Escanciano, 2016](#)). Further, VaR and CVaR are also used in studies on the cross-section of returns ([Atilgan et al., 2020](#)).

This approach translates into a strategy that increases the weight of the risky asset if the risky asset's tail risk is expected to be low and vice versa. [Basak and Shapiro \(2001\)](#), [Alexander and Baptista \(2004\)](#), [Cuoco et al. \(2008\)](#), [Agarwal and Naik \(2004\)](#) and [Wang et al. \(2012\)](#) demonstrate the benefits of downside risk timing instead of volatility timing in portfolio allocations. To compare the economic of volatility and downside risk targeting, we follow the literature and assess the economic value of risk targeting for a mean-variance investor. Further, to incorporate preferences for higher moments, like skewness and kurtosis, we additionally assess the economic value for a CRRA investor ([Dreyer and Hubrich, 2019](#), [Jondeau and Rockinger, 2012](#)). Finally, since most investors weight losses higher than gains, we assess the economic value of risk targeting for loss-averse investors. We find that tail risk targeting strategies deliver high utility gains compared to a static portfolio allocation. This is in line with [Cuoco et al. \(2008\)](#) who find that frequently reallocating portfolio weights based on estimates of downside risk is superior to static portfolio allocations. In particular, an investor should manage portfolio risk based on a conditional risk model using a dynamic volatility model, like the GARCH(1,1) or EWMA model. Simple risk estimation models like the Historical Standard Deviation, as used in [Moreira and Muir \(2017\)](#), [Barroso and Maio \(2017\)](#) and [Barroso and Santa-Clara \(2015\)](#), or Historical Simulation typically fail to significantly increase an investor's utility and produce lower Sharpe Ratios than the conditional approaches. Moreover, we find that the economic value of CVaR timing is significantly higher than the economic value of volatility timing, especially when investors are highly risk- or loss-averse and in times of bear markets. This finding also holds during the recent corona crisis. Interestingly, even mean-variance investors should manage CVaR instead of volatility. For example, we find that a mean-variance investor is willing to pay a fee of about 0.8% per year to have access to a volatility targeting strategy. However, the same investor would even pay 4.253% per year to have access to the CVaR managed strategy. In contrast, a loss-averse investor is not willing to pay a positive fee for volatility targeting, but the same investor would pay up to 18% per year to have access to the CVaR targeting strategy.

Since estimating downside risk is more sophisticated than estimating volatility, we additionally show how the target VaR and target CVaR strategies can be approximated by a target

volatility strategy. Further, we demonstrate how the accuracy of the target volatility, target VaR and target CVaR strategies can be backtested. In order to assess the accuracy of volatility targeting, we resort to the approaches of [Diebold and Mariano \(1995\)](#), [White \(2000\)](#), [Hansen \(2005\)](#), [Romano and Wolf \(2005\)](#), [Hansen et al. \(2003\)](#), [Hansen et al. \(2011\)](#), [Hsu et al. \(2010\)](#), [Barras et al. \(2010\)](#) and [Bajgrowicz and Scaillet \(2012\)](#) that test for equal or superior predictive ability. In order to assess the accuracy of VaR and CVaR targeting, we use the VaR backtest of [Christoffersen \(1998\)](#) and the CVaR backtests of [McNeil and Frey \(2000\)](#) and [Embrechts et al. \(2005\)](#). With these backtests in hand, we assess the accuracy of approximating a target VaR or target CVaR strategy by a target volatility strategy, i.e. we answer the question if controlling volatility is sufficient when downside risk is targeted. We find that for investors who are interested in targeting a constant VaR or CVaR over time, controlling volatility is not sufficient. Similarly, in order to target a constant level of volatility, an investor should manage volatility directly instead of downside risk. Generally, risk should be managed by a dynamic risk model, based on a dynamic volatility model, like the EWMA or GARCH(1,1) model. In contrast, using a static risk model, like the Historical Standard Deviation or Historical Simulation, fails to target the portfolio risk at a constant level, achieves a worse risk-adjusted performance and produces lower utilities for investor. In line with [Bollerslev et al. \(2018\)](#), we find a positive relation between forecasting accuracy, and hence a more constant portfolio risk, and risk-adjusted performance and utility gains (see also [Perchet et al. \(2016\)](#) and references therein).

Finally, we develop strategies that switch between volatility and CVaR targeting, based on an estimate of the expected market regime. If the market is expected to be in a down-market, CVaR targeting is used, whereas the portfolio's risk is managed by volatility if an up-market is expected. To determine up- and down-markets, we use technical trading rules ([Bajgrowicz and Scaillet, 2012](#), [Moskowitz et al., 2012](#)) and the portfolio's expected volatility. We find that these switching strategies further increase the risk-adjusted return and the investors' utility. For example, a mean-variance investor is willing to pay 5.689% per year to switch to a strategy that dynamically switches between volatility and CVaR targeting. Further, a loss-averse investor is even willing to pay 21.135% per year to have access to this strategy. The

benefits of this switching approach also hold during the recent corona crisis and in the long-run. Over the last 88 years, a 100\$ investment in the market would result into a portfolio value of 357,591\$. By using the volatility targeting strategy, this amount can be raised to 4,420,160\$. However, by switching between volatility and CVaR targeting, the wealth would even increase to 48,535,249\$. Thus, switching between volatility and CVaR targeting is much more profitable than volatility targeting alone. Similarly, we find that volatility targeting produces a negative alpha, once we control for CVaR targeting or the switching strategy. In contrast, the returns of CVaR targeting and the switching strategy cannot be explained by volatility targeting.

This paper is structured as follows. In Section 2, we present the target volatility framework and review the literature on volatility targeting. Section 3 presents the target VaR and CVaR strategies and shows how VaR and CVaR are estimated. Furthermore, we show how the target VaR and CVaR strategies can be approximated by a target volatility strategy. Section 4 demonstrates how the accuracy of volatility, VaR and CVaR targeting can be tested. Section 5 shows the empirical results and Section 6 concludes the paper. The Appendix contains additional results and shows how the strategies have performed in the long-run and during the recent corona crisis.

2 Target Volatility Strategy

Throughout the paper, we consider a risky asset, e.g. an equity index, with price process $\{S_t\}_{t \in \{0, \dots, T\}}$ over the period $[0, T]$, $T \in \mathbb{N}$, and we define the return of the risky asset over the period $[t-1, t]$, representing one day, as

$$R_t := \frac{S_t}{S_{t-1}} - 1. \quad (1)$$

Further, we consider a riskless asset with returns $\{R_t^f\}_{t \in \{0, \dots, T\}}$. R_t^f describes the return of the riskless asset over the period $[t-1, t]$ and we assume that R_t^f is known at time $t-1$.⁶ The day t return R_t^P of the portfolio that invests a weight w_t in the risky asset and $1 - w_t$ in the riskless

⁶More formally, we assume that R_t^f is measurable with respect to $\mathcal{F}_{t-1}$, where $\mathcal{F}_{t-1}$ is the σ -algebra generated by the variables that are observed up to time $t-1$ (Hansen and Lunde, 2005, p. 875). Hence, $\mathcal{F}_{t-1}$ contains all relevant information available at time $t-1$.

asset is then given by

$$R_t^P := w_t \cdot R_t + (1 - w_t) \cdot R_t^f. \quad (2)$$

The aim of the target volatility strategy is to determine the weight w_t for each day t such that the portfolio volatility is constant over time and equals a predefined value. We denote the *portfolio volatility*, i.e. the (conditional) standard deviation of the portfolio return R_t^P conditioned on the information $\mathcal{F}_{t-1}$ that is available at time $t - 1$, by $\sigma_t^P := \sqrt{\text{var}(R_t^P | \mathcal{F}_{t-1})}$, where $\text{var}(R_t^P | \mathcal{F}_{t-1})$ denotes the (conditional) portfolio variance (see Hansen and Lunde (2005, p. 875)). In order to achieve a constant volatility level σ^{target} for σ_t^P over time, the weight of the risky asset has to be chosen as

$$w_t = \frac{\sigma^{\text{target}}}{\sigma_t}, \quad (3)$$

where σ^{target} is the desired volatility target and $\sigma_t := \sqrt{\text{var}(R_t | \mathcal{F}_{t-1})}$ is the (conditional) volatility of the risky asset at day t (see Bollerslev et al. (2018, p. 2757) for example). By construction, the day t weight w_t is known at day $t - 1$ since σ_t is $\mathcal{F}_{t-1}$ -measurable. The use of volatility targeting has several advantages, which are summarized in Appendix A.⁷

To implement a target volatility strategy, the volatility of the risky asset σ_t in Equation (3) is needed, which is unobservable in practice. Therefore, based on the available information at time $t - 1$, the volatility for day t has to be forecasted.⁸ We denote this (one-step ahead) forecast by $\hat{\sigma}_t$. Based on this volatility forecast, the weight w_t of the risky asset is given by

$$w_t = \frac{\sigma^{\text{target}}}{\hat{\sigma}_t}. \quad (4)$$

Consequently, the success of the target volatility strategy strongly depends on the quality of the volatility forecast $\hat{\sigma}_t$.⁹ Benson et al. (2014) show that a target volatility strategy with perfect foresight, i.e. a strategy that knows the next period's volatility, outperforms the benchmark by

⁷Volatility targeting is also quite frequently used by practitioners. See, for example, the fund offered by Man AHL (<https://www.man.com/ahltargetrisk>).

⁸See Bollerslev et al. (1992), Taylor (2005, Sec. 2), Poon and Granger (2003) and Hansen and Lunde (2005) for surveys on volatility forecasting.

⁹Obviously, the volatility of the target volatility strategy is only constant over time and equals σ^{target} if and only if the volatility forecast $\hat{\sigma}_t$ equals the true (ex-post) realized volatility σ_t on each day t .

more than 10% per year with a lower volatility and is successful in producing a constant volatility, indicated by an almost zero volatility of volatility (see also [Bollerslev et al. \(2018\)](#)). Moreover, [Moreira and Muir \(2017\)](#) and [Bollerslev et al. \(2018\)](#) show that using advanced volatility forecasting models in a volatility targeting strategy improves the risk-adjusted performance and heightens utility gains compared to simple and less accurate forecasting models. [Taylor \(2014\)](#) and [Fleming et al. \(2003\)](#) find a similar observation for a multivariate volatility timing strategy. In particular, [Bollerslev et al. \(2018\)](#) find a positive relation between forecasting accuracy and utility gains for investors who use volatility targeting. Furthermore, [Dreyer and Hubrich \(2019\)](#) find that a stabilization of portfolio volatility is linked to a lower crash risk of the portfolio. Thus, a more constant portfolio volatility is rewarded by higher risk-adjusted returns and utility gains as well as lower drawdowns (see also [Perchet et al. \(2016\)](#) and references therein).¹⁰ Similarly, [Marquering and Verbeek \(2004\)](#) find that periods where volatility can be predicted well correspond to periods where volatility timing generates high utility gains. In total, target volatility strategies based on more accurate volatility forecasting models should obtain better risk-return profiles than strategies that use inaccurate volatility forecasts. In particular, simply using past volatility, as is it is frequently done in the literature, is not sufficient for volatility targeting. This is also confirmed by the finding of [Dopfel and Ramkumar \(2013, p. 31\)](#). The authors find that high volatility regimes concurrently occur with negative returns and significantly lower Sharpe Ratios compared to regimes with a normal volatility. However, the authors also show that this result reverses when returns of regimes with a high or normal volatility in the *previous* period are compared.¹¹ Similarly, [Dachraoui \(2018\)](#) finds a negative relation between σ_t and R_t but no relation between σ_{t-1} and R_t . This result highlights that accurately *forecasting* future volatility σ_t by $\hat{\sigma}_t$ is crucial when portfolio volatility should be managed, since simply measuring today's volatility is not sufficient to determine tomorrow's weight of

¹⁰In a cross-sectional setting, [Baltussen et al. \(2018\)](#) find that assets with a high volatility of volatility (vol-of-vol) underperform assets with a more constant volatility. This especially holds during downturns, where high vol-of-vol assets underperform low vol-of-vol assets by 0.83% per month. Further, high vol-of-vol assets also exhibit higher downside risk.

¹¹Interestingly, although returns of periods following a high volatility period are higher than returns following a low volatility period, Sharpe Ratios are slightly higher for periods following a low volatility period. Thus, the higher volatility is not compensated by an adequately higher return and these periods are unappealing for investors ([Moreira and Muir, 2017](#)).

the risky asset.¹² Due to the importance of using accurate forecasting models and achieving a stable portfolio volatility, we present methods that test the accuracy of different target volatility strategies in Section 4.1.

For practical implementations, simple forecasting methods, like the Historical Standard Deviation (HSD) or the Exponential Weighted Moving Average (EWMA), which was proposed by the RiskMetricsTM group, can be used. Nevertheless, more advanced – and potentially more accurate – methods, like the GARCH(1,1) model of [Bollerslev \(1986\)](#), could also be interesting. In this paper, we compare these three volatility models, where the HSD statically measures today's volatility, which is used as a forecast for tomorrow's volatility. Hence, this model does not consider the aforementioned issue of *forecasting* next day's volatility. In contrast, the EWMA and GARCH(1,1) models dynamically forecast next day's volatility, and thus should result in a more constant portfolio volatility and a higher risk-adjusted performance.¹³ All three models have several advantages and disadvantages. A possible extension could be to combine several forecasting models as suggested by [Taylor \(2014\)](#).

The day t volatility using the HSD is estimated by

$$\hat{\sigma}_t = \sqrt{\frac{1}{m} \sum_{i=1}^m (R_{t-i} - \hat{\mu}_t)^2}, \quad (5)$$

where $\hat{\mu}_t = \frac{1}{m} \sum_{i=1}^m R_{t-i}$ is an estimate of the expected mean return. For the EWMA and the GARCH(1,1) models, it is assumed that the day t return of the risky asset can be described by

$$R_t = \sigma_t \cdot Z_t, \quad (6)$$

where Z_t is iid with mean zero, variance one and cumulative distribution function F_Z (see [McNeil and Frey \(2000, p. 275\)](#)). As usual, when working with daily returns, we assume that

¹²As mentioned in Appendix A, volatility targeting is still advantageous when there is no negative risk-return relation between σ_{t-1} and R_t . In other words, volatility targeting is still advantageous even when volatility is simply measured by past realized volatility up to day $t - 1$. However, a strategy based on an accurate forecast of σ_t further enhances the risk-return profile of volatility targeting.

¹³In the EWMA and GARCH(1,1) model, past negative and positive returns have the same impact on future volatility. A well-known stylized fact, the so-called *leverage effect*, states that past negative returns influence future volatility more than past positive returns. We also used the GJR-GARCH model of [Glosten et al. \(1993\)](#) and the EGARCH model of [Nelson \(1991\)](#) that account for the leverage effect, but results were quite similar to the results of the EWMA and GARCH(1,1) model. This is in line with [Taylor \(2014\)](#) who comes to the same conclusion in a multivariate setting. See also [Poon and Granger \(2003\)](#) and [Hansen and Lunde \(2005\)](#) for a comparison of different volatility forecasting models.

the expected mean return is zero. This is a quite weak assumption, since (absolute) daily returns are close to zero. Further, accurately estimating the expected daily return is not feasible (see [Merton \(1980\)](#), [Fleming et al. \(2001, p. 332\)](#), [Fleming et al. \(2003, p. 476\)](#), [Kirby and Ostdiek \(2012\)](#) among others). [Christoffersen and Diebold \(2006\)](#) show that the conditional mean is not forecastable, since returns R_t conditioned on $\mathcal{F}_{t-1}$ do not fluctuate over time. Further, [Hansen and Lunde \(2005\)](#) compare different mean specifications and find that all lead to an almost identical performance of the volatility models.

For the EWMA model, the volatility forecast $\hat{\sigma}_t$ is given by

$$\hat{\sigma}_t = \sqrt{(1 - \lambda) \cdot R_{t-1}^2 + \lambda \cdot \hat{\sigma}_{t-1}^2}, \quad (7)$$

where λ is typically chosen as 0.94 when working with daily returns ([Christoffersen, 2012, p. 70](#)). The advantage of the EWMA model is that no parameters have to be estimated, what makes this model interesting for practical applications ([Halbleib and Pohlmeier, 2012](#)). However, frequently re-estimating the model parameters, as it is done for the GARCH(1,1) model, should also result in a more accurate volatility forecast. The volatility forecast in the GARCH model is given by

$$\hat{\sigma}_t = \sqrt{\hat{\omega} + \hat{\alpha} R_{t-1}^2 + \hat{\beta} \hat{\sigma}_{t-1}^2}, \quad (8)$$

where the parameters $\hat{\omega}, \hat{\alpha}, \hat{\beta}$ are estimated via Quasi Maximum Likelihood, i.e. we assume that the innovations Z_t in Equation (6) are iid standard normally distributed.¹⁴

Another field of current research, that could be of high interest in the context of target volatility strategies, is forecasting volatility based on the theory of *realized volatility* that measures volatility using high-frequency-data (see [Andersen et al. \(2001\)](#) for example). Due to its simplicity, the Heterogeneous Autoregressive model of Realized Volatility (HAR-RV), proposed by [Corsi \(2009\)](#), fits well to the target volatility framework (see [Taylor \(2014\)](#) who finds good results of the HAR model in a multivariate setting). [Bollerslev et al. \(2018\)](#) extend the

¹⁴The GARCH(p, q) model is defined for any lag order p and q . [Bollerslev et al. \(1992, p. 22\)](#) state that small lag orders are sufficient to model the volatility of equity returns in empirical applications (see also [Kellner and Rösch \(2016\)](#) in the context of VaR and CVaR forecasting). Since target volatility strategies are of high interest for practical implementations, we restrict ourselves to the lag orders $p = 1$ and $q = 1$.

HAR-RV model in several directions and use these modifications in a volatility targeting framework. The authors find good results of these models compared to models that rely on daily data. For example, an investor who uses a volatility targeting strategy would pay an annualized fee of 0.46% to switch from a simple strategy to a high-frequency-data based strategy. Similarly, [Fleming et al. \(2003\)](#) examine the economic value of high-frequency-data based estimates of daily volatility. They find that using high-frequency-data based volatility measures instead of daily data based measures can substantially increase the economic value of volatility timing in a multivariate mean-variance context ([Fleming et al., 2003](#), p. 495-496).¹⁵ This again demonstrates that the quality of a volatility targeting strategy strongly depends on the accuracy of the inherent volatility forecasting model.

For the implementation of the target volatility strategy, we follow [Barroso and Santa-Clara \(2015\)](#) who use an annualized volatility target σ^{target} of 12%. Typically, target volatility levels used in the literature range from 5% to 40% as annualized volatility target. Clearly, the higher the volatility target the higher the exposure to the risky asset. Therefore, risk-averse or loss-averse investors will prefer a lower target volatility level, whereas risk-seeking investors will choose a high volatility target.¹⁶ [Bollerslev et al. \(2018\)](#) show how the volatility target can be derived as a function of the investor's risk aversion. An appealing alternative to choosing a fixed target volatility level was introduced by [Wang et al. \(2012\)](#) in a slightly different setting. The authors propose to switch between two target levels, where a high (low) target level is used when a low (high) risk regime is expected. Another alternative would be to use a time-varying volatility target that equals the long-term volatility ([Dreyer and Hubrich, 2019](#)).

¹⁵The authors use a mean-variance framework with a constant mean, which essentially translates into a volatility timing strategy, i.e. the weights of the assets are determined by estimates of conditional volatility and correlation solely. Although the authors use a multivariate setting – based on stocks, bonds, gold and cash – their findings, i.e. that volatility timing adds economic value and is positively influenced by more accurate models, is highly related to our approach based on stocks and cash.

¹⁶By choosing a quite low volatility target, the target volatility strategy usually does not need leverage. Since some investors are leveraged constrained, the weight w_t is often capped by a maximum allowed weight. [Strub \(2013\)](#), [Moreira and Muir \(2017\)](#) and the S&P Dow Jones Risk Control Indices use a cap of 150% ([Banerjee et al., 2016](#)). [Das and Uppal \(2004\)](#) and [Liu et al. \(2003\)](#) find that investors should face potential jump risk by not leveraging the risky asset, i.e. they should choose an equity cap of 100% (see also [Poon et al. \(2004\)](#)). We focus in this paper on uncapped target risk strategies since we are also interested in the accuracy of different forecasting methods. Using an equity cap would distort this examination. However, we show additional results for leverage constrained investors in Appendix D.4. See also [Moreira and Muir \(2017\)](#) on how an equity cap of 100% and 150% affects the utility gains of a mean-variance investor compared to the unconstrained strategy. The authors find substantial utility gains of volatility timing, even after a tight equity cap is set.

Since the volatility of the risky asset is usually not constant over time, the weight of the risky asset has to be rebalanced every day, which can lead to high transaction costs. However, [Dreyer and Hubrich \(2019\)](#), [Moreira and Muir \(2017\)](#) and [Bollerslev et al. \(2018\)](#) find that volatility targeting is still beneficial, even when realistic transaction costs are considered. [Harvey et al. \(2018, Exhibit 8\)](#) also find that transaction costs hardly influence the Sharpe Ratio of volatility targeting. Similarly, [Marquering and Verbeek \(2004\)](#) find that transaction costs only marginally impact the utility gains of dynamic trading strategies if short sales and leverage in the risky asset are not allowed. By definition, risk targeting is a long-only strategy and by choosing a moderate target volatility level, the strategy is seldom leveraged. Thus, in order to better assess the accuracy of different risk models, we reallocate the weight of the risky asset on a daily basis as it is also done in [Dreyer and Hubrich \(2019\)](#). Nevertheless, several possibilities can be used to lower the turnover and, as a consequence, the transaction costs of risk targeting (see for example [Kirby and Ost diek \(2012\)](#), [Moreira and Muir \(2017\)](#) and [Bollerslev et al. \(2018\)](#)).¹⁷ We follow the idea of [Perchet et al. \(2016\)](#) and show additional results of risk targeting for three different reallocation buffers in Section D.3.

3 Targeting a Constant Level of Tail Risk: Target VaR and CVaR Strategies

3.1 Managing Volatility versus Managing Tail Risk

As motivated in the previous section, due to the risk-averse nature of most investors, the demand for risk-managed investment strategies is very high. Risk management emerged as a major topic within the financial industry and is becoming more important for portfolio managers ([Berkowitz and O'Brien, 2002](#), [Christoffersen and Diebold, 2000](#)). In Appendix A, we have summarized

¹⁷One possibility to reduce transaction costs is to reallocate the weight less frequently, e.g. monthly or quarterly. [Moreira and Muir \(2017, Sec II.F\)](#) compare several rebalancing intervals and find a superior performance of the volatility timing strategy when portfolio weights are adjusted monthly instead of quarterly or annually. [Zakamulin \(2015, Exhibit 2\)](#) and [Perchet et al. \(2016\)](#) also find better results for strategies with shorter rebalancing intervals. However, in an earlier version of their paper, [Bollerslev et al. \(2018\)](#) find a trade-off between forecasting accuracy and transaction costs and conclude that it may be better not to trade every change in the optimal weight. The authors find better utility gains for the strategies that adjust the weight less frequently, especially when transaction costs are high and/or when models induce high day-to-day changes in the optimal weight. [Kirby and Ost diek \(2012\)](#) also find good results of volatility timing strategies that react less sensitive to volatility changes. In order to reduce transaction costs, [Taylor \(2014\)](#) presents a method that decreases the changes in the optimal weight.

several justifications and advantages of volatility targeting as a tool to manage a portfolio's risk. Thus, dynamically scaling the exposure to a portfolio of risky asset is an appealing portfolio risk management tool. However, managing volatility does not necessarily mean managing risk (Poon and Granger, 2003, Szegö, 2002). In this section, we summarize several disadvantages of volatility as a risk measure and we argue why managing a portfolio's downside risk is more appropriate.

Return distributions are typically skewed and fat-tailed (see Farinelli et al. (2008) among others).¹⁸ A negatively skewed return distribution coincides with a higher probability of extremely negative returns, whereas a positively skewed return distribution coincides with a higher probability of extremely positive returns. A fat-tailed distribution implies that extreme (positive or negative) returns are more likely than would be expected if returns were normally distributed (see Campbell and Hentschel (1992) among others). Hence, a negatively skewed and fat-tailed distribution makes extremely negative returns much more likely than anticipated by a normal distribution. Interestingly, Gormsen and Jensen (2017) find that skewness becomes more negative when kurtosis increases, i.e. both moments comove in directions that increase left tail risk. Furthermore, the authors find that times of a negative skewness and/or high kurtosis are typically followed by low future returns. Managing volatility, which is at least implicitly based on the assumption of normally distributed returns, can thus lead to too low weights in times of huge positive returns and too high weights in times of extremely negative returns. For that reason, Harvey and Siddique (2000, p. 1293) suggest that, instead of a mean-variance framework, a mean-variance-skewness framework should be used in an asset allocation analysis (see also Ghysels et al. (2016)). Guidolin and Timmermann (2008) also show that accounting for higher moments, like skewness and kurtosis, strongly affects the investor's asset allocation. Thus, higher moments should be incorporated in asset allocation decisions (see also Patton (2004), Ang et al. (2006a) and Jondeau and Rockinger (2012)). This is also confirmed by Xiong and Idzorek (2011) who highlight that accounting for skewness and kurtosis is crucial and superior

¹⁸Campbell and Hentschel (1992) explain the existence of negatively skewed and fat-tailed return distributions by the volatility feedback effect and the arrival of news. More precisely, news, both positive and negative, increase volatility and thus lower stock prices. Negative news additionally cause a stock price decline, whereas positive news dampen the volatility feedback induced stock price decline. Hence, the combination of these effects produces a negative skewness and a high excess kurtosis.

to mean-variance optimization, especially in times of extremely negative returns. A simple way to incorporate higher moments in asset allocation decisions is to use downside risk measures as portfolio risk management tool. Downside risk measures typically increase when skewness becomes more negative and/or kurtosis increases ([Bali et al., 2009](#)). Managing a portfolio's risk based on downside risk measures has the advantage that higher moments are considered, without relying on noisy estimates of skewness and kurtosis ([Ghysels et al., 2016](#)). Downside risk based portfolio allocation methods have been frequently used in the financial literature and have been compared to mean-variance optimized portfolios. For example, [Farinelli et al. \(2008\)](#) show that maximizing the Sharpe Ratio, i.e. maximizing the mean-variance trade-off, leads to a lower portfolio performance than maximizing the mean-downside risk trade-off. [Jarrow and Zhao \(2006\)](#) compare mean-variance optimized portfolios with mean-downside risk optimized portfolios and find huge differences in both portfolios when asset return distributions are non-normally distributed. Similarly, using hedge fund data, [Agarwal and Naik \(2004\)](#) compare a mean-downside risk framework with the mean-variance framework. The authors demonstrate that the mean-variance framework significantly underestimates the funds' downside risk and produces much higher losses during downturn periods. Generally, managing volatility is only suitable if asset returns are normally distributed or if investors have quadratic utility (see [Agarwal and Naik \(2004\)](#) and [Bali et al. \(2009\)](#) and references therein). This is confirmed by [Jondeau and Rockinger \(2006\)](#) who show that mean-variance portfolios and portfolio allocations that account for higher moments are nearly indistinguishable when returns are approximately normally distributed. However, both approaches produce significantly diverse allocations for non-normally distributed returns. [Packham et al. \(2017\)](#) manage a portfolio's risk using the difference between Value at Risk (VaR) forecasts based on a normality assumption and distributions that account for fat tails and skewness. The authors find huge improvements of this approach compared to buy-and-hold and other tail risk-protection strategies. Thus, portfolio allocation methods that incorporate the assets' non-normalities are superior to volatility based allocation methods. Furthermore, [Campbell and Hentschel \(1992\)](#), [Jondeau and Rockinger \(2003\)](#), [Harvey and Siddique \(1999\)](#) and [Bali et al. \(2008\)](#) show that conditional skewness

and kurtosis are highly time-varying. Hence, frequently reallocating the risky asset's weight based on an estimate of the current downside risk, and hence incorporating the time-variation in higher moments, seems crucial. Volatility based portfolios or simple static portfolio allocations do not incorporate the time-variation in higher moments. [Jondeau and Rockinger \(2012\)](#) show that incorporating the time-variation in skewness and kurtosis is crucial in portfolio selection problems and that higher moment timing outperforms volatility timing. Similarly, [Cuoco et al. \(2008\)](#) demonstrate that portfolio weights should be readjusted frequently based on estimates of the assets' tail risk.

Besides the existence of skewed and fat-tailed return distributions and the importance of incorporating this observation in asset allocation decisions, [Scott and Horvath \(1980\)](#) theoretically show that, under some assumptions, investors have preferences for higher (or positive) skewness and lower kurtosis (see also [Guidolin and Timmermann \(2008\)](#) and [Bali et al. \(2009\)](#)).¹⁹ More generally, investors have a preference for odd moments, e.g. higher returns and positive skewness, but dislike even moments like variance and kurtosis. [Bali et al. \(2009\)](#) show that higher downside risk predicts lower future skewness. Similarly, [Kelly and Jiang \(2014\)](#) show that an increase of tail risk predicts higher kurtosis and lower skewness of future returns. Hence, investors with preferences as in [Scott and Horvath \(1980\)](#) should lower the exposure to the risky asset if tail risk – not necessarily volatility – increases. Generally, downside risk measures increase if the return distribution is leptokurtic or negatively skewed ([Bali et al., 2009](#), [Ghysels et al., 2016](#)). Consequently, by managing downside risk instead of volatility, a higher kurtosis and/or a more negative skewness of the risky asset's return distribution induces a lower weight of the risky asset and fits well to these investors' preferences. Furthermore, investors do not only have preferences for higher moments, but are also concerned about the occurrence of extreme crashes. [Bollerslev and Todorov \(2011, p. 2187\)](#) find that the compensation of tail risk –

¹⁹See also [Kraus and Litzenberger \(1976\)](#), [Harvey and Siddique \(2000\)](#) and [Patton \(2004\)](#) on the preference of positive skewness. [Kraus and Litzenberger \(1976\)](#) extend the traditional CAPM to a three moment CAPM that includes mean, variance and skewness. [Harvey and Siddique \(2000\)](#) extend this model to a conditional version. See also Section I.C in [Harvey and Siddique \(2000\)](#) on the geometry of the three moment efficient portfolios, where investors demand higher expected returns for holding negatively skewed assets. [Guidolin and Timmermann \(2008\)](#) examine the optimal asset allocation under four-moment preferences and regime switching and demonstrate that the asset allocation under four-moment preferences differs from the asset allocation of a mean-variance investor. See also [Jondeau and Rockinger \(2006\)](#), [Jondeau and Rockinger \(2012\)](#) and [Lempérière et al. \(2017\)](#) and references therein on preferences for higher moments and implications on asset allocation decisions.

called “crash-o-phobia” by the authors – is extremely high and much higher than the compensation for volatility, i.e. investors fear tail risk much more than volatility (see also [Bollerslev et al. \(2015\)](#) and [Chabi-Yo et al. \(2018\)](#)). This is also confirmed by the earlier work of [Lee and Rao \(1988\)](#) who find that investors are more concerned about downside risk and that managing volatility is only sufficient when asset returns follow a symmetric distribution (see also [Szegö \(2002\)](#) and [Strub \(2013\)](#)).²⁰ Hence, investors are not concerned about return deviations from a mean but about extremely negative returns, which are described by higher moments and rare tail events (see [Lempérière et al. \(2017\)](#) and references therein). Similarly, in a utility based setting, [Bali et al. \(2009, p. 892\)](#) find that “investors dislike VaR”. Furthermore, most investors are loss-averse, i.e. they weight losses higher than gains ([Benartzi and Thaler, 1995](#)). Loss-averse investors have a high demand for portfolio insurance methods that avoid huge losses and these investors seek for risk reduction methods, especially in times of high market downturns ([Aït-Sahalia and Brandt, 2001](#), [Ang et al., 2006a](#), [Bollerslev and Todorov, 2011](#), [Chabi-Yo et al., 2018](#)). Consequently, for loss-averse investors, controlling downside risk instead of volatility, i.e. controlling negative returns instead of return deviations, is crucial in order to increase their utility (see [Aït-Sahalia and Brandt \(2001, p. 1298\)](#), [Ang et al. \(2005\)](#), [Ang et al. \(2006a\)](#) and references therein). [Aït-Sahalia and Brandt \(2001, p. 1315\)](#) state that the theory of loss aversion is related to the literature on downside risk-based investment decisions. Similarly, [Jarrow and Zhao \(2006\)](#) motivate that loss-averse investors should manage downside risk instead of volatility when asset return distributions are non-normally distributed. Timing downside risk instead of volatility also fits well to safety-first investors who are concerned about avoiding financial disasters (see [Arzac and Bawa \(1977\)](#), [Bali et al. \(2009\)](#), [van Oordt and Zhou \(2016\)](#) and references therein).

As summarized in Appendix A, due to the increase of correlations in bear markets, assets typically crash together. This reduces the benefits of diversification just when it is most needed, i.e. diversification is an inappropriate portfolio risk management tool ([Ang and Chen, 2002](#), [Butler and Joaquin, 2002](#), [Karolyi and Stulz, 1996](#), [Longin and Solnik, 2001](#), [Poon et al., 2004](#)).

²⁰For most investors, “risk” is associated with low or even negative returns. Describing risk by volatility does not differentiate between positive or negative returns (see [Lee and Rao \(1988, p. 452\)](#) or [Poon and Granger \(2003, p. 480\)](#)).

For example, [Chabi-Yo et al. \(2018\)](#) find a stronger asymptotic dependence in the left tail of stock returns than in the right tail, i.e. stocks tend to crash simultaneously. In particular, the left tail dependence increases in periods of market crashes ([Chabi-Yo et al., 2018](#), Figure 2). Therefore, lowering the exposure to the risky asset in bear markets is crucial in order to manage the risk of the portfolio. However, by using Extreme Value Theory (EVT) and thus measuring tail risk, [Longin and Solnik \(2001\)](#) show that increases in correlations do not necessarily coincide with increases in volatility, but with the occurrence of huge negative returns. Thus, the portfolio's exposure in bear markets should be better determined by the portfolio's tail risk instead of the portfolio's volatility. This is line with [Poon et al. \(2004\)](#) and [Kelly and Jiang \(2014\)](#) who find that volatility standardized returns still exhibit significant tail dependency and tail risk. Additionally to the simultaneous increase of correlations, [Jondeau and Rockinger \(2003\)](#) show that skewness and kurtosis of different risky assets also comove, i.e. large (negative) returns in different risky assets tend to occur simultaneously. As a consequence, simply combining several risky assets or managing volatility does not necessarily reduce the occurrence of extremely negative returns. Similarly, [Gormsen and Jensen \(2017\)](#) find that skewness and kurtosis of the same asset typically co-move, i.e. kurtosis increases when skewness becomes more negative, which eventually increases the asset's crash risk. These periods often occur when market volatility is low, i.e. in low volatile periods, risk "hides in the tails". [Ghysels et al. \(2016\)](#) also find that skewness is typically hidden in the tails and that skewness in the tails has a high impact on portfolio allocations. Based on this finding, [Gormsen and Jensen \(2017\)](#) show that volatility targeting strategies still exhibit a high tail risk, i.e. managing volatility does not mean managing extremely negative returns. Thus, volatility targeting is an inappropriate crash risk mitigation tool. This especially holds for periods with an extremely high crash risk. For example, [Liu et al. \(2003\)](#) and [Das and Uppal \(2004\)](#) find that in times of huge price jumps, like the global financial crisis, negative skewness and kurtosis are higher than in normal times, which is again not captured by volatility targeting. Similarly, [Jarrow and Zhao \(2006\)](#) show that the portfolio allocation of volatility and downside risk managed strategies can be vastly different when returns exhibit price jumps. These rare tail events are not predictable and cannot be completely avoided

by managing risk dynamically ([Bollerslev and Todorov, 2011](#)).²¹ However, rare events occur in the tail of the loss distribution and are often accompanied with changes in moments higher than volatility ([Poon et al., 2004](#), p. 582). To better manage the potential event risk, combining an estimation method that reflects the current market condition, measured by a dynamic volatility model, combined with an estimation method that directly models the tail of the distribution, like EVT, should be used instead of volatility alone ([Longin, 2000](#)).

Another advantage of volatility targeting is that choosing the weight of the risky asset inversely to the asset's volatility is a simple tool to reduce left tail risk and drawdowns ([Dreyer and Hubrich, 2019](#), [Harvey et al., 2018](#)). [Benson et al. \(2014, p. 96\)](#) state that, even in the absence of a negative risk-return relation, the mitigation of drawdowns is a main driver of the profitability of volatility targeting.²² In order to achieve a high long-term performance, due to the asymmetric behavior of compounded returns, mitigating highly negative returns is more crucial than achieving highly positive returns.²³ This is also confirmed by the results of [Barroso and Santa-Clara \(2015\)](#) who find that the superior performance of the volatility managed momentum strategy is significantly driven by the huge drawdown reduction (see also [Moreira and Muir \(2017\)](#) and [Barroso and Maio \(2017\)](#)). Thus, drawdown reduction is a main driver of the superior performance of the target volatility strategy. Since asset returns are usually non-elliptically distributed, managing volatility underestimates the potential of extreme losses ([Szegö, 2002](#), p. 1255). Consequently, managing downside risk instead of volatility should be more successful in mitigating drawdowns and should eventually result in an even better (risk-adjusted) performance compared to the target volatility and buy-and-hold strategies.²⁴

²¹Systematic event risk, like unpredictable jumps, affect the allocation between the risky and the riskless asset (see [Poon et al. \(2004, p. 602\)](#) and [Das and Uppal \(2004\)](#)). In order to account for the potential of unpredictable price jumps, [Liu et al. \(2003\)](#) suggest that investors should avoid leveraged positions.

²²[Benson et al. \(2014\)](#) compare arithmetic and geometric returns and find that the enhanced risk-return profile of volatility targeting comes from avoiding huge negative returns and not from a negative relation between risk and future returns. [Dachraoui \(2018\)](#) theoretically shows that a negative risk-return relation is not needed to provide an enhanced risk-return profile of risk targeting.

²³For example, a return of -50% has to be compensated by a return of 100% to obtain a compounded return of zero. In contrast, returns of -25% and 50% produce a compounded return of $12,5\%$. Thus, avoiding extremely high negative returns is beneficial for long-term investors, since high losses have to be compensated by even higher returns. As stated above, this also fits well to the loss aversion of most investors.

²⁴This is confirmed by [Strub \(2013, p. 6\)](#) who finds that "larger than normal tail risk is partly responsible for the outsized drawdowns experienced in market downturns [...], thus being able to accurately measure and control it is likely to yield significant improvements in risk adjusted performance" (see also [Hocquard et al. \(2013\)](#)).

As summarized above, the demand for tail risk hedging strategies is high, since these strategies fit well to the preferences of most investors and, by reducing left tail risk and drawdowns, these approaches deliver an enhanced risk-return profile. The main approaches to reduce the tail risk of a portfolio are derivative based and cash based strategies (see [Strub \(2013, p. 1\)](#), [Asvanunt et al. \(2015\)](#) and [Happersberger et al. \(2019\)](#)). Derivative based strategies achieve downside risk protection by buying or selling derivatives, e.g. options or futures, on the risky asset. Cash based strategies, as the here presented target risk strategies, dynamically allocate the wealth invested in the risky and riskless asset, based on the expected risk of the risky asset and are related to portfolio insurance strategies like CPPI ([Happersberger et al., 2019](#)). [Asvanunt et al. \(2015, Exhibit 4\)](#) find that strategies that dynamically readjust their portfolio allocation outperform strategies that use options as tail risk hedging instrument. The reason for this finding is that an option-based hedging approach works well during crises, but is too expensive in periods following a crisis ([Asvanunt et al., 2015, Exhibit 9](#)). Thus, cash based tail risk hedging strategies are appealing for practical implementations. However, so far, almost all studies on cash based tail risk hedging strategies focused on allocating money based on a forecast of volatility instead of a forecast of tail risk.²⁵ Similarly, there exists a huge literature on the economic value of volatility timing, whereas the economic value of *downside risk timing* is hardly examined ([Basak and Shapiro, 2001](#)). [Cuoco et al. \(2008\)](#) find that dynamically reallocating the amount invested in several assets based on downside risk is beneficial and superior to static approaches or approaches that do not account for downside risk. Therefore, we will later assess the economic value of downside risk timing and compare it to the economic value of volatility timing.

To account for the above mentioned drawbacks of the target volatility strategy, we next present the target Value at Risk (target VaR) and target Conditional Value at Risk (target CVaR) strategies. These strategies aim to achieve a constant VaR or CVaR of the portfolio over time.

²⁵[Strub \(2013\)](#) and [Happersberger et al. \(2019\)](#) use a cash based tail risk hedging strategy that relies on a similar weighting as in the target volatility strategy, but replaces the volatility in Equation (4) by an estimate of the risky asset's downside risk. Essentially, as we will see later, these strategies do not aim to target a constant level of portfolio risk over time, and hence do not belong to the class of risk targeting strategies. See also [Basak and Shapiro \(2001\)](#), [Alexander and Baptista \(2004\)](#), [Cuoco et al. \(2008\)](#) and [Packham et al. \(2017\)](#) for other tail risk based investment strategies.

VaR is a widely used tool to measure market risk (Alexander and Baptista, 2004, Bali et al., 2008, Berkowitz et al., 2011, Berkowitz and O'Brien, 2002, Cuoco et al., 2008). However, CVaR has recently become more important from a regulatory and practical view (Du and Escanciano, 2016). By construction, both strategies, the target VaR and target CVaR strategy, automatically manage the downside risk of the risky asset, and thus correct for the drawbacks of the target volatility strategy.²⁶

3.2 Target Value at Risk Strategy

This section derives the optimal weight for the target VaR strategy and shows how the VaR of the risky asset can be estimated. We again consider an investor who invests w_t in a risky asset and $1 - w_t$ in a riskless asset. The goal of the target VaR strategy is to determine w_t such that the portfolio achieves a constant Value at Risk over time. By definition, the VaR at a significance level α is the maximum loss, defined as the negative daily return, that is only exceeded with a probability of $100 \cdot \alpha\%$ (see Szegö (2002), Yamai and Yoshida (2005) among others). In order to achieve a constant Value at Risk level $\text{VaR}_\alpha^{\text{target}}$, the investor specifies the desired (daily) Value at Risk level, i.e. the critical loss or loss threshold the investor is willing to accept, as well as the corresponding significance level α , i.e. the exceedance probability. For example, a target VaR level $\text{VaR}_\alpha^{\text{target}}$ of 1% with a corresponding significance level α of 5% translates into a strategy, where daily returns below -1% only occur with a probability of 5%.²⁷ In other words, daily returns should be higher than -1% with a probability of 95%.²⁸ Hence, the target

²⁶These tail risk targeting strategies are similar to the approach of Basak and Shapiro (2001, p. 376) and Cuoco et al. (2008) who incorporate downside risk measures in an asset allocation framework, but instead of targeting a constant level of tail risk, the authors require the downside risk to be below some prespecified limit (see also Wang et al. (2012) and Alexander and Baptista (2004)). Similar to our tail risk targeting strategies, this downside risk managed strategy also allocates wealth between a riskless asset and an (optimal) portfolio of risky assets (see Cuoco et al. (2008, Remark 3) for example). Basak and Shapiro (2001, p. 376) call this approach a softer form of portfolio insurance. We will motivate in Section 3.4 that the target VaR and target CVaR strategies can be derived as optimal trading strategies under downside risk limits.

²⁷By defining the target VaR strategy in terms of a potential loss of wealth, this strategy can be easily interpreted by retail investors, whereas a given volatility target level should be less clear for most retail investors. VaR based investment strategies are already available for retail investors. See, for example, the strategies offered by Scalable Capital (<http://www.scalable.capital>).

²⁸By choosing low values of $\text{VaR}_\alpha^{\text{target}}$ and α , this strategy can also be used by hedge fund managers as an alternative to absolute return strategies. These strategies typically have absolute return targets that are independent of the current market environment, whereas most mutual fund managers have relative return targets that are compared to a benchmark asset (see Fung and Hsieh (1997) and Agarwal and Naik (2004)). For example, assuming 250 trading days per year and by choosing $\text{VaR}_\alpha^{\text{target}} = 0.5\%$ and $\alpha = 0.4\%$, a daily return below -0.5%

VaR strategy has the advantage that it manages extreme losses instead of loss deviations. In particular, investors can choose a target VaR strategy that fits well to their acceptable loss limit, where the choice of α and $\text{VaR}_\alpha^{\text{target}}$ strongly depends on the investor's preferences and degree of risk aversion (Alexander and Baptista, 2004).

Since tail risk measures are typically defined based on loss variables, we define the daily portfolio loss at day t as

$$L_t^P := -R_t^P. \quad (9)$$

Similarly, the day t loss of the risky asset is defined as $L_t := -R_t$. Thus, the portfolio loss can be written as

$$L_t^P = w_t \cdot L_t - (1 - w_t) \cdot R_t^f. \quad (10)$$

The day t VaR of the portfolio for a significance level α , denoted by $\text{VaR}_\alpha^{P,t}$, is defined through the relation²⁹

$$\mathbb{P}(L_t^P \leq \text{VaR}_\alpha^{P,t} \mid \mathcal{F}_{t-1}) = 1 - \alpha. \quad (11)$$

Therefore, the portfolio VaR is given by the $(1 - \alpha)$ -quantile of the (conditional) portfolio loss distribution, denoted by $F_{L_t^P \mid \mathcal{F}_{t-1}}^{-1}(1 - \alpha)$, i.e. $\text{VaR}_\alpha^{P,t} = F_{L_t^P \mid \mathcal{F}_{t-1}}^{-1}(1 - \alpha)$. In Appendix B, we show that the portfolio VaR is given by

$$\text{VaR}_\alpha^{P,t} = w_t \cdot \text{VaR}_\alpha^t - (1 - w_t) \cdot R_t^f, \quad (12)$$

where $\text{VaR}_\alpha^t := F_{L_t \mid \mathcal{F}_{t-1}}^{-1}(1 - \alpha)$ denotes the day t VaR of the risky asset.³⁰ In order to achieve a constant portfolio VaR level $\text{VaR}_\alpha^{\text{target}}$ over time, i.e. $\text{VaR}_\alpha^{P,t} = \text{VaR}_\alpha^{\text{target}}$ for all t , the weight

should only occur once a year (see Figure III in Appendix D for a performance chart of this strategy). Thus, the target VaR strategy aims to constantly produce returns with limited downside risk, regardless of the current market environment. This strategy should be advantageous to other hedge funds strategies since some hedge funds strategies exhibit huge losses during market downturns and bear significant tail risk (Agarwal and Naik, 2004). Similarly, Agarwal and Naik (2004) find low correlations between hedge funds and the market in upmarkets, but high positive correlations during market downturn periods and that hedge funds often resemble a short put payoff profile. Investors who are interested in an absolute return strategy should therefore better use a conservative tail risk targeting strategy.

²⁹Throughout the paper, we assume that the loss variables L_t^P and L_t are continuously distributed.

³⁰The representation in Equation (12) can be directly seen by positive homogeneity and translation invariance of VaR (Szegö, 2002, p. 1259-1260).

of the risky asset has to be chosen as

$$w_t = \frac{\text{VaR}_\alpha^{\text{target}} + R_t^f}{\text{VaR}_\alpha^t + R_t^f}. \quad (13)$$

By construction, since VaR_α^t and R_t^f are $\mathcal{F}_{t-1}$ -measurable, the weight w_t is known at time $t-1$. Furthermore, the weight of the risky asset is increased if the downside risk of the risky asset, measured by its VaR, is expected to be low and vice versa. Thus, the tail risk of the portfolio is managed by allocating money between the risky and the riskless asset, based on the risky asset's VaR.³¹ When a market crash becomes more likely, the amount invested in the risky asset is reduced. When market risk declines, the amount invested in the risky asset is subsequently increased.³²

Similar to the volatility, the VaR of the risky asset is not observable, and hence a forecast of the risky asset's VaR is needed.³³ We use again several models to estimate the risky asset's VaR. As first approach, we use the unconditional Historical Simulation (HS) based on a rolling window of n days, i.e. we estimate VaR_α^t by the empirical $(1-\alpha)$ -quantile of the past n daily losses (see [Kuester et al. \(2006, p. 56-57\)](#) or [Halbleib and Pohlmeier \(2012\)](#)). More formally, for a sample $l_{t-n}, \dots, l_{t-1}$ of n realized losses, the day t VaR is given by

$$\widehat{\text{VaR}}_\alpha^t = l_{([n(1-\alpha)]), t-1}, \quad (14)$$

where $l_{(1), t-1} \leq \dots \leq l_{(n), t-1}$ denotes the order statistics of the sample $l_{t-n}, \dots, l_{t-1}$. This estimation method is frequently used by banks and practitioners ([Berkowitz et al., 2011](#), [Berkowitz](#)

³¹Many tail risk hedging strategies that aim to reduce a portfolio's tail risk only work well when markets exhibit huge drawdowns. In upmarkets, these tail risk hedging strategies usually perform worse than a simple buy-and-hold strategy, which translates into a worse overall performance ([Hocquard et al., 2013](#)). The target VaR strategy has the advantage that this strategy increases the weight of the risky asset as downside shrinks, and hence captures the upside potential while downside risk is managed ([Wang et al., 2012, p. 38](#)). [Dopfel and Ramkumar \(2013\)](#) show that the periods following high risk periods are the most attractive ones (see also [Muir \(2017\)](#)). Hence, similar to portfolio insurance strategies, risk targeting delivers an option-like return profile (see also [Fung and Hsieh \(1997\)](#) who found a similar behavior for dynamic trading strategies used by hedge fund managers).

³²The target VaR strategy manages a portfolio's crash risk by dynamically scaling the exposure to the risky asset. Another approach to manage a portfolio's crash risk would be to dynamically change the portfolio allocation. For example, [Chabi-Yo et al. \(2018\)](#) show in a cross-sectional setting that assets with a lower crash sensitivity outperform during times of market distress but underperform when markets are calm (see also [van Oordt and Zhou \(2016\)](#)). In particular, assets with a high crash sensitivity exhibit higher returns after huge market declines. Thus, during a crash period, the amount invested in crash-sensitive assets should be decreased and should then subsequently be increased when the crash risk declines. This strategy could additionally be combined with the target VaR strategy (see [Rickenberg \(2020b\)](#) who uses a similar approach for long-short strategies).

³³See [Taylor \(2005, Sec. 3\)](#) and [Kuester et al. \(2006\)](#) for a survey of VaR estimation models.

and O'Brien, 2002). Furthermore, [Atilgan et al. \(2020\)](#) also use this VaR estimation method and find that stocks with a higher VaR underperform stocks with a lower VaR. Thus, in a cross-sectional setting, there exists a negative VaR-return relation.³⁴

Historical Simulation relies on the assumption that the loss distribution can be estimated by the empirical distribution of past losses ([McNeil and Frey, 2000](#), p. 273). Hence, Historical Simulation is based on the assumption that losses are iid. In practice, this assumption does not hold for losses of most risky assets, since asset returns (or losses respectively) are known to exhibit a time-varying volatility and volatility clustering ([Pritsker, 2006](#), p. 563). Further, [Pritsker \(2006\)](#) shows that Historical Simulation does not respond to the 1987 crash. Thus, Historical Simulation is an inappropriate tool to manage extreme market crashes. Generally, most VaR estimation models that are frequently used in the financial industry, like Historical Simulation, work well in calm periods but fail to produce accurate risk forecast in times of high downside risk, just when reliable forecasts are most needed ([Berkowitz et al., 2011](#), [Halbleib and Pohlmeier, 2012](#)). As a consequence, using such static estimation models as portfolio risk management tool can translate in high probabilities of extreme losses ([Cuoco et al., 2008](#)). Hence, using Historical Simulation in the context of a target VaR strategy can translate in a high exposure to the risky asset in times when financial markets are very risky, although a good risk-managed investment strategy should exhibit a low weight in the risky asset during times of high market risk. For that reason, a fast adapting estimation model that reflects the current market environment is crucial for a good performance of the target VaR strategy (see also [Taylor \(2014\)](#) and [Bollerslev et al. \(2018\)](#) who find a similar result for volatility managed portfolios). However, estimating VaR by Historical Simulation is easy, straightforward and is the current industry standard for estimating VaR (see [Berkowitz et al. \(2011\)](#) and references therein). Thus, this approach is in particular interesting for index providers and practitioners who are interested in simple target VaR strategies. Consequently, the target VaR strategy based on Historical Simulation deals as a benchmark strategy for more complex target VaR strategies.

Due to the disadvantages of the HS approach, we use three other VaR forecasting models

³⁴[Bali et al. \(2009\)](#) cannot confirm this finding in a time-series setting. However, as mentioned earlier, a negative risk-return relation is not needed for the risk targeting strategy.

that are based on a volatility forecast of the EWMA or GARCH(1,1) model. When estimating quantile risk measures, [McNeil and Frey \(2000, p. 273-274\)](#) propose to reflect the current volatility background, estimated by a dynamic volatility model, and to account for heavy tails in the conditional loss distribution (see also [Longin \(2000\)](#)). [Christoffersen and Diebold \(2000\)](#) find that volatility is highly forecastable for short horizons of less than 10 days. Thus, these volatility forecasts are highly valuable and should be incorporated when short-term risk is managed. In order to incorporate the risky asset's volatility in the VaR estimation, we assume that returns are given as in Equation (6). Under this assumption, the day t VaR of the risky asset is given by

$$\text{VaR}_\alpha^t = \sigma_t \cdot F_{L^*}^{-1}(1 - \alpha), \quad (15)$$

where $L_t^* := -Z_t$ is a random variable representing a standardized loss with expectation zero, variance one and $F_{L^*}^{-1}(1 - \alpha)$ denotes the $(1 - \alpha)$ -quantile of L_t^* .³⁵ We also denote dynamic risk models that account for the current volatility as conditional models and static models, like Historical Simulation or HSD, that are based on the assumption that returns are iid as unconditional models ([Longin, 2000](#)). The forecast $\widehat{\text{VaR}}_\alpha^t$ for the day t VaR based on the information at time $t - 1$ is then given by

$$\widehat{\text{VaR}}_\alpha^t = \hat{\sigma}_t \cdot \hat{F}_{L^*,t}^{-1}(1 - \alpha), \quad (16)$$

where $\hat{F}_{L^*,t}^{-1}(1 - \alpha)$ denotes the estimator of $F_{L^*}^{-1}(1 - \alpha)$, given the available information at day $t - 1$. $\widehat{\text{VaR}}_\alpha^t$ is then estimated in the following way using a two-stage approach as described by [McNeil and Frey \(2000, p. 277\)](#). In the first stage, we estimate the volatility $\hat{\sigma}_t$ using the EWMA or the GARCH(1,1) model given in Equation (7) or (8), respectively. The parameters of the GARCH(1,1) model are estimated using a Quasi Maximum Likelihood (QML) approach, i.e. assuming a standard normal distribution for the innovations Z_t . In the second stage, the standardized losses, i.e. $l_t^* = -R_t/\hat{\sigma}_t$ are calculated, which are used to calculate $\hat{F}_{L^*,t}^{-1}(1 - \alpha)$.³⁶

³⁵By assumption, the quantile $F_{L^*}^{-1}(1 - \alpha)$ of the standardized loss $L_t^* := -Z_t = -R_t/\sigma_t$ does not depend on t ([McNeil and Frey, 2000, p. 276](#)). As an extension of this assumption, we follow [Jondeau and Rockinger \(2003\)](#) and [Bali et al. \(2008\)](#) and use a more sophisticated approach below that does not assume that Z_t is iid.

³⁶The volatility σ_t , as described in Section 2, is calculated using daily returns and represents a forecast for the return volatility. However, from $\text{var}(R_t|\mathcal{F}_{t-1}) = \text{var}(-R_t|\mathcal{F}_{t-1})$ it follows that the volatility forecast $\hat{\sigma}_t$ can be directly used as a forecast for the volatility of the losses.

In this context, VaR is often estimated by assuming a standard normal distribution for Z_t , i.e. $\hat{F}_{L^*,t}^{-1}(1-\alpha)$ is given by the $(1-\alpha)$ -quantile of the standard normal distribution. However, even after standardizing returns or losses by a time-varying volatility, these observations still exhibit a non-zero skewness and fatter tails than the normal distribution (see [Campbell and Hentschel \(1992\)](#), [Bollerslev et al. \(1992\)](#), [Glosten et al. \(1993\)](#), [Harvey and Siddique \(1999\)](#), [Ghysels et al. \(2016\)](#), [Jondeau and Rockinger \(2003\)](#) and [Bali et al. \(2008\)](#)).³⁷ Similarly, [Kelly and Jiang \(2014\)](#) find that volatility standardized returns still exhibit significant tail risk. Therefore, we estimate $\hat{F}_{L^*,t}^{-1}(1-\alpha)$ based on a sample of n past standardized losses, denoted by $l_{t-n}^*, \dots, l_{t-1}^*$, using three different methods that account for this stylized fact.

First, we use the Filtered Historical Simulation (FHS) approach ([Barone-Adesi et al., 2008, 1999](#)), i.e. we estimate $\hat{F}_{L^*,t}^{-1}(1-\alpha)$ by the empirical $(1-\alpha)$ -quantile of the standardized losses $l_{t-n}^*, \dots, l_{t-1}^*$ ([Kuester et al., 2006](#), p. 57). The estimator for the day t VaR of the risky asset is then given by

$$\widehat{\text{VaR}}_{\alpha}^t = \hat{\sigma}_t \cdot l_{([n(1-\alpha)]),t-1}^*, \quad (17)$$

where $l_{(1),t-1}^* \leq \dots \leq l_{(n),t-1}^*$ denotes the order statistics of the sample $l_{t-n}^*, \dots, l_{t-1}^*$.³⁸ The FHS approach easily combines the conditional heteroscedasticity and the non-normality of asset returns in a simple estimation method without relying on any distributional assumption ([Giannopoulos and Tunaru, 2005](#), p. 983).

As second estimation method, we use the Extreme Value Theory (EVT) approach of [McNeil and Frey \(2000\)](#).³⁹ The EVT approach is based on the assumption that the tail of the distribution of the standardized losses can be described by a Generalized Pareto Distribution (GPD).⁴⁰ The

³⁷This stylized fact holds for standardized equity returns but not for foreign exchange rates ([Bollerslev et al., 1992](#), p. 38).

³⁸[Pritsker \(2006\)](#) states that the choice of n is not straightforward. However, $n = 1000$ is frequently used in applications (see [Kuester et al. \(2006\)](#) or [Christoffersen \(2012\)](#)). See also [Halbleib and Pohlmeier \(2012\)](#) on how the window size impacts estimation results of VaR forecasts.

³⁹See also [McNeil et al. \(2015, Section 5.2.6\)](#) for a survey on estimating quantile risk measures when a GARCH volatility model is used in the first stage and [Kuester et al. \(2006, Section 1.4\)](#) for a good summary on how to estimate VaR using EVT. EVT is also used by [Allen et al. \(2012\)](#), [Poon et al. \(2004\)](#), [Longin and Solnik \(2001\)](#), [Kelly and Jiang \(2014\)](#), [Longin \(2000\)](#) and [van Oordt and Zhou \(2016\)](#) in other related financial topics.

⁴⁰The EVT approach assumes that losses are iid. Hence, when working with short horizons, as done in this paper, standardizing losses by a time-varying volatility is crucial for the EVT approach. A well-known stylized fact states that daily losses are far away from iid, whereas the iid assumption fits quite well to standardized losses ([Kuester et al., 2006](#), p. 62). When working with longer time horizons, the EVT approach can be directly applied to the non-standardized losses ([McNeil and Frey, 2000](#)).

tail of this distribution is defined in terms of a threshold u .⁴¹ Based on this threshold, the EVT approach assumes that standardized losses above the threshold u follow a GPD that is defined as

$$G_{\xi,\beta}(y) = \begin{cases} 1 - (1 + \xi y/\beta)^{-1/\xi}, & \text{if } \xi \neq 0 \\ 1 - \exp(-y/\beta), & \text{if } \xi = 0, \end{cases} \quad (18)$$

with $\beta > 0$. The support of this distribution is given by $y \geq 0$ if $\xi \geq 0$ and $0 \leq y \leq -\beta/\xi$ if $\xi < 0$ (McNeil and Frey, 2000, p. 280). The parameter ξ is usually called the *shape* parameter and β is called the *scale* parameter (McNeil et al., 2015, p. 147). For the estimation of $\hat{F}_{L^*,t}^{-1}(1 - \alpha)$, we again assume that the sample contains n standardized losses. The estimator $\hat{F}_{L^*,t}^{-1}(1 - \alpha)$ is then given by

$$\hat{F}_{L^*,t}^{-1}(1 - \alpha) = u + \frac{\hat{\beta}}{\hat{\xi}} \left(\left(\frac{\alpha n}{N_u} \right)^{-\hat{\xi}} - 1 \right), \quad (19)$$

where $\hat{\beta}$ and $\hat{\xi}$ are the Maximum Likelihood estimates and N_u denotes the number of standardized losses that exceed the threshold u (McNeil et al., 2015, p. 154 and 349). The VaR forecast is then given by Equations (16) and (19).

We will next use a further extension of the models presented above. The Historical Simulation approach assumes that returns are iid, which is not realistic in practice. In contrast, by assuming that only volatility standardized returns are iid, the EVT and FHS approaches are more realistic. However, several studies show that even this assumption is too restrictive, since even volatility standardized returns exhibit autoregressive patterns in the conditional skewness and kurtosis (Bali et al., 2008, Harvey and Siddique, 1999, Jondeau and Rockinger, 2003). Thus, we follow Jondeau and Rockinger (2003) and Bali et al. (2008) and use the EWMA and GARCH based approach combined with the skewed t distribution of Hansen (1994), where conditional skewness and kurtosis are modeled autoregressively. Similar to Equation (6), we assume that

⁴¹One drawback of EVT is the choice of the threshold u (Kellner and Röscher, 2016). If u is chosen too high, the estimation of the parameters is based on only few exceedance observations, making the estimation less precise. Choosing u too low contradicts the approximation in Equation (18), since this approximation only holds for the tails of the distribution (see Longin and Solnik (2001, Sec. II.A), Kuester et al. (2006, p. 62) and Yamai and Yoshida (2005, p. 1008)). Longin and Solnik (2001, Appendix 1) show how u can be optimally chosen based on Monte Carlo Simulations. Packham et al. (2017, p. 740) find that their VaR-based tail risk protection strategy is robust against changes in u . As in Kellner and Röscher (2016), we choose the threshold u as the 90%-quantile.

the daily return can be described by

$$R_t = \sigma_t \cdot Z_t, \quad Z_t \sim stsk(\eta_t, \lambda_t), \quad (20)$$

where $Z_t \sim stsk(\eta_t, \lambda_t)$ means that Z_t is skewed t distributed with mean zero, variance one and time-varying parameters η_t and λ_t . The skewed t distribution of [Hansen \(1994\)](#) is characterized by the pdf

$$f_{stsk}(z | \eta_t, \lambda_t) = \begin{cases} bc \left(1 + \frac{1}{\eta_t - 2} \left(\frac{bz+a}{1-\lambda_t} \right)^2 \right)^{-(\eta_t+1)/2} & \text{if } z < -\frac{a}{b} \\ bc \left(1 + \frac{1}{\eta_t - 2} \left(\frac{bz+a}{1+\lambda_t} \right)^2 \right)^{-(\eta_t+1)/2} & \text{if } z \geq -\frac{a}{b} \end{cases}, \quad (21)$$

where

$$a := 4\lambda_t c \frac{\eta_t - 2}{\eta_t - 1}, \quad b^2 := 1 + 3\lambda_t^2 - a^2, \quad c := \frac{\Gamma\left(\frac{\eta_t+1}{2}\right)}{\sqrt{\pi(\eta_t - 2)}\Gamma\left(\frac{\eta_t}{2}\right)}.$$

The parameters of this distribution are restricted to $\eta_t > 2$ and $-1 < \lambda_t < 1$ (see [Hansen \(1994, p. 710\)](#) and [Jondeau and Rockinger \(2003, p. 1702\)](#)). Further, for $\lambda_t = 0$, this distribution is symmetric and equals the standardized t distribution. For $\lambda_t > 0$ ($\lambda_t < 0$), the distribution is positively (negatively) skewed ([Hansen, 1994](#)). Moreover, skewness exists for $\eta_t > 3$ and kurtosis exists for $\eta_t > 4$ ([Jondeau and Rockinger, 2003](#)). [Jondeau and Rockinger \(2003\)](#) show that, although η_t is often referred as the parameter that determines kurtosis and λ_t is referred as the skewness parameter, both parameters, η_t and λ_t , affect both, skewness and kurtosis. In particular, the relation between the parameters and higher moments is highly non-linear. The parameters of the skewed t distribution are then modeled autoregressively, where we first model unrestricted parameters by

$$\tilde{\eta}_t = a_1 + b_1 R_{t-1} + c_1 \tilde{\eta}_{t-1}, \quad (22)$$

$$\tilde{\lambda}_t = a_2 + b_2 R_{t-1} + c_2 \tilde{\lambda}_{t-1}. \quad (23)$$

To guarantee that the standardized skewed t distribution is well defined, the parameters have to be rescaled to fulfill the conditions $\eta_t > 2$ and $-1 < \lambda_t < 1$. We follow [Jondeau and Rockinger \(2003\)](#) and [Bali et al. \(2008\)](#) and use a logistic transformation to guarantee that these restrictions hold. The parameters η_t and λ_t are then given by

$$\eta_t = 2 + \exp(\tilde{\eta}_t) \quad (24)$$

$$\lambda_t = \frac{2}{1 + \exp(-\tilde{\lambda}_t)} - 1. \quad (25)$$

The α -quantile of the skewed t distribution is given by

$$F_{stsk}^{-1}(\alpha | \eta_t, \lambda_t) = \begin{cases} \frac{1}{b} \left((1 - \lambda_t) \sqrt{\frac{\eta_t - 2}{\eta_t}} F_t^{-1}\left(\frac{\alpha}{1 - \lambda_t} | \eta_t\right) - a \right) & \text{if } \alpha < \frac{1 - \lambda_t}{2} \\ \frac{1}{b} \left((1 + \lambda_t) \sqrt{\frac{\eta_t - 2}{\eta_t}} F_t^{-1}\left(\frac{\alpha + \lambda_t}{1 + \lambda_t} | \eta_t\right) - a \right) & \text{if } \alpha \geq \frac{1 - \lambda_t}{2}, \end{cases} \quad (26)$$

where $F_t^{-1}(z | \eta_t)$ is the inverse of the t distribution's cdf $F_t(z | \eta_t) = \int_{-\infty}^z f_t(u | \eta_t) du$ (Jondeau and Rockinger, 2003). The t distribution's pdf with η_t degrees of freedom is given by

$$f_t(z | \eta_t) = \frac{\Gamma\left(\frac{\eta_t + 1}{2}\right)}{\Gamma\left(\frac{\eta_t}{2}\right) \sqrt{\pi \eta_t}} \left(1 + \frac{z^2}{\eta_t}\right)^{-(\eta_t + 1)/2}, \quad (27)$$

where $\Gamma(\cdot)$ denotes the Gamma function. The VaR forecast for day t is then again given by Equation (16), where

$$\hat{F}_{L^*,t}^{-1}(1 - \alpha) = -F_{stsk}^{-1}\left(\alpha | \hat{\eta}_t, \hat{\lambda}_t\right), \quad (28)$$

and $\hat{\eta}_t$ and $\hat{\lambda}_t$ denote the Maximum Likelihood estimates of η_t and λ_t .⁴²

Kuester et al. (2006) compare several VaR forecasting approaches and find that the GARCH-based EVT, FHS and skewed t distribution approaches always belong to the best conditional models, where the authors only use the skewed t distribution with constant parameters instead of time-varying parameters. The authors state that unconditional models, like Historical Simulation, fail to produce accurate VaR forecast and that only conditionally heteroskedastic models deliver acceptable VaR forecasts. Therefore, the authors prefer a conditional approach that accounts for the volatility dynamics. Furthermore, the authors find that the VaR violations of dynamic models are reasonably independent over time, which usually does not hold for Historical Simulation. This demonstrates the importance of a fast adapting model, which is crucial for the target risk strategies, since wrong risk timing translates into a high exposure to the risky asset when the market's downside risk is high and vice versa.⁴³ Moreover, McNeil and Frey (2000, p. 283) state that using a symmetric distribution, like the normal or the t -distribution,

⁴²The minus one in front of the quantile and the significance level α results since we fit the skewed t distribution to standardized returns instead of standardized losses.

⁴³We will come back to this point in Section 4.2, when we show how the accuracy of the target VaR strategies can be backtested.

underestimates the loss potential (see also [Szegö \(2002\)](#)). Similarly, [Kellner and Rösch \(2016\)](#) find that only models that account for fat tails and/or skewness deliver accurate VaR forecasts. Since standardized losses typically follow an asymmetric distribution, using an approach like EVT, FHS or skewed t is a better choice for modeling the right tail of the standardized losses ([Xiong and Idzorek, 2011](#)). In particular, directly modeling the right tail of the loss distribution, instead of modeling the whole distribution, usually gives a better fit and is superior when tail risk is forecasted ([Longin, 2000](#)). [McNeil and Frey \(2000, p. 290-291\)](#) compare the GARCH-EVT approach with the unconditional EVT, the GARCH-normal and the GARCH- t models and find that, in most cases, the GARCH-EVT model is superior to the benchmark models and that the GARCH-EVT model is the only model that is not rejected in all 15 cases. [Packham et al. \(2017\)](#) use several alternative generalized distributions for Z_t and find that only the GPD works well for tail risk management. [Halbleib and Pohlmeier \(2012\)](#) also find convincing results of combining a dynamic volatility model, like the GARCH(1,1) model, with an estimate of $F_{L*}^{-1}(1 - \alpha)$ that accounts for skewness, like the EVT approach or the skewed t -distribution. Similarly [Bali et al. \(2008\)](#) find good results of combining a dynamic volatility model with the skewed t -distribution. As a consequence, combining a dynamic volatility model with the EVT, FHS or skewed t approach should result in a portfolio VaR that is closer to the desired target VaR level than the portfolio VaR of the simple Historical Simulation approach.⁴⁴ As stated above, since a more stable portfolio risk is typically rewarded by higher performance gains, we expect a better risk-return profile for the conditional approaches. We answer the question which model delivers the most accurate portfolio VaR later, when we assess the accuracy of several target VaR strategies using the backtesting method described in Section 4.2.

Since R_t^f is (typically) small compared to $\text{VaR}_\alpha^{\text{target}}$ and VaR_α^t , the weight w_t of the target VaR strategy can be approximated by⁴⁵

$$w_t \approx \frac{\text{VaR}_\alpha^{\text{target}}}{\text{VaR}_\alpha^t}. \quad (29)$$

⁴⁴A possible extension to further improve the accuracy and robustness of VaR targeting in a simple manner could be to combine several VaR forecasting models as presented in [Halbleib and Pohlmeier \(2012\)](#). [Allen et al. \(2012\)](#) also use the average of different VaR forecasts made with different forecasting methods.

⁴⁵For our sample, the mean of R_t^f and VaR_α^t is 0.007% and 4.0766%, respectively. Thus, for our $\text{VaR}_\alpha^{\text{target}}$ of 1.947% and these averages, the weight based on Equation (13) would be 0.4785, whereas the approximated weight would be 0.4776.

The structure in Equation (29) is similar to the weight of the target volatility strategy, given in Equation (3), but the volatility of the risky asset is replaced by the VaR of the risky asset. The weighting in Equation (29) was also used by [Happersberger et al. \(2019\)](#). By using the approximation in Equation (29) and the decomposition of the VaR in Equation (15), the weight of the risky asset of the target VaR strategy can be approximated by

$$w_t \approx \frac{\text{VaR}_\alpha^{\text{target}}}{\text{VaR}_\alpha^t} = \frac{F_{L^*}^{-1}(1 - \alpha) \cdot \frac{\text{VaR}_\alpha^{\text{target}}}{F_{L^*}^{-1}(1 - \alpha)}}{\sigma_t \cdot F_{L^*}^{-1}(1 - \alpha)} = \frac{\sigma^{\text{target}}}{\sigma_t}, \quad (30)$$

with $\sigma^{\text{target}} := \text{VaR}_\alpha^{\text{target}} / F_{L^*}^{-1}(1 - \alpha)$. Therefore, the weight of a target VaR strategy can be approximated by the weight of a target volatility strategy, where the target volatility level is determined by the target VaR level and the $(1 - \alpha)$ -quantile of the standardized losses. Hence, one could argue that every target VaR strategy, based on the decomposition in Equation (15), can be approximated by a target volatility strategy.⁴⁶ Further, a constant portfolio volatility can also be achieved by controlling VaR instead of volatility. [Taylor \(2005\)](#) shows that incorporating higher moments in volatility forecasts is beneficial. This results since the shape of the conditional return distribution is not fix over time as shown by [Jondeau and Rockinger \(2003\)](#) and [Bali et al. \(2008\)](#). However, for several reasons, the approximation in Equation (30) has to be regarded carefully. First, Equation (29) is just an approximation, which is only exact for $R_t^f = 0$. Second, to transform a target VaR strategy into the corresponding target volatility strategy, the distribution of the standardized losses, or at least the quantile $F_{L^*}^{-1}(1 - \alpha)$, has to be known. In practice, both are unknown and this transformation is not directly feasible. As a rough approximation, quantiles of the standard normal distribution can be used. In this case, the target VaR strategy can be approximated by the target volatility strategy, using a target volatility level of

$$\sigma^{\text{target}} = \frac{\text{VaR}_\alpha^{\text{target}}}{N_{1-\alpha}}, \quad (31)$$

where $N_{1-\alpha}$ denotes the $(1 - \alpha)$ -quantile of the standard normal distribution.⁴⁷

⁴⁶This is a contrarian approach to [Taylor \(2005\)](#) who uses VaR forecasts based on the CAViaR model and Historical Simulation to obtain estimates of conditional volatility.

⁴⁷In our notation, the volatility target σ^{target} is denoted as an annualized volatility, whereas the VaR target is chosen as a daily loss. Thus, a target VaR strategy with $\alpha = 5\%$ and $\text{VaR}_\alpha^{\text{target}} = 1\%$ can be approximated by a target volatility strategy with an (annualized) volatility target of $\sigma^{\text{target}} = (0.01/1.645) \cdot \sqrt{252} = 9.6\%$.

3.3 Target CVaR Strategy

The target VaR strategy, presented in the previous section, has the advantage that the weight of the risky asset is a function of the expected downside risk instead of expected volatility. As stated above, focusing on downside risk management instead of volatility management has several advantages. VaR is the current industry standard when downside risk is measured and managed ([Bali et al., 2008](#), [Berkowitz et al., 2011](#), [Berkowitz and O'Brien, 2002](#)). However, the Conditional Value at Risk (CVaR) is becoming more important and is establishing as the more relevant risk measure for managing market risk and from a regulatory perspective ([Du and Escanciano, 2016](#), [Kellner and Rösch, 2016](#)). The reason for this development is that the CVaR corrects for several drawbacks of VaR and the CVaR is often claimed as the better risk measure.⁴⁸ For example, the VaR has been criticized in the academic literature due to its lack of subadditivity and the disregarding of extreme losses, which are of main interest in risk management (see [Artzner et al. \(1999\)](#), [Giannopoulos and Tunaru \(2005, p. 980\)](#), [McNeil and Frey \(2000, p. 291-292\)](#) or [Yamai and Yoshiba \(2005, p. 998\)](#)). More precisely, VaR only contains information on a certain quantile, whereas CVaR contains information on the whole right tail of the loss distribution ([Du and Escanciano, 2016, p. 942](#)). [Berkowitz and O'Brien \(2002\)](#), examining the VaR models of six commercial banks, demonstrate that the size of a VaR violation can be surprisingly large. This is confirmed by the study of [Du and Escanciano \(2016\)](#) who find that VaR responds less to extreme losses, such as those experienced during the recent financial crisis. Hence, VaR may underestimate risk in times of market stress, i.e. times of high asset price fluctuations ([Yamai and Yoshiba, 2005, p. 998](#)). Similarly, in a portfolio context, managing VaR means that only the exceedance probability is managed instead of the expected loss magnitude (see [Basak and Shapiro \(2001, p. 385\)](#) and [Aït-Sahalia and Brandt \(2001\)](#)). In contrast, by managing CVaR, both the exceedance probability and the size of extreme losses are managed. [Basak and Shapiro \(2001\)](#) and [Alexander and Baptista \(2004\)](#) find that managing CVaR is superior to managing VaR in an asset allocation context, especially if a risk-free asset

⁴⁸See [Yamai and Yoshiba \(2005\)](#) for a good comparison of VaR and CVaR. Moreover, see [Szegö \(2002, p. 1261\)](#) for a list of drawbacks of VaR. See [Du and Escanciano \(2016\)](#) for a good motivation of why CVaR is becoming the more relevant risk measure for managing downside risk.

is available.⁴⁹ Due the drawbacks of the VaR, we next extend the target VaR strategy to the target CVaR strategy, that aims to have a constant portfolio CVaR over time. As before, the weight of the risky asset is then given as a function of the expected CVaR of the risky asset, i.e. if the CVaR of the risky asset is expected to be low, the weight of the risky asset is increased and vice versa.

The CVaR is defined as the average loss in the worst $100 \cdot \alpha\%$ cases, i.e. the cases where the loss exceeds the VaR (see [Acerbi and Tasche \(2002, p. 1488\)](#) or [Yamai and Yoshida \(2005, p. 999\)](#)). More formally, we define the portfolio CVaR, denoted by $\text{CVaR}_\alpha^{P,t}$, as

$$\text{CVaR}_\alpha^{P,t} = \mathbb{E}(L_t^P \mid L_t^P \geq \text{VaR}_\alpha^{P,t}, \mathcal{F}_{t-1}). \quad (32)$$

In Appendix B.2, we show that the portfolio CVaR is given by

$$\text{CVaR}_\alpha^{P,t} = w_t \cdot \text{CVaR}_\alpha^t - (1 - w_t) \cdot R_t^f, \quad (33)$$

where $\text{CVaR}_\alpha^t := \mathbb{E}(L_t \mid L_t \geq \text{VaR}_\alpha^t, \mathcal{F}_{t-1})$ denotes the day t CVaR of the risky asset.⁵⁰ In order to achieve a constant portfolio CVaR of $\text{CVaR}_\alpha^{\text{target}}$ over time, i.e. $\text{CVaR}_\alpha^{P,t} = \text{CVaR}_\alpha^{\text{target}}$ for all t , the weight of the risky asset has to be chosen as

$$w_t = \frac{\text{CVaR}_\alpha^{\text{target}} + R_t^f}{\text{CVaR}_\alpha^t + R_t^f}. \quad (34)$$

Due to the definition of the CVaR, the target CVaR strategy manages expected losses, where the acceptable loss magnitude can be governed by the investor who chooses the values α and $\text{CVaR}_\alpha^{\text{target}}$. For example, a target CVaR level $\text{CVaR}_\alpha^{\text{target}}$ of 2% with a corresponding significance level α of 5% translates into a strategy with an average loss of 2% on the 5% worst days. In other words, the target CVaR strategy's average return on the worst 5 out of 100 days will be -2%.⁵¹ The choices of α and $\text{CVaR}_\alpha^{\text{target}}$ again strongly depend on the individual investor's preferences and risk aversion ([Alexander and Baptista, 2004](#)).

⁴⁹[Basak and Shapiro \(2001\)](#) demonstrate that, in the context of asset allocation decisions, losses of VaR managed portfolios can exceed the desired VaR extremely (see also [Alexander and Baptista \(2004\)](#), [Berkowitz et al. \(2011\)](#), [Cuoco et al. \(2008\)](#) and references therein). However, [Cuoco et al. \(2008\)](#) show that this observation does no longer hold once risk is managed dynamically, i.e. taking the actual information into account and reevaluating the risk level dynamically, instead of managing risk by a static model (see also [Berkowitz et al. \(2011\)](#)). This highlights the importance of managing portfolio risk dynamically, as done by risk targeting.

⁵⁰The representation in Equation (33) again follows by positive homogeneity and translation invariance of CVaR.

⁵¹By choosing $\text{CVaR}_\alpha^{\text{target}}$ and α adequately, this strategy can also be an alternative to absolute return or other hedge fund strategies as examined in [Fung and Hsieh \(1997\)](#) and [Agarwal and Naik \(2004\)](#).

As in the previous sections, since the CVaR of the risky asset is not observable, a forecast of CVaR_α^t is needed. We use the same estimation methods that we also use for the VaR of the risky asset. First, and especially interesting for practical implementations, we use Historical Simulation. For this method, we again assume that a data set of n realized losses $l_{t-1}, \dots, l_{t-n}$ with order statistics $l_{(1),t-1} \leq l_{(2),t-1} \leq \dots \leq l_{(n),t-1}$ exists. Based on the ordered losses, we estimate CVaR_α^t by ⁵²

$$\widehat{\text{CVaR}}_\alpha^t = \frac{1}{n - [n(1 - \alpha)] + 1} \cdot \sum_{j=[n(1-\alpha)]}^n l_{(j),t-1}. \quad (35)$$

This estimator is motivated by [Acerbi and Tasche \(2002, Proposition 4.1\)](#),⁵³ who demonstrate that the estimator in Equation (35) is only unbiased for n converging to infinity. For small n , the estimator in Equation (35) is biased. Methods that account for this estimation bias are presented in [Ko et al. \(2009\)](#) among others. As before, the quality of the target CVaR strategy strongly depends on the accuracy of the CVaR estimation.⁵⁴ However, since Historical Simulation deals as a benchmark model that is particularly interesting for practitioners, we keep the estimation as simple as possible.

For the return decomposition of Equation (6), the CVaR of the risky asset is given by

$$\text{CVaR}_\alpha^t = \sigma_t \cdot \text{CVaR}_\alpha^*, \quad (36)$$

where we define $\text{CVaR}_\alpha^* := \mathbb{E}(L^* \mid L^* \geq F_{L^*}^{-1}(1 - \alpha))$ and L^* is again a continuously distributed random variable, representing a standardized loss, with expectation zero, variance one and $F_{L^*}^{-1}(1 - \alpha)$ denotes the $(1 - \alpha)$ -quantile of L^* (see [McNeil and Frey \(2000, p. 276\)](#) for example). As for the VaR, we next present the estimation of the CVaR based on a two-stage approach, where the volatility is estimated in a first stage by one of the volatility models presented in Equation (7) or (8). In the second stage, we again consider a sample $l_{t-n}^*, \dots, l_{t-1}^*$ of n standardized losses with order statistics $l_{(1),t-1}^* \leq \dots \leq l_{(n),t-1}^*$. We denote the estimator

⁵²See [Ko et al. \(2009, p. 719\)](#) or [Giannopoulos and Tunaru \(2005, p. 985-986\)](#).

⁵³In this paper, we assume that losses are continuously distributed. This is confirmed by [Giannopoulos and Tunaru \(2005, p. 982\)](#) who state that only continuous probability distributions are used in practice. In this case, the CVaR, defined as Tail Conditional Expectation (TCE) in [Acerbi and Tasche \(2002, Definition 2.3\)](#), is equal to the Expected Shortfall (ES), defined in [Acerbi and Tasche \(2002, Corollary 5.3\)](#).

⁵⁴This is in line with [Yamai and Yoshida \(2005, p. 999\)](#) who find that the “effectiveness of expected shortfall, however, depends on the accuracy of estimation.”

for $\text{CVaR}_\alpha^* = \mathbb{E}(L^* \mid L^* \geq F_{L^*}^{-1}(1 - \alpha))$ based on the available information at day $t - 1$ by $\widehat{\text{CVaR}}_\alpha^{t,*}$. Hence, the estimator for the CVaR of the risky asset on day t , denoted by $\widehat{\text{CVaR}}_\alpha^t$, is given by

$$\widehat{\text{CVaR}}_\alpha^t = \hat{\sigma}_t \cdot \widehat{\text{CVaR}}_\alpha^{t,*}. \quad (37)$$

By using the FHS approach, the estimator $\widehat{\text{CVaR}}_\alpha^{t,*}$ is given by Equation (35), where the j -th order statistic $l_{(j),t-1}$ of the loss variables is replaced by the j -th order statistic $l_{(j),t-1}^*$ of the standardized losses (see [Giannopoulos and Tunaru \(2005\)](#) for example). By using the EVT approach, the estimator $\widehat{\text{CVaR}}_\alpha^{t,*}$ is given by

$$\widehat{\text{CVaR}}_\alpha^{t,*} = \frac{\hat{F}_{L^*,t}^{-1}(1 - \alpha)}{1 - \hat{\xi}} + \frac{\hat{\beta} - \hat{\xi}u}{1 - \hat{\xi}}, \quad (38)$$

where the estimator $\hat{F}_{L^*,t}^{-1}(1 - \alpha)$ is given in Equation (19), u denotes the predetermined threshold and the parameters $\hat{\xi}$ and $\hat{\beta}$ are the QML estimators (see [McNeil and Frey \(2000, p. 293\)](#) or [McNeil et al. \(2015, p. 154\)](#)).

Finally, we also use the skewed t distribution with time-varying parameters to forecast next day's CVaR. [Christoffersen \(2012\)](#) and [Rickenberg \(2020a, Appendix A\)](#) show that, for a random variable $Z_t \sim \text{stsk}(\eta_t, \lambda_t)$, we have

$$\begin{aligned} & \mathbb{E}(Z_t \mid Z_t < F_{\text{stsk}}^{-1}(\alpha \mid \eta_t, \lambda_t)) \\ &= \begin{cases} \frac{1}{\alpha} \frac{(1 - \lambda_t)^2}{b} \left(f_{st}(z^{(-)} \mid \eta_t) \cdot \frac{\eta_t - 2 + (z^{(-)})^2}{1 - \eta_t} - \frac{a \cdot F_{st}(z^{(-)} \mid \eta_t)}{1 - \lambda_t} \right) & \text{for } F_{\text{stsk}}^{-1}(\alpha \mid \eta_t, \lambda_t) < -\frac{a}{b} \\ \frac{1}{\alpha} \frac{(1 + \lambda_t)^2}{b} \left(f_{st}(z^{(+)} \mid \eta_t) \cdot \frac{\eta_t - 2 + (z^{(+)})^2}{1 - \eta_t} + \frac{a \cdot (1 - F_{st}(z^{(+)} \mid \eta_t))}{1 + \lambda_t} \right) & \text{for } F_{\text{stsk}}^{-1}(\alpha \mid \eta_t, \lambda_t) \geq -\frac{a}{b}, \end{cases} \end{aligned} \quad (39)$$

where $z^{(-)}$ and $z^{(+)}$ are given by

$$z^{(-)} = \frac{b \cdot F_{\text{stsk}}^{-1}(\alpha \mid \eta_t, \lambda_t) + a}{1 - \lambda_t}, \quad z^{(+)} = \frac{b \cdot F_{\text{stsk}}^{-1}(\alpha \mid \eta_t, \lambda_t) + a}{1 + \lambda_t}.$$

Further, $f_{st}(z \mid \eta_t)$ and $F_{st}(z \mid \eta_t) = \int_{-\infty}^z f_{st}(u \mid \eta_t) du$ correspond to the pdf and cdf of the *standardized t distribution* with mean zero and variance one. The pdf of the standardized t distribution is given by (see [Bollerslev \(1987, p. 543\)](#) and [Hansen \(1994, p. 709\)](#))

$$f_{st}(z \mid \eta_t) = \frac{\Gamma\left(\frac{\eta_t + 1}{2}\right)}{\Gamma\left(\frac{\eta_t}{2}\right) \sqrt{\pi(\eta_t - 2)}} \left(1 + \frac{z^2}{\eta_t - 2}\right)^{-(\eta_t + 1)/2}. \quad (40)$$

For the cdf of the t and standardized t distributions it holds $F_{st}(z|\eta_t) = F_t\left(\sqrt{\frac{\eta_t}{\eta_t-2}}z|\eta_t\right)$.⁵⁵ Bollerslev (1987) uses the standardized t distribution in the context of the GARCH(1,1) model and finds that the GARCH(1,1)- t model is superior to both, the GARCH(1,1)-normal and the unconditional t distribution. The forecast of day t 's CVaR is then given by Equation (37). In this case, $\widehat{\text{CVaR}}_\alpha^{t,*}$ is given by

$$\widehat{\text{CVaR}}_\alpha^{t,*} = -\mathbb{E}\left(Z_t | Z_t < F_{stsk}^{-1}\left(\alpha | \hat{\eta}_t, \hat{\lambda}_t\right)\right), \quad (41)$$

where $\hat{\eta}_t$ and $\hat{\lambda}_t$ are again the Maximum Likelihood estimates of η_t and λ_t .⁵⁶

McNeil and Frey (2000, p. 292) find that the quality of the CVaR estimation strongly depends on the approach that is used to model the tails of the loss distribution. Correctly modeling the tails, which is also crucial for the estimation of the VaR, becomes even more important when CVaR is estimated (Yamai and Yoshida, 2005). Therefore, correctly modeling the tails of the loss distribution is a central issue in achieving a constant portfolio CVaR over time. As for the VaR, Kellner and Rösch (2016) find that only models that account for fat tails and/or skewness are able to produce accurate CVaR forecasts. As stated above, a more accurate risk forecast, and hence a more constant portfolio risk, is typically rewarded with a superior risk-return profile and high utility gains.

As in Equation (29), since R_t^f is (typically) small compared to the CVaR values, the weight of the risky asset can be approximated by

$$w_t \approx \frac{\text{CVaR}_\alpha^{\text{target}}}{\text{CVaR}_\alpha^t}. \quad (42)$$

By using Equation (42), similar to Equation (30), we can approximate a target CVaR strategy by a target volatility strategy. The weight of the risky asset for this target volatility strategy is then given by

$$w_t \approx \frac{\text{CVaR}_\alpha^{\text{target}}}{\text{CVaR}_\alpha^t} = \frac{\mathbb{E}(L^* | L^* \geq F_{L^*}^{-1}(1 - \alpha)) \cdot \frac{\text{CVaR}_\alpha^{\text{target}}}{\mathbb{E}(L^* | L^* \geq F_{L^*}^{-1}(1 - \alpha))}}{\mathbb{E}(L^* | L^* \geq F_{L^*}^{-1}(1 - \alpha)) \cdot \sigma_t} = \frac{\sigma^{\text{target}}}{\sigma_t}, \quad (43)$$

⁵⁵This relation is advantageous since the cdf of the t distribution is often available in most software packages, whereas the cdf of the standardized t distribution is not available (Jondeau and Rockinger, 2003).

⁵⁶The minus one in front of the expectation in Equation (41) again results, since we fit the skewed t distribution to standardized returns instead of standardized losses. From Equations (28) and (41), it directly follows $-\mathbb{E}(Z_t | Z_t < F_{stsk}^{-1}(\alpha | \hat{\eta}_t, \hat{\lambda}_t)) = \mathbb{E}(L_t^* | L_t^* \geq F_{L_t^*}^{-1}(1 - \alpha))$, where $L_t^* = -Z_t$.

with $\sigma^{\text{target}} = \text{CVaR}_\alpha^{\text{target}} / \mathbb{E}(L^* \mid L^* \geq F_{L^*}^{-1}(1 - \alpha))$. However, since the distribution of the standardized losses, or at least $\mathbb{E}(L^* \mid L^* \geq F_{L^*}^{-1}(1 - \alpha))$, is not known in practice, the volatility target σ^{target} cannot be directly calculated. An approximation can be done by using a standard normal distribution for L^* . In this case, $\mathbb{E}(L^* \mid L^* \geq F_{L^*}^{-1}(1 - \alpha))$ is given by $\frac{\varphi(N_{1-\alpha})}{\alpha}$, where φ and $N_{1-\alpha}$ denote the density function and the $(1 - \alpha)$ -quantile of the standard normal distribution, respectively. The volatility target is then given by

$$\sigma^{\text{target}} = \frac{\text{CVaR}_\alpha^{\text{target}}}{\varphi(N_{1-\alpha})/\alpha}. \quad (44)$$

For example, a target CVaR strategy with a significance level of $\alpha = 5\%$ and desired CVaR target $\text{CVaR}_\alpha^{\text{target}} = 2\%$ can be approximated by a target volatility strategy with an annualized volatility target of $\sigma^{\text{target}} = \frac{0.02}{2.063} \cdot \sqrt{252} = 15.4\%$. Further, a constant volatility of 15.4% can be achieved by managing the risky asset's CVaR, and thus incorporating information on the risky asset's skewness and kurtosis. Moreover, by Equations (15) and (42), we obtain

$$w_t \approx \frac{\text{CVaR}_\alpha^{\text{target}}}{\text{CVaR}_\alpha^t} = \frac{\text{CVaR}_\alpha^{\text{target}}}{\sigma_t \cdot F_{L^*}^{-1}(1 - \alpha) \cdot \frac{\mathbb{E}(L^* \mid L^* \geq F_{L^*}^{-1}(1 - \alpha))}{F_{L^*}^{-1}(1 - \alpha)}} = \frac{\text{VaR}_\alpha^{\text{target}}}{\text{VaR}_\alpha^t}, \quad (45)$$

with $\text{VaR}_\alpha^{\text{target}} = \frac{F_{L^*}^{-1}(1-\alpha)}{\mathbb{E}(L^* \mid L^* \geq F_{L^*}^{-1}(1-\alpha))} \cdot \text{CVaR}_\alpha^{\text{target}}$. Therefore, a target CVaR strategy can also be approximated by a target VaR strategy with an adjusted target VaR level.⁵⁷ However, the approximation of the target CVaR strategy by a target volatility strategy, given in Equation (43), is appealing since forecasting volatility is much easier than forecasting CVaR, but this argument is only partly valid for the approximation by a target VaR strategy, given in Equation (45).⁵⁸ Nevertheless, Equation (45) is helpful for comparing target VaR and target CVaR strategies. By assuming a standard normal distribution for L^* , we obtain the comparable target VaR level

$$\text{VaR}_\alpha^{\text{target}} = \frac{N_{1-\alpha}}{\varphi(N_{1-\alpha})/\alpha} \cdot \text{CVaR}_\alpha^{\text{target}}. \quad (46)$$

⁵⁷This approach is similar to [Cuoco et al. \(2008\)](#) who show how a VaR limit can be transformed in a CVaR limit and vice versa.

⁵⁸[Yamai and Yoshida \(2005, p. 1012\)](#) state that the estimation error for CVaR is larger than for VaR, especially when the return distribution exhibits fat tails. Similarly, [Kellner and Rösch \(2016\)](#) find that the model risk for CVaR is higher than for VaR, which is mainly driven by fat tails in the return distribution. Especially in times of financial market turmoils, like the recent global financial crisis or corona crisis, CVaR forecasts among different models are more volatile than VaR forecasts. Thus, the approximation of a target CVaR strategy by a target VaR strategy could be appealing for extremely fat-tailed asset return distributions.

A target CVaR strategy with $\alpha = 5\%$ and $\text{CVaR}_\alpha^{\text{target}} = 2\%$ should then be compared to a target VaR strategy with $\alpha = 5\%$ and $\text{VaR}_\alpha^{\text{target}} = \frac{1.645}{2.063} \cdot 0.02 = 1.6\%$, which is again approximated by a target volatility strategy with $\sigma^{\text{target}} = \frac{0.016}{1.645} \cdot \sqrt{252} = 15.4\%$.

This approach of comparing a target CVaR strategy with a target volatility or target VaR strategy is similar to the approach of [Strub \(2013\)](#). The author starts with a predefined volatility target and transforms this volatility target to a CVaR target by using a normality assumption. The weight of the risky asset is then obtained as the ratio of the transformed CVaR target and the forecast of the risky asset's CVaR. [Strub \(2013, p. 16\)](#) finds that the volatility and CVaR managed strategies offer a substantial drawdown protection, especially in the years when the underlying index suffers the most. By comparing a volatility managed strategy with a CVaR managed strategy, [Strub \(2013, p. 17\)](#) finds that managing CVaR translates into a better risk-adjusted performance and lower drawdowns. Moreover, the author finds that, even after transaction costs, the risk managed strategies still deliver convincing performances, which makes the risk managed strategies appealing for practical applications and an interesting alternative to hedge fund strategies as examined in [Fung and Hsieh \(1997\)](#) and [Agarwal and Naik \(2004\)](#). Similarly, the CVaR managed strategy of [Wang et al. \(2012\)](#) reduces drawdowns without sacrificing returns, and hence captures the upside potential while downside risk is limited ([Wang et al., 2012, p. 38](#)).⁵⁹ This is in line with [Basak and Shapiro \(2001\)](#) who also find convincing results of managing expected losses, as done by managing CVaR, and the authors conclude that managing expected losses is superior to managing exceedance probabilities, as done by managing VaR (see also [Aït-Sahalia and Brandt \(2001, p. 1316\)](#)).

3.4 VaR and CVaR Targeting as Optimal Trading Strategies under Risk Limits

In this section, we motivate the VaR and CVaR targeting strategies from another perspective as optimal trading strategies when a trader faces an absolute risk limit. This is similar to the examination of [Basak and Shapiro \(2001\)](#), [Wang et al. \(2012\)](#), [Cuoco et al. \(2008\)](#) and [Alexander](#)

⁵⁹[Wang et al. \(2012\)](#) call their strategy a target CVaR strategy. However, the authors do not target a constant level of risk over time, but allow a maximum level of risk (see also [Basak and Shapiro \(2001\)](#) and [Alexander and Baptista \(2004\)](#)).

and Baptista (2004).⁶⁰ We again consider a trader who invests w_t in the risky and $1 - w_t$ in the riskless asset and define the trader's portfolio value by $W_t := W_{t-1} \cdot (1 + R_t^P)$, $W_0 > 0$. Further, we define the absolute loss in t by $L_t^{abs} := W_{t-1} - W_t = -W_{t-1} \cdot R_t^P$. We now consider a portfolio optimization problem under an absolute risk limit $\overline{\text{VaR}}_t$, given by

$$\max_{w_t} \mathbb{E}(R_t^P \mid \mathcal{F}_{t-1}) \quad \text{s.t.} \quad \text{VaR}_\alpha^{t,abs} \leq \overline{\text{VaR}}_t, \quad (47)$$

where $\text{VaR}_\alpha^{t,abs} = W_{t-1} \cdot \text{VaR}_\alpha^{P,t}$ denotes the VaR of the absolute loss L_t^{abs} . The risk limit $\text{VaR}_\alpha^{t,abs} \leq \overline{\text{VaR}}_t$ can then be rewritten as $\text{VaR}_\alpha^{P,t} \leq \overline{\text{VaR}}_t / W_{t-1}$. From Equation (12) it follows that the risk limit holds if

$$w_t \leq \frac{\overline{\text{VaR}}_t / W_{t-1} + R_t^f}{\text{VaR}_\alpha^t + R_t^f}. \quad (48)$$

Under the assumption $\mathbb{E}(R_t \mid \mathcal{F}_{t-1}) > R_t^f$, which is typically fulfilled in practice (Benartzi and Thaler, 1995), the expected portfolio return $\mathbb{E}(R_t^P \mid \mathcal{F}_{t-1})$ is increasing in w_t . Hence, the trader chooses the highest possible equity exposure that still fulfills the risk limit $\text{VaR}_\alpha^{t,abs} \leq \overline{\text{VaR}}_t$. Thus, w_t is given by

$$w_t = \frac{\overline{\text{VaR}}_t / W_{t-1} + R_t^f}{\text{VaR}_\alpha^t + R_t^f}. \quad (49)$$

Consequently, by choosing a constant relative risk limit $\overline{\text{VaR}}_t / W_{t-1} = \text{VaR}_\alpha^{\text{target}}$, the target VaR strategy follows as optimal dynamic trading strategy under a downside risk limit. By the same arguments as above, the weighting for the target CVaR strategy can be obtained if an investor faces a CVaR limit.

4 Assessing the Accuracy of Risk Targeting

In this section, we present methods that can be used to test the accuracy of the target risk strategies, i.e. we present tests that can be used to test if the different risk models are successful in targeting a constant level of portfolio risk over time. A constant portfolio risk is important for several reasons. First, a constant risk of the strategies should be achieved by definition of risk targeting. Second, an investor who chose a fund that targets a volatility level that fits to

⁶⁰I thank Peter Albrecht and Markus Huggenberger for this helpful comment.

the investor's risk preferences would sell this fund if the fund achieves a significantly higher volatility than expected. Similarly, an investor who expects only a limited number of days where the portfolio return is smaller than $-\text{VaR}_\alpha^{\text{target}}$ would also divest the fund if the fund exhibits too many extremely negative returns. Third, risk-averse investors are willing to pay for hedges against a changing portfolio volatility ([Adrian and Rosenberg, 2008](#), [Ang et al., 2006b](#), [Bollerslev and Todorov, 2011](#)). These investors are willing to pay higher fees for strategies with a more constant portfolio risk. Fourth, having a constant level of portfolio risk over time is frequently used by practitioners ([Barroso and Santa-Clara, 2015](#), p. 112). Fifth, several studies show that a higher forecasting accuracy, and hence a more constant portfolio risk, coincides with higher (risk-adjusted) performance and utility gains ([Bollerslev et al., 2018](#), [Marquering and Verbeek, 2004](#), [Moreira and Muir, 2017](#), [Perchet et al., 2016](#), [Taylor, 2014](#)). Consequently, a fund that fails to target a constant risk over time typically achieves a suboptimal risk-return profile. For example, [Bollerslev et al. \(2018, p. 2732\)](#) write:

“the investor achieves the maximum utility by successfully targeting a constant risk level, while the utility decreases with the volatility-of-volatility. Hence, risk models that help the investor achieve more accurate volatility forecasts are associated with higher levels of utility”

[Bollerslev et al. \(2018\)](#) find that an investor who uses volatility targeting is willing to pay a fee of 0.48% per year to switch from an inaccurate to a more accurate volatility model. The authors find that there exists a positive, non-linear relation between forecasting accuracy of volatility models and utility benefits. Further, they find that a model with perfect foresight, i.e. a model that produces a totally constant portfolio volatility over time, exhibits the highest utility benefit (see also [Benson et al. \(2014\)](#)). Moreover, [Dreyer and Hubrich \(2019\)](#) find that a portfolio volatility stabilization, i.e. a lower volatility of volatility, is linked to a higher tail risk reduction. Similarly, in a cross-sectional setting, [Baltussen et al. \(2018\)](#) find that assets with a high volatility of volatility (vol-of-vol) underperform assets with a more constant volatility. Further, higher vol-of-vol assets also exhibit higher downside risk and the vol-of-vol is linked to an asset's kurtosis. This especially holds during down-markets where high vol-of-vol assets

underperform low vol-of-vol assets by 0.83% per month. Consequently, forecasting accuracy is an important driver of the investor's benefit from risk targeting and should therefore be tested. Besides backtesting the accuracy of volatility targeting, we additionally show how the accuracy of VaR and CVaR targeting can be tested. To assess the accuracy of several volatility models, [Bollerslev et al. \(2018\)](#) use the R^2 as well as the DM-test of [Diebold and Mariano \(1995\)](#), which tests for equal predictive ability. However, both methods have several disadvantages. Therefore, in order to assess the accuracy of volatility targeting, we use more powerful tools that will be presented in the next section.

4.1 Assessing the Accuracy of Volatility Targeting

Although several studies on volatility targeting have been made, only a few studies statistically assess if it is possible to achieve the desired volatility target over time. To assess the accuracy of volatility targeting for some set of models $\mathcal{M}$, we measure the portfolio variance of model k on day t by $RV_{k,t}^2 := w_{k,t}^2 \cdot RV_t^2$, where $w_{k,t}$ is the weight of strategy k , $k \in \mathcal{M}$, on day t and RV_t denotes the Realized Volatility on day t of the risky asset (see [Andersen et al. \(2001\)](#), [Patton \(2011\)](#) or [Bollerslev et al. \(2018\)](#) for a definition of RV_t).⁶¹ Motivated by [Hansen and Lunde \(2005\)](#) and [Patton \(2011\)](#), we define the QLIKE loss function of model k on day t by

$$L_{k,t} := L(RV_{k,t}^2, \sigma_{\text{target,d}}^2) := \frac{RV_{k,t}^2}{\sigma_{\text{target,d}}^2} - \ln \left(\frac{RV_{k,t}^2}{\sigma_{\text{target,d}}^2} \right) - 1, \quad (50)$$

where the daily volatility target is given by $\sigma_{\text{target,d}} = \sigma^{\text{target}} / \sqrt{252}$ (see [Christoffersen \(2012, p. 85\)](#) and [Taylor \(2014, p. 475\)](#)). [Patton \(2011\)](#) shows that the QLIKE and the MSE loss functions are robust against noise in the volatility proxy. The MSE relies on the absolute forecast error, whereas the QLIKE relies on the relative forecast error. We choose the QLIKE instead of the MSE since the QLIKE penalizes models that underestimate risk, and hence produce a portfolio volatility that is too high. We follow [Christoffersen \(2012\)](#) and use a slightly different representation than the definition used by [Hansen and Lunde \(2005\)](#) and [Patton \(2011\)](#). This

⁶¹The Realized Volatility data are downloaded from the Oxford Man Realized Library (<https://realized.oxford-man.ox.ac.uk/>). We follow [Hansen and Lunde \(2005\)](#) and scale the Realized Volatility to a measure of the close-to-close volatility of day t . [Bollerslev et al. \(2018\)](#) simply add the squared overnight return to the Realized Volatility to obtain a measure for the whole day's variance. However, both methods deliver similar results.

representation has the advantage that $L(RV_{k,t}^2, \sigma_{\text{target,d}}^2) = L(RV_t^2, \sigma_t^2)$ holds, which is the usual choice in the volatility evaluation literature. Moreover, our loss function is normalized in the sense that $L_{k,t} = 0$ holds if the portfolio volatility on day t equals the desired volatility target, whereas the representation of [Hansen and Lunde \(2005\)](#) and [Patton \(2011\)](#) is not normalized. In particular, by choosing $C(z) = \frac{1}{z}$, $\tilde{C}(z) = \log(z)$ and $B(z) = -\log(z)$, our representation still fulfills Proposition 1 of [Patton \(2011\)](#). Thus, our representation is a robust loss function in the sense of [Patton \(2011, Definition 1\)](#), and hence is robust against noise in the volatility proxy. [Patton \(2011\)](#) shows that using squared daily returns instead of Realized Volatility leads to quite similar conclusions. We also used squared daily returns instead of the Realized Volatility and found similar results for both methods.

In order to assess the accuracy of the volatility models, we define the relative loss between model i and j as $X_{ij,t} = L_{i,t} - L_{j,t}$ and the average relative loss between these models as

$$\bar{X}_{ij} = \frac{1}{T} \sum_{t=1}^T X_{ij,t}. \quad (51)$$

The basic idea of testing for predictive accuracy is that a positive value of $\bar{X}_{ij}$ indicates that model j is more accurate than model i , i.e. model j is more successful in targeting a constant level of volatility.

To test for the accuracy of the different target volatility strategies, we first use the test for equal predictive ability (DM-test) of [Diebold and Mariano \(1995\)](#), which was also used by [Patton \(2011\)](#) and [Bollerslev et al. \(2018\)](#). Further, we use the Reality Check (RC-test) of [White \(2000\)](#) and [Sullivan et al. \(1999\)](#) and its extension, the test for Superior Predictive Ability (SPA-test) of [Hansen \(2005\)](#) and [Hansen and Lunde \(2005\)](#). In contrast to the DM-test, both tests, the RC- and SPA-test, test for superior predictive ability and can also be applied to more than two models simultaneously. Both tests test the null-hypothesis that a chosen benchmark model is more accurate than all the remaining models. Moreover, we use the stepwise extensions of the RC-test and SPA-test that are presented in [Romano and Wolf \(2005\)](#) and [Hsu et al. \(2010\)](#). We denote these stepwise extensions by Step-RC and Step-SPA. The stepwise approaches can be used to construct sets of models that are superior to a chosen benchmark model. Similarly, we also use the algorithm based on the False Discovery Rate (FDR) that is presented in [Barras](#)

et al. (2010) and Bajgrowicz and Scaillet (2012). The authors show how the FDR can be used to identify models that are superior to a chosen benchmark model. The set of superior models, constructed by the stepwise approaches or the FDR approach, contains the models that produce a more constant portfolio volatility than the chosen benchmark model. Finally, we use the Model Confidence Set (MCS) of Hansen et al. (2011) and Hansen et al. (2003), where we mainly follow Hansen et al. (2003) who also applied this algorithm to assess the accuracy of volatility models. The MCS also identifies a set of superior models and has the advantage that no benchmark model is needed. A short summary of the tests can be found in Rickenberg (2020a, Appendix C).

For the DM-test, the Step-RC, the Step-SPA and the FDR approach, a certain benchmark model has to be chosen. These tests then compare the accuracy of all other models to this benchmark model. As benchmark model we choose the easiest one, which is the HSD. This model is frequently used in the volatility targeting literature and is similar to the model used by Barroso and Santa-Clara (2015), Barroso and Maio (2017), Cederburg et al. (2020) and Moreira and Muir (2017). When applying the RC- and SPA-test, we choose each model once as the benchmark and test if this benchmark is outperformed by at least one other model. Romano and Wolf (2005) and Bajgrowicz and Scaillet (2012) argue that applying the RC- and SPA-test to different benchmarks has several disadvantages that are corrected by the Step-RC, Step-SPA, the MCS and the FDR approaches.

4.2 Assessing the Accuracy of VaR Targeting

In Section 3, we have presented several VaR forecasting methods and we have shown how these VaR forecasting methods can be used to derive the weight w_t of the risky asset in a target VaR strategy. Moreover, we have shown how a target VaR strategy can be approximated by a target volatility strategy with an adjusted target volatility level given in Equation (30). Next, we want to assess the quality of the different forecasting methods and we want to compare the “true” target VaR strategies, based on a proper VaR forecast for the risky asset, with the approximated target VaR strategies, based on the risky asset’s volatility solely. In other words, we want to assess if the different target VaR strategies succeed to produce a constant portfolio VaR over

time and if controlling volatility is sufficient for this task. Similarly, [Christoffersen and Diebold \(2000\)](#) show how VaR backtesting methods can be used to backtest the accuracy of volatility models.

To assess the quality of the target VaR strategies, we define the hit variables

$$H_t^P = \begin{cases} 1, & \text{if } L_t^P > \text{VaR}_\alpha^{\text{target}} \\ 0, & \text{if } L_t^P \leq \text{VaR}_\alpha^{\text{target}}, \end{cases} \quad (52)$$

i.e. H_t^P is equal to one if the portfolio loss is higher than the VaR target $\text{VaR}_\alpha^{\text{target}}$, called a hit, and zero else. An accurate target VaR strategy should exhibit two abilities. First, the percentage of days when the portfolio loss is higher than the predefined VaR target, i.e. the proportion of hits in the hit-series $\{H_t^P\}_{t=1}^T$, should be equal to the desired significance level α . Second, the days when the portfolio loss is higher than the VaR target should occur randomly over time and should not be clustered (see [Berkowitz and O'Brien \(2002, p. 1101\)](#) and [Berkowitz et al. \(2011, p. 2217\)](#)). Assume that the hits of a target VaR strategy occur clustered on many subsequent days, i.e. the portfolio losses are higher than the predefined VaR target on every day in a certain period. As a consequence, investors would remove money from a fund using this strategy, since this strategy seems to fail in having a constant VaR over time.⁶²

To test these two abilities, we resort to the VaR backtesting method of [Christoffersen \(1998\)](#), which is one of the most widely used VaR backtests in the academic literature ([Du and Escanciano, 2016](#)).⁶³ In Appendix C.1, we show that the variable H_t^P is equivalent to

$$H_t = \begin{cases} 1, & \text{if } L_t > \text{VaR}_\alpha^t \\ 0, & \text{if } L_t \leq \text{VaR}_\alpha^t, \end{cases} \quad (53)$$

i.e. H_t^P is equal to the hit variable based on the losses and VaRs of the risky asset solely, which is used in the backtest of [Christoffersen \(1998\)](#). Consequently, the backtesting approach of [Christoffersen \(1998\)](#) can be directly adopted for the variables H_t^P . Moreover, this result directly provides critical values which allow us to draw conclusions on the accuracy of the target

⁶²Besides this economic importance of independent hits, this ability should also hold by definition of VaR. See, for example, [McNeil et al. \(2015, Lemma 9.5\)](#) who show that the process of hit variables is a process of iid Bernoulli random variables with probability α (see also [Christoffersen \(1998\)](#), [Berkowitz and O'Brien \(2002\)](#) and [Berkowitz et al. \(2011\)](#)).

⁶³See also [Berkowitz and O'Brien \(2002\)](#), [Berkowitz et al. \(2011, p. 2217\)](#) and [Kuester et al. \(2006, Sec. 2\)](#) for a short overview of this backtesting procedure.

VaR strategies.⁶⁴ The backtest of the target VaR strategy is then formed with the variables

$$\hat{H}_t^P = \begin{cases} 1, & \text{if } l_t^P > \text{VaR}_\alpha^{\text{target}} \\ 0, & \text{if } l_t^P \leq \text{VaR}_\alpha^{\text{target}}, \end{cases} \quad (54)$$

where l_t^P is the realized portfolio loss on day t . The first above mentioned ability, i.e. the correct hit proportion, is then tested with the unconditional coverage test. The second ability, i.e. the independence of the hits, is tested with the test of independence and both abilities are simultaneously tested by the conditional coverage test ([Christoffersen, 1998](#)).

4.3 Assessing the Accuracy of CVaR Targeting

In order to assess the accuracy of the target CVaR strategies, we again use backtesting methods that were developed for evaluating different CVaR forecasting methods. For backtesting CVaR, there does not exist a common backtesting procedure ([Du and Escanciano, 2016](#)). Furthermore, backtesting CVaR is more challenging than backtesting VaR. Therefore, we will use two different CVaR backtesting procedures that help us to draw more sound conclusions on the accuracy of the target CVaR strategies.⁶⁵ For this purpose, as first CVaR backtest, we use the CVaR backtesting procedure described in [McNeil and Frey \(2000, Section 4.3\)](#). This backtesting method compares the loss of the risky asset with the CVaR of the risky asset and is based on the result that, under the condition that the loss L_t exceeds VaR_α^t , the variables

$$X_t = \frac{L_t - \text{CVaR}_\alpha^t}{\sigma_t} = L_t^* - \text{CVaR}_\alpha^{t,*} \quad (55)$$

are iid with expectation zero. Based on this result, a backtesting procedure using a distribution free bootstrap is derived. However, in this paper, we are interested in the (normalized) difference between the portfolio loss and target CVaR level $\text{CVaR}_\alpha^{\text{target}}$, i.e. we are interested in the ratio

$$X_t^P = \frac{L_t^P - \text{CVaR}_\alpha^{\text{target}}}{\sqrt{\text{var}(R_t^P | \mathcal{F}_{t-1})}}, \quad (56)$$

where we normalize these differences by the portfolio volatility. In Appendix C.2, we show that X_t^P equals X_t and thus, given $L_t^P - \text{VaR}_\alpha^{t,P} > 0$, X_t^P should be iid with expectation zero as

⁶⁴[Christoffersen \(1998\)](#) shows that, under the null hypothesis, the test statistic asymptotically follows a χ^2 distribution.

⁶⁵Both backtests used in this paper are unconditional backtests, which are less powerful than conditional backtests ([Du and Escanciano, 2016](#)). However, as opposed to the VaR backtesting literature, there does not exist a widely used conditional CVaR backtesting method.

well. Hence, we can adopt the backtesting procedure of [McNeil and Frey \(2000, Section 4.3\)](#) for the variables X_t^P . The backtest is then implemented by using the realizations

$$x_t^P = \frac{l_t^P - \text{CVaR}_\alpha^{\text{target}}}{w_t \cdot \hat{\sigma}_t}, \quad (57)$$

where l_t^P denotes the realized day t portfolio loss based on the weight w_t . If the weight w_t of the risky asset is estimated correctly, the sample

$$\left\{ x_t^P : t = 1, \dots, T, l_t^P > \widehat{\text{VaR}}_\alpha^{t,P} \right\} \quad (58)$$

should behave like an iid sample with mean zero.⁶⁶

As second target CVaR backtesting procedure, we use the backtest derived in [Embrechts et al. \(2005\)](#). We again consider the days where the portfolio loss is higher than the portfolio VaR, i.e. we consider the days where $L_t^P > \text{VaR}_\alpha^{t,P}$ holds. In these cases, stemming from the definition of the CVaR, the mean between the portfolio loss and the portfolio CVaR should be zero. Since the portfolio CVaR should be equal to $\text{CVaR}_\alpha^{\text{target}}$ over time, the measure

$$V_1 = \frac{\sum_{t=1}^T (L_t^P - \text{CVaR}_\alpha^{\text{target}}) \cdot \mathbb{1}_{\{L_t^P > \text{VaR}_\alpha^{t,P}\}}}{\sum_{t=1}^T \mathbb{1}_{\{L_t^P > \text{VaR}_\alpha^{t,P}\}}} \quad (59)$$

should exhibit a low absolute value ([Embrechts et al., 2005, p. 72](#)).⁶⁷ Nevertheless, [Embrechts et al. \(2005\)](#) argue that the measure V_1 has the drawback that it relies on an estimate of the portfolio VaR. In the definition of the measure V_1 , the worst cases are defined as the days when the portfolio loss exceeds the estimated portfolio VaR. If the risky asset's VaR forecast, and hence by Equation (12) the portfolio VaR, is not credible, the validity of the measure V_1 is doubtful. To account for this observation, the authors propose a second measure V_2 that does not rely on a VaR forecast. The motivation of this measure stems from the interpretation

⁶⁶We standardize the strategies that rely on the Historical Simulation by the HSD volatility. Moreover, backtesting the target CVaR strategies for which a proper VaR forecast, and hence a portfolio VaR, exists is straightforward. For the strategies that are only based on a volatility forecast, the time series of the portfolio VaR is not available. Since we calculate the target volatility level by assuming a normal distribution for Z_t in these cases, we solve this problem in the following way. If a target CVaR strategy relies on a volatility forecast $\hat{\sigma}_t$ solely, we estimate the corresponding VaR by $\widehat{\text{VaR}}_\alpha^t = \hat{\sigma}_t \cdot N_{1-\alpha}$, i.e. again assuming that Z_t follows a standard normal distribution. In this case, the portfolio VaR is given by $\widehat{\text{VaR}}_\alpha^{t,P} = w_t \cdot \hat{\sigma}_t \cdot N_{1-\alpha} - (1 - w_t) \cdot R_t^f$. An alternative would be to use the VaR target $\text{VaR}_\alpha^{\text{target}}$ as proxy for the portfolio VaR, i.e. $\widehat{\text{VaR}}_\alpha^{t,P} = \text{VaR}_\alpha^{\text{target}}$.

⁶⁷For the volatility based strategies, we again use $\widehat{\text{VaR}}_\alpha^{t,P} = w_t \cdot \hat{\sigma}_t \cdot N_{1-\alpha} - (1 - w_t) \cdot R_t^f$ as forecast for the portfolio VaR in this backtest.

that the CVaR is the expected loss in the $100 \cdot \alpha\%$ “worst” cases. Therefore, we denote by $D_t := L_t^P - \text{CVaR}_\alpha^{\text{target}}$ the difference between the portfolio loss and the CVaR target. Based on these differences, we define the worst cases as the $100 \cdot \alpha\%$ highest differences D_t , i.e. we define the worst cases as the cases when the target CVaR level is exceeded the most. This has the advantage that the worst cases do not depend on an estimate of the VaR, where we do not know if this estimate is credible. We denote the $(1 - \alpha)$ -quantile of $\{D_t\}_{t=1}^T$ by $D^{1-\alpha}$ and calculate V_2 by

$$V_2 = \frac{\sum_{t=1}^T D_t \cdot \mathbb{1}_{\{D_t > D^{1-\alpha}\}}}{\sum_{t=1}^T \mathbb{1}_{\{D_t > D^{1-\alpha}\}}}. \quad (60)$$

As before, the absolute value of V_2 should be low for a successful target CVaR strategy. As a third measure, denoted by V , [Embrechts et al. \(2005, p. 72\)](#) combine the measures V_1 and V_2 and the authors define

$$V = \frac{|V_1| + |V_2|}{2}, \quad (61)$$

which should again be low for a good target CVaR strategy.

5 Empirical Results

5.1 Data

To evaluate the performance of the different target risk strategies and to backtest the ability of achieving a constant level of portfolio risk over time, we use data for the DAX Performance Index as risky asset. As proxy for the risk-free rate, we use the three month Euribor.⁶⁸ The data range from 01.01.2000 to 31.12.2018 and are obtained from Datastream. Although many studies on investment or fund strategies use monthly data, we use daily data, since daily data better capture the dynamics of the financial markets and are more closely to the manner how funds are managed (see [Busse \(1999, p. 1015\)](#) and [Karolyi and Stulz \(1996, p. 952\)](#)). Further, even long-term investors typically have short evaluation horizons ([Benartzi and Thaler, 1995](#))

⁶⁸This is similar to [Marquering and Verbeek \(2004\)](#) who use the S&P 500 as risky asset and the three month US T-bill rate as risk-free asset to examine the economic value of volatility timing in the US market. For risk targeting, it is crucial to use a highly liquid asset as underlying risky asset since times of increasing volatility, which induce a portfolio reallocation, typically coincide with times of lower market liquidity ([Ang et al., 2006b](#)).

and should also time short-term volatility ([Moreira and Muir, 2019](#)). Moreover, in order to better manage potential extreme events, focusing on daily return data is also beneficial, since extreme price changes can occur during short time intervals ([Longin, 2000](#), p. 1104). Most studies on risk targeting – or more precisely volatility targeting – use data for the S&P 500, whereas risk targeting for German stocks is only examined rarely so far.⁶⁹ Some additional results for US data and small caps, proxied by the S&P 500 and the German small cap index SDAX, are given in Appendix D.

The chosen period from 2000 to 2018 is marked by changing periods of low and high risk and contains the collapse of the tech bubble, the global financial crisis and the European debt crisis, but also times of continuously uptrending markets. This illustrates how risk targeting works in different market environments and whether the models are successful in adapting to changing market regimes. [Dopfel and Ramkumar \(2013\)](#) demonstrate the importance of portfolio risk management during in the global financial crisis, where volatility managing delivers higher returns with lower volatility. However, the authors also show that a good strategy should increase the equity exposure immediately after the crisis. Thus, a good strategy should adapt quite fast to the changes in market risk in our sample. In total, a well performing strategy should limit the downside while the upside potential is captured, as it is found for many hedge fund strategies ([Fung and Hsieh, 1997](#)). Additionally to our main results, Appendix D.6 shows how risk targeting works for a longer data set that covers about 88 years. Further, Appendix D.7 shows out-of-sample results for the recent corona crisis.

As in [Kellner and Rösch \(2016\)](#), we use an estimation window of $n = 1000$ days for Historical Simulation, FHS, EVT, the skewed t distribution and for estimating the GARCH(1,1) parameters. The HSD is estimated with an estimation window of $m = 30$ days. As benchmark portfolios for the risk targeting strategies, we use two buy-and-hold investment strategies. The first benchmark strategy is fully invested in the risky asset, i.e. $w_t = 1$ for all t . The second strategy initially invests $w_0 = 60\%$ of wealth in the risky asset and the remaining $1 - w_0 = 40\%$

⁶⁹[Packham et al. \(2017\)](#) examine data for German stocks as well, but in a slightly different setting. [Barroso and Santa-Clara \(2015\)](#) examine volatility targeting for a momentum portfolio consisting of German stocks. [Ang et al. \(2009\)](#) examine the (cross-sectional) low volatility anomaly for an international data set that also includes Germany.

in the risk-free asset, without rebalancing the weights over time. [Benartzi and Thaler \(1995\)](#) state that portfolios that contain approximately 50% stocks and 50% bonds are optimal for loss-averse investors. Similarly, [Ang et al. \(2005, Fig. 3\)](#) find that such portfolios are also held by moderately risk-averse investors. Further, 60/40 portfolios are frequently used by pension fund managers ([Benartzi and Thaler, 1995](#), p. 87) for whom risk targeting can be an interesting alternative. In particular, a 60/40 portfolio should also be in line with the risk profile of an average investor ([Asvanunt et al., 2015](#), Footnote 2).

In order to better manage extreme losses and to reduce drawdowns, we choose a low significance level of $\alpha = 0.5\%$ for the target VaR and CVaR strategies. Low significance levels are frequently used in practice and are also important from a regulatory perspective. For example, the Bank of Internal Settlements has set the significance level to 1% for measuring market risk and to only 0.1% for credit risk. Further, a significance level of $\alpha = 0.5\%$ is also set to calculate the Solvency Capital Requirement under Solvency II. [Bali et al. \(2008\)](#) also use a significance level of 0.5% in a VaR forecasting setting. In particular, [Happersberger et al. \(2019\)](#) find better result for downside risk managed strategies when a lower α is chosen. Further, [Ghysels et al. \(2016\)](#) find that skewness information is hidden in the distribution's tails and that this "tail skewness" is important to determine the optimal portfolio allocation. Thus, in order to better capture skewness risk, lower significance levels should be chosen. Additional results for significance levels of $\alpha = 1\%$, 2.5% and 5% are given in Appendix D.2. Finally, as in [Barroso and Santa-Clara \(2015\)](#) and [Barroso and Maio \(2017\)](#), we choose an annualized volatility target of $\sigma^{\text{target}} = 12\%$. By using Equations (31) and (46), we obtain VaR and CVaR target levels of $\text{VaR}_{\alpha}^{\text{target}} = 1.9471\%$ and $\text{CVaR}_{\alpha}^{\text{target}} = 2.1861\%$ for a significance level of $\alpha = 0.5\%$.⁷⁰

⁷⁰We have chosen the same α for both, the target VaR and target CVaR strategies, but different target risk levels. Another possibility would be to choose the same target level, but different significance levels as it is done in [Alexander and Baptista \(2004, p. 1262\)](#), i.e. $\text{VaR}_{\alpha}^{\text{target}} = \text{CVaR}_{\tilde{\alpha}}^{\text{target}}$ with $\alpha < \tilde{\alpha}$. [Du and Escanciano \(2016\)](#) suggest that the significance level for CVaR should be about twice the significant level of VaR, i.e. $2\alpha \approx \tilde{\alpha}$. For example, to guarantee that both strategies have the same target risk level when Equations (31) and (46) are used, a significance level of $\tilde{\alpha} = 5\%$ for the target CVaR strategy requires a significance level of about $\alpha = 1.96\%$ for the target VaR strategy.

5.2 Testing the Accuracy of Risk Targeting

We start the empirical part by assessing the accuracy of the different target risk strategies. By definition, the aim of risk targeting is to achieve a predefined level of portfolio risk constantly over time. In particular, we are interested in the question if more advanced models produce a more constant portfolio risk over time and what kind of risk – volatility, VaR or CVaR – an investor should manage if the investor targets a predefined level of volatility, VaR or CVaR. Further, we are interested in the question if managing volatility is sufficient or if incorporating higher moments, as done by managing VaR and CVaR, leads to a higher accuracy as found by [Taylor \(2005\)](#) in a different setting. Testing the accuracy of the different risk models is also important from an economical perspective, since previous studies have shown that a higher forecasting accuracy coincides with a higher economic value in terms of a higher risk-adjusted performance and high utility gains ([Bollerslev et al., 2018](#), [Fleming et al., 2003](#), [Marquering and Verbeek, 2004](#), [Moreira and Muir, 2017](#), [Taylor, 2014](#)). Further, a more constant portfolio risk is also related to a lower tail risk ([Dreyer and Hubrich, 2019](#)). Consequently, a high forecasting accuracy, and hence a more constant portfolio risk, is beneficial for risk targeting investors. We first test the accuracy of the strategies when the investor's aim is to target a certain level of portfolio volatility over time. Whenever a benchmark model is needed, we choose the HSD model as benchmark, which we denote by model 0. Thus, we assess if more advanced models are more successful in targeting a constant level of volatility than the model used in [Barroso and Santa-Clara \(2015\)](#), [Moreira and Muir \(2017\)](#), [Dreyer and Hubrich \(2019\)](#) and [Barroso and Maio \(2017\)](#). This model is then tested against the remaining models $k = 1, \dots, 16$. [Bollerslev et al. \(2018\)](#) find that more advanced models produce more accurate forecasts and higher utility gains than static forecasting models, like the HSD model.

Tables I and II show the results for the tests presented in Section 4.1, where Table I shows the results for the DM-, RC- and SPA-test. The first column of Table I contains the average loss of all models, normalized by the average loss of the HSD model, i.e. $\bar{L}_k^{norm} = \frac{\frac{1}{T} \sum_{t=1}^T L_{k,t}}{\frac{1}{T} \sum_{t=1}^T L_{0,t}}$, $k = 1, \dots, 16$. A normalized loss smaller than 100% indicates that model k is (on average) more accurate than the HSD model, whereas values greater than 100% indicate that the HSD model

Table I. Testing the Accuracy of Volatility Targeting

This table contains the results of the tests of predictive accuracy presented in Section 4.1. $\bar{L}_k^{norm} = \frac{\frac{1}{T} \sum_{t=1}^T L_{k,t}}{\frac{1}{T} \sum_{t=1}^T L_{0,t}}$ defines the average loss of model k , normalized by the loss of model 0 and is given in percent. DM-test stands for the test statistic of the [Diebold and Mariano \(1995\)](#) test. The null-hypothesis of equal predictive ability is rejected for $|\text{DM-test}| > 1.64$, where positive values indicate that model k is more accurate than the HSD model. Bold numbers of DM-test indicate that the model is significantly superior to the HSD model. $p^{RC,n}$ and p^{RC} stand for the naive p -value and p -value of the RC-test ([Sullivan et al., 1999](#), [White, 2000](#)). $p^{SPA,n}$ and $p^{SPA,c}$ stand for the naive p -value and p -value of the SPA-test ([Hansen, 2005](#), [Hansen and Lunde, 2005](#)). $p^{SPA,l}$ and $p^{SPA,u}$ give lower and upper bounds for the p -value of the SPA-test. Bold numbers of these tests indicate that the null-hypothesis that model k is the best model cannot be rejected at a test level of 10%. All p -values are given in percent.

Model	$\bar{L}_k^{norm}$	DM-test	$p^{RC,n}$	p^{RC}	$p^{SPA,n}$	$p^{SPA,l}$	$p^{SPA,c}$	$p^{SPA,u}$
Vola Hist	100.00	0.00	0.00	0.32	0.00	0.00	0.00	0.00
Vola EWMA	83.47	6.28	2.73	63.60	2.73	2.73	2.73	12.30
Vola GARCH	80.88	4.32	100.00	100.00	100.00	100.00	100.00	100.00
VaR Hist	278.33	-9.90	0.00	0.00	0.00	0.00	0.00	0.00
VaR EWMA FHS	98.44	0.31	0.00	0.66	0.00	0.00	0.00	0.00
VaR EWMA EVT	108.13	-1.36	0.00	0.00	0.00	0.00	0.00	0.00
VaR EWMA Stsk	111.72	-2.47	0.00	0.00	0.00	0.00	0.00	0.00
VaR GARCH FHS	89.25	2.00	0.00	12.18	0.00	0.00	0.00	0.00
VaR GARCH EVT	98.76	0.20	0.00	0.32	0.00	0.00	0.00	0.00
VaR GARCH Stsk	119.79	-2.88	0.00	0.00	0.00	0.00	0.00	0.00
CVaR Hist	273.18	-10.63	0.00	0.00	0.00	0.00	0.00	0.00
CVaR EWMA FHS	119.44	-2.83	0.00	0.00	0.00	0.00	0.00	0.00
CVaR EWMA EVT	124.49	-3.47	0.00	0.00	0.00	0.00	0.00	0.00
CVaR EWMA Stsk	142.35	-6.71	0.00	0.00	0.00	0.00	0.00	0.00
CVaR GARCH FHS	100.38	-0.06	0.00	0.10	0.00	0.00	0.00	0.00
CVaR GARCH EVT	106.77	-1.03	0.00	0.00	0.00	0.00	0.00	0.00
CVaR GARCH Stsk	149.67	-6.90	0.00	0.00	0.00	0.00	0.00	0.00

is more successful in achieving a constant portfolio volatility. Table I clearly shows that the dynamic volatility models, i.e. the EWMA and GARCH based target volatility strategies, are the most accurate models, whereas managing VaR or CVaR typically leads to a less accurate portfolio volatility. In particular, when the aim is to target a constant level of volatility, the Historical Simulation managed strategies (VaR-HS and CVaR-HS) are the least accurate models. Further, the DM-test indicates that most of the CVaR models are significantly less accurate in targeting a constant level of volatility, indicated by values of less than -1.64 . When using the RC-test, only three strategies – EWMA, GARCH and VaR-GARCH-FHS – cannot be rejected at a test level of 10%. The RC-test tests if a chosen benchmark model is at least as accurate as all the remaining models. If the null-hypothesis of a model cannot be rejected, i.e. the p -value is higher than the chosen test level of 10%, there is no indication that any other model is more successful in targeting a constant level of portfolio volatility over time than this model. The SPA-test, which extends the RC-test by using a studentized test statistic and a sample dependent null distribution, is typically more powerful in determining inferior models ([Hansen, 2005](#),

Hansen and Lunde, 2005). This is confirmed by our results, since more null-hypotheses are rejected. The SPA-test rejects all null-hypotheses of superior predictive ability, except for the null-hypothesis when the GARCH model is used as benchmark model. As a conclusion, Table I shows that the dynamic volatility models produce the most accurate portfolio volatility.

Table II. Sets of Accurate Volatility Targeting Models

This table contains the results of the stepwise RC-test of Romano and Wolf (2005), the stepwise SPA-test of Hsu et al. (2010), the MCS of Hansen et al. (2003) and Hansen et al. (2011) as well as the FDR method of Barras et al. (2010) and Bajgrowicz and Scaillet (2012). p^R and p^{SQ} stand for the p -values of the MCS and are given in percent. Bold values indicate that the model is contained in the MCS for a test level of 10%. Step-RC and Step-RCst show the step in which the model is added to the set of superior models using the stepwise multiple testing of Romano and Wolf (2005), where Step-RCst uses a studentized test statistic. Step-SPA and Step-SPAst show the step in which the model is added to the set of superior models using the stepwise multiple testing of Hsu et al. (2010), where Step-SPAst uses a studentized test statistic. A value of zero means that the model is not added to the set of superior models. The tests are performed for a test-level of 10%. The last column contains the step in which the model is added to the set of superior models targeting an FDR^+ of 10%. A value of zero indicates that the model is not contained in the superior set.

Model	p^R	p^{SQ}	Step-RC	Step-RC st	Step-SPA	Step-SPA st	$FDR^+ = 10\%$
Vola Hist	0.00	0.00	-	-	-	-	-
Vola EWMA	4.90	4.90	1	1	1	1	1
Vola GARCH	100.00	100.00	1	1	1	1	2
VaR Hist	0.00	0.00	0	0	0	0	0
VaR EWMA FHS	0.00	0.00	0	0	0	0	0
VaR EWMA EVT	0.00	0.00	0	0	0	0	0
VaR EWMA Stsk	0.00	0.00	0	0	0	0	0
VaR GARCH FHS	0.00	0.00	1	1	1	1	3
VaR GARCH EVT	0.00	0.00	0	0	0	0	0
VaR GARCH Stsk	0.00	0.00	0	0	0	0	0
CVaR Hist	0.00	0.00	0	0	0	0	0
CVaR EWMA FHS	0.00	0.00	0	0	0	0	0
CVaR EWMA EVT	0.00	0.00	0	0	0	0	0
CVaR EWMA Stsk	0.00	0.00	0	0	0	0	0
CVaR GARCH FHS	0.00	0.00	0	0	0	0	0
CVaR GARCH EVT	0.00	0.00	0	0	0	0	0
CVaR GARCH Stsk	0.00	0.00	0	0	0	0	0

Table II shows the sets of superior models identified by the stepwise RC-test, the stepwise SPA-test, the MCS and the FDR approach. Whenever a benchmark model is needed, we choose the HSD model as benchmark strategy. The MCS has the advantage that no benchmark model has to be chosen. The MCS, which is an extension of the SPA-test, produces similar results to the SPA-test for all reasonable test levels. For both approaches, only the GARCH model is indicated as an accurate model. The Step-RC and Step-SPA produce larger sets than the MCS and these sets contain the EWMA, GARCH and VaR-GARCH-FHS models. This result is similar to the result of the RC-test. In particular, there are no differences between the sets

of the Step-RC and Step-SPA test. Further, studentizing does not lead to different results. The FDR approach, which is known to typically produce sets of superior models that are at least as large as the sets of the Step-RC and Step-SPA approaches, chooses the same models as superior models. For this approach, the two volatility models (EWMA and GARCH) are chosen in the first two steps, which strengthens the earlier findings that the dynamic volatility models produce the most constant portfolio volatility.

To summarize the results of Tables I and II, we find convincing results of the EWMA, GARCH and VaR-GARCH-FHS model, where the GARCH model delivers the best results. Hence, an investor who wants to achieve a constant portfolio volatility over time should manage volatility directly by a dynamic risk model. Managing downside risk typically fails to target a constant level of volatility. Further, unconditional models, i.e. HSD or Historical Simulation, produce a portfolio volatility that significantly deviates from the desired volatility target. Since a higher forecasting accuracy typically coincides with a higher risk-adjusted performance and high utility gains, we expect better performance results for conditional models. [Bollerslev et al. \(2018\)](#) also find that static models, like the HSD model, are inaccurate and, due to their inaccuracy, produce lower utility gains for an investor who targets a constant level of volatility.

Table III reports results for the VaR backtest of [Christoffersen \(1998\)](#) that is summarized in Section 4.2, where we report p -values for the unconditional and conditional coverage test for significance levels of $\alpha = 0.5\%$, 1% , 2.5% and 5% . These significance levels are also frequently used in the literature on VaR forecasting (see [Bali et al. \(2008\)](#) for example). The VaR backtesting results demonstrate that, for all significance levels, controlling volatility is not sufficient when an investor's aim is to target a constant portfolio VaR over time. In contrast, managing CVaR is feasible for an investor who targets a constant VaR over time. However, VaR based strategies are more successful in targeting a constant portfolio VaR over time than strategies that manage CVaR. Only two of the VaR based strategies that rely on a conditional volatility model can be rejected for a significance level of $\alpha = 0.5\%$ and a test level of 10% . Further, for higher significance levels of α , unconditional models based on the Historical Simulation also fail to target a constant VaR, whereas these models cannot be rejected for low significance levels. A

possible explanation for this result is that low significance levels produce only a limited number of hits. Historical Simulation is known for producing adequate hit ratios, but these hits are usually clustered over time. Hence, when testing for unconditional coverage, Historical Simulation usually delivers convincing results. However, due to the failure of producing independent hits, Historical Simulation is often rejected once the independence or conditional coverage test is applied (Kuester et al., 2006). Since the independence test of Christoffersen (1998) only regards successive hits, low significance levels, and hence only very few hits over the whole sample, imply that the independence test fails to detect the lack of independence. This explains why Historical Simulation seems to perform well for low levels of α . Pritsker (2006) also finds that VaR backtests fail to identify inferior models when only few exceedances occurred over the sample.

Table III. VaR Backtesting Results

The table reports the backtesting results of the Christoffersen (1998) VaR backtest for significance levels of $\alpha = 0.5\%, 1\%, 2.5\%$ and 5% . p_{uc} and p_{cc} denote the p -values for the unconditional coverage and conditional coverage test and are given in percent. Bold numbers mark the models that are not rejected at a test level of 10% .

	$\alpha = 0.5\%$		$\alpha = 1\%$		$\alpha = 2.5\%$		$\alpha = 5\%$	
Model	p_{uc}	p_{cc}	p_{uc}	p_{cc}	p_{uc}	p_{cc}	p_{uc}	p_{cc}
Vola Hist	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.00
Vola EWMA	0.01	0.03	0.00	0.00	0.00	0.01	0.02	0.04
Vola GARCH	1.46	2.90	2.09	6.87	0.10	0.40	5.98	18.22
VaR Hist	82.09	25.17	68.80	27.71	55.22	0.00	51.14	0.00
VaR EWMA FHS	43.75	27.08	41.01	63.71	49.33	25.13	64.51	59.47
VaR EWMA EVT	27.94	10.06	22.13	30.36	67.98	71.83	15.16	8.21
VaR EWMA Stsk	66.34	21.64	79.56	80.76	3.53	5.07	0.17	0.34
VaR GARCH FHS	56.13	29.04	58.70	74.75	61.45	86.73	64.51	90.32
VaR GARCH EVT	19.21	6.95	28.47	37.41	96.32	99.90	39.42	72.83
VaR GARCH Stsk	19.21	6.95	12.52	18.43	74.79	85.83	26.64	50.57
CVaR Hist	19.21	39.97	4.45	1.13	0.61	0.08	0.00	0.00
CVaR EWMA FHS	27.94	10.06	12.52	18.43	6.43	12.41	2.57	4.24
CVaR EWMA EVT	12.57	4.51	1.96	3.17	0.32	1.12	0.99	2.10
CVaR EWMA Stsk	19.21	6.95	12.52	18.43	28.10	52.96	69.05	14.61
CVaR GARCH FHS	27.94	10.06	35.88	44.91	11.92	26.42	8.03	19.06
CVaR GARCH EVT	4.57	1.55	3.00	4.80	1.46	5.08	5.06	12.92
CVaR GARCH Stsk	4.57	1.55	0.15	0.24	0.16	0.68	0.99	2.55

Table IV shows the backtesting results for the two CVaR backtests that were summarized in Section 4.3. We again choose the four significance levels $\alpha = 0.5\%, 1\%, 2.5\%$ and 5% . The p -value of the backtest of McNeil and Frey (2000) is denoted by p_{CVaR} . Nearly all target volatility and target VaR strategies fail to accurately target the portfolio CVaR and are rejected at a test level of 10% . In contrast, only two of the target CVaR strategies can be rejected for significance

levels of $\alpha = 0.5\%$ and $\alpha = 5\%$. This indicates that, by controlling the CVaR of the risky asset, it is possible to achieve a constant portfolio CVaR over time. This finding is also supported by the results of the CVaR backtest of Embrechts et al. (2005), which exhibits the lowest values for the CVaR managed strategies. Further, the values V of the dynamically managed target CVaR strategies are systematically lower than the values of the remaining models, indicating that the CVaR of the risky asset should be managed dynamically. Interestingly, the CVaR-HS approach cannot be rejected by the backtest of McNeil and Frey (2000). However, this backtest is an unconditional backtest which only tests if the produced CVaR is correct on average (Du and Escanciano, 2016). As mentioned above, Historical Simulation is typically rejected once a conditional backtest is applied. The lower values of V of the conditionally managed strategies indicate a higher accuracy of conditional models.

Table IV. CVaR Backtesting Results

This table reports the backtesting results of the McNeil and Frey (2000) and Embrechts et al. (2005) CVaR backtests for significance levels of $\alpha = 0.5\%$, 1% , 2.5% and 5% . V denotes the measure of the Embrechts et al. (2005) backtest, given in Equation (61). Bold numbers mark the lowest value of V . $pCVaR$ denotes the p -value of the backtest of McNeil and Frey (2000) and is given in percent. Bold numbers mark the models that are not rejected at a test level of 10% .

	$\alpha = 0.5\%$		$\alpha = 1\%$		$\alpha = 2.5\%$		$\alpha = 5\%$	
Model	V	$pCVaR$	V	$pCVaR$	V	$pCVaR$	V	$pCVaR$
Vola Hist	0.5825	1.11	0.4257	0.39	0.2840	0.04	0.2097	0.00
Vola EWMA	0.4934	1.85	0.3164	1.93	0.2068	0.42	0.1527	0.09
Vola GARCH	0.3337	10.03	0.2150	7.84	0.1163	9.48	0.0917	1.66
VaR Hist	0.5699	3.30	0.4179	3.69	0.3849	0.20	0.3169	0.00
VaR EWMA FHS	0.3621	2.30	0.2282	2.62	0.0843	11.46	0.0891	1.05
VaR EWMA EVT	0.3145	3.94	0.1842	3.94	0.0898	8.12	0.0957	1.16
VaR EWMA Stsk	0.3356	5.53	0.2182	3.53	0.1506	1.37	0.1472	0.06
VaR GARCH FHS	0.2825	10.42	0.1369	13.93	0.0681	16.24	0.0688	3.30
VaR GARCH EVT	0.2318	14.41	0.1094	16.87	0.0637	14.92	0.0738	2.50
VaR GARCH Stsk	0.1955	26.53	0.1219	13.24	0.0732	12.96	0.0867	1.60
CVaR Hist	0.3052	12.97	0.1376	51.87	0.0848	77.77	0.0526	15.62
CVaR EWMA FHS	0.1814	28.01	0.0772	63.80	0.0200	93.16	0.0085	90.92
CVaR EWMA EVT	0.2305	5.81	0.0857	17.86	0.0209	35.76	0.0186	14.09
CVaR EWMA Stsk	0.1913	27.28	0.0894	34.94	0.0390	54.87	0.0384	47.23
CVaR GARCH FHS	0.1602	44.38	0.0691	57.07	0.0133	99.67	0.0070	94.12
CVaR GARCH EVT	0.1936	17.85	0.0508	41.94	0.0211	49.32	0.0139	32.10
CVaR GARCH Stsk	0.0634	70.90	0.0387	73.45	0.0455	14.36	0.0315	4.93

The backtesting results presented in this section demonstrate two important issues. First, if an investor is interested in targeting portfolio risk in terms of volatility, VaR or CVaR, the investor should directly manage volatility, VaR or CVaR, respectively. In particular, when the aim is to target a certain level of tail risk, it is not sufficient to manage volatility. Second,

when portfolio risk is managed, the investor should use a fast-adapting dynamic risk model instead of unconditional models, like HSD or Historical Simulation, as it is done by [Barroso and Santa-Clara \(2015\)](#), [Barroso and Maio \(2017\)](#) and [Moreira and Muir \(2017\)](#). The next section examines the performance of risk targeting. Based on the backtesting results from this section, we expect a superior performance for the strategies that are based on a conditional risk model.

5.3 Performance of Risk Targeting

We next assess the performance of the different risk targeting strategies and the two benchmark portfolios. Results of this performance analysis over the whole sample are given in Table V. All target risk strategies, except for the Historical Simulation managed target VaR strategy, exhibit higher returns than the 60/40 portfolio with a risk, measured by volatility, drawdown, VaR or CVaR, that is lower – or comparable in the case of the volatility managed strategies – than the risk of the 60/40 portfolio. Further, the risk targeting strategies deliver higher returns with lower risk compared to the DAX. Therefore, dynamically managing portfolio risk can significantly reduce the portfolio's risk without simultaneously sacrificing returns (see also [Fung and Hsieh \(1997\)](#) who found a similar behavior for dynamic trading strategies used by hedge funds). This is also reflected by higher Sharpe Ratios for the dynamically managed target risk strategies compared to the Sharpe Ratios of the two benchmark portfolios. Moreover, within the (dynamically) managed target risk strategies, returns are quite similar. However, the downside risk managed strategies take significantly less risk than the volatility managed strategies. The highest Sharpe Ratio is found for the CVaR-EWMA-Stsk strategy, which is about 287.5% higher than the Sharpe Ratio of the DAX and 51.96% higher than the Sharpe Ratio of the HSD model. The Sharpe Ratio of the best volatility managed strategy is still 205% higher than the Sharpe Ratio of the DAX. The Sharpe Ratio of the best CVaR managed strategy is $0.155/0.122 - 1 = 27.05\%$ higher than the Sharpe Ratio of the best volatility managed strategy. In particular, Sharpe Ratios of the dynamically managed CVaR strategies are all higher than the Sharpe Ratios of the volatility managed strategies. This can also be seen by the modified Sharpe Ratio, which measures the risk-adjusted annualized excess return (see [Jondeau and Rockinger \(2012\)](#) for a

definition of the modified Sharpe Ratio). The best results in terms of the Sharpe Ratios are found for the strategies that are based on the skewed t distribution of [Jondeau and Rockinger \(2003\)](#) and [Bali et al. \(2008\)](#). Generally, in order to increase the risk-adjusted performance, risk should be managed by a dynamic risk model and not by a static risk model, like HSD or Historical Simulation. The Sharpe Ratios of the statically managed strategies are significantly lower than the Sharpe Ratios of the dynamically managed strategies. This finding is in line with [Bollerslev et al. \(2018\)](#) since models that produce a more constant portfolio risk over time, as shown in Section 5.2, also yield an enhanced risk-adjusted performance.

Although the differences in the Sharpe Ratio seem small, results of Table V indicate significant performance gains of portfolio risk management, especially when downside risk is managed. This is the case, since our strategies are highly correlated, which results in very small standard errors for the relative Sharpe Ratios as highlighted in [Kirby and Ostdiek \(2012\)](#). For example, the average correlation between all risk targeting models is 97.34% and the maximum correlation between two strategies is 99.98%. This high correlation between the risk targeting strategies demonstrates that even small differences in the Sharpe Ratios indicate a striking improvement in the tail risk targeting strategies' performance. For example, [Kirby and Ostdiek \(2012\)](#) find Sharpe Ratios for their strategies in the range of 0.47 to 0.49, compared to the benchmark's Sharpe Ratio of 0.46, and they conclude that, due to the high correlation of the strategies, "[t]hese differences translate into significant performance gains". The performance gains of CVaR targeting compared to volatility targeting are even higher in magnitude than the gains found by [Kirby and Ostdiek \(2012\)](#), demonstrating the vast performance gains of CVaR targeting compared to volatility targeting. To test if any model produces a statistically higher Sharpe Ratio than the HSD managed model, we use the corrected version of the Sharpe Ratio test of [Jobson and Korkie \(1981\)](#), which is also used by [DeMiguel et al. \(2009, p. 1928\)](#) and [Cederburg et al. \(2020\)](#). This test could also be applied to more than one strategy simultaneously as shown by [Jobson and Korkie \(1981, II.C\)](#). However, in Section 5.4, we use more sophisticated approaches to test for higher performance gains of all portfolios simultaneously. For that reason, we only test each strategy's Sharpe Ratio against the Sharpe Ratio of the HSD

model. The test of [Jobson and Korkie \(1981\)](#) indicates that only the VaR-EWMA-Stsk model exhibits a Sharpe Ratio that is significantly higher than the Sharpe Ratio of the HSD model when using a test level of 10%.

Table V. Performance Results of Risk Targeting

This table shows the performance results of all target risk strategies and the two benchmark portfolios over the whole period. Return and Volatility stand for the annualized return and volatility, respectively. SR stands for the annualized Sharpe Ratio, z_{JK} stands for the test statistic of the corrected version of the test of [Jobson and Korkie \(1981\)](#) and mSR is the modified Sharpe Ratio defined in [Jondeau and Rockinger \(2012\)](#). MDD and Δ MDD stand for the maximum drawdown and the reduction of the maximum drawdown compared to the maximum drawdown of the DAX. Calmar stands for the drawdown-adjusted return and is defined as in [Farinelli et al. \(2008\)](#) and [Eling and Schuhmacher \(2007\)](#). VaR and CVaR are the in-sample VaR and CVaR, which are estimated with Historical Simulation using all data. Min and Max stand for the minimum and maximum daily return, respectively. Return, Volatility, MDD, Δ MDD, Min and Max are given in percent. Bold numbers of z_{JK} correspond to positive values that are significant at the 10% level, i.e. $z_{JK} \geq 1.6449$.

Model	Return	Volatility	SR	z_{JK}	mSR	MDD	Δ MDD	Calmar	VaR	CVaR	Min	Max
Vola Hist	3.12	12.84	0.102	-	1.49	41.23	41.46	0.032	1.38	1.82	-5.64	5.14
Vola EWMA	3.34	12.50	0.122	0.96	1.97	40.52	42.46	0.038	1.34	1.76	-5.28	4.98
Vola GARCH	3.19	11.96	0.116	0.39	1.82	39.23	44.29	0.035	1.29	1.67	-5.29	4.03
VaR Hist	2.07	9.69	0.029	-0.68	-0.25	31.62	55.10	0.009	0.97	1.44	-4.99	5.29
VaR EWMA FHS	3.23	11.09	0.128	0.87	2.12	37.09	47.33	0.038	1.19	1.56	-4.33	4.24
VaR EWMA EVT	3.23	10.43	0.136	1.29	2.30	35.28	49.91	0.040	1.11	1.47	-4.24	4.01
VaR EWMA Stsk	3.43	10.83	0.150	1.69	2.63	35.07	50.20	0.046	1.16	1.53	-4.76	4.11
VaR GARCH FHS	3.11	11.33	0.115	0.34	1.81	38.32	45.59	0.034	1.22	1.59	-4.67	3.52
VaR GARCH EVT	3.13	10.63	0.125	0.62	2.04	36.55	48.10	0.036	1.15	1.49	-4.63	3.37
VaR GARCH Stsk	3.34	10.39	0.147	1.10	2.58	34.68	50.75	0.044	1.11	1.46	-4.84	3.32
CVaR Hist	2.43	9.26	0.069	-0.33	0.70	28.51	59.52	0.022	0.95	1.37	-5.12	4.17
CVaR EWMA FHS	3.22	10.16	0.139	1.13	2.38	34.35	51.22	0.041	1.07	1.43	-4.01	3.79
CVaR EWMA EVT	3.23	9.93	0.143	1.32	2.48	33.80	52.00	0.042	1.05	1.40	-3.90	3.74
CVaR EWMA Stsk	3.38	10.14	0.155	1.53	2.77	33.12	52.97	0.048	1.09	1.44	-4.67	3.70
CVaR GARCH FHS	3.23	10.68	0.134	0.79	2.25	37.00	47.47	0.039	1.14	1.49	-4.58	3.26
CVaR GARCH EVT	3.26	10.33	0.141	0.99	2.42	35.51	49.58	0.041	1.10	1.44	-4.41	3.19
CVaR GARCH Stsk	3.21	9.74	0.145	0.94	2.51	33.02	53.11	0.043	1.04	1.38	-4.75	2.98
DAX	2.73	23.46	0.040	-0.57	-	70.42	-	0.013	2.36	3.46	-8.49	11.40
60/40	2.37	11.94	0.048	-0.59	0.21	40.07	43.10	0.014	1.24	1.74	-4.32	5.10

We now turn to the drawdown protection ability of risk targeting. Several studies demonstrate that volatility targeting is an easy but successful drawdown and tail risk reduction method (see, for example, [Benson et al. \(2014\)](#), [Barroso and Santa-Clara \(2015\)](#), [Dreyer and Hubrich \(2019\)](#), [Harvey et al. \(2018\)](#) and [Moreira and Muir \(2017\)](#)). As expected, all risk targeting strategies and the 60/40 portfolio are successful in reducing the DAX's drawdown. The maximum drawdown (MDD) of the risk targeting strategies and the 60/40 portfolio is about half of the maximum drawdown of the DAX. Furthermore, the risk targeting strategies exhibit a slightly higher drawdown reduction than the 60/40 portfolio. This can be seen by Δ MDD,

which measures the percentage drawdown reduction compared to the drawdown of the DAX. Managing downside risk, especially managing CVaR, instead of volatility results in a higher drawdown reduction without simultaneously sacrificing returns. Interestingly, the Historical Simulation managed strategies exhibit the highest drawdown reduction. However, this superior drawdown protection is accompanied with significantly lower returns. This is confirmed by the Calmar Ratio, which measures the drawdown-adjusted return and takes the highest values for the dynamically managed target CVaR strategies, whereas the Calmar Ratios of the Historical Simulation managed strategies are significantly lower.⁷¹

In line with the results of the maximum drawdown, the highest (least negative) minimum daily return is achieved by the CVaR managed strategies. The minimum return of the VaR managed strategies is also comparable to the minimum return of the CVaR managed strategies, but slightly lower. In contrast, the minimum return of the volatility managed strategies is significantly more negative than the minimum return of the downside risk managed strategies. Further, the minimum daily return of the Historical Simulation based strategies is significantly more negative than the minimum return of the dynamically managed strategies. This is somewhat surprising, since the Historical Simulation based strategies exhibit the lowest drawdowns, which indicates that these models are the most conservative. Further, the Historical Simulation based strategies also have the lowest average equity exposure, which is not shown here. However, the lower average equity exposure of the Historical Simulation managed strategies is consistent with [Berkowitz and O'Brien \(2002\)](#) who find that banks, who often use Historical Simulation as risk measurement tool, typically exhibit too conservative, i.e. too high, risk estimates that translate in lower equity weights of the strategies that are managed by Historical Simulation. Further, [Berkowitz and O'Brien \(2002\)](#) find that, although commercial banks' in-

⁷¹Since asset returns are usually non-normally distributed, performance measurement based on the Sharpe Ratio solely can lead to wrong conclusions (see [Farinelli et al. \(2008\)](#) or [Eling and Schuhmacher \(2007\)](#) for example). The Sharpe Ratio is only suitable for elliptical distributions, a class of distributions that contains the normal distribution ([Eling and Schuhmacher, 2007](#), p. 2633). Therefore, besides the Sharpe Ratio, we additionally use the Calmar Ratio. This measure replaces the volatility in the Sharpe Ratio by the maximum drawdown. See [Eling and Schuhmacher \(2007\)](#) for definitions and a motivation of enhanced risk-adjusted performance measures. Further, [Jobson and Korkie \(1981, Sec. I\)](#) gives an overview on several widely used performance measures. We also used other performance measures that incorporate the non-normalities of the strategies' returns. However, results were quite similar to the Sharpe Ratio and Calmar Ratio and we do not show these measures here. This finding is in line with [Eling and Schuhmacher \(2007\)](#) who also find a similar ranking order across hedge funds when several risk-adjusted performance measures are used.

ternal risk models produce more conservative risk estimates than a GARCH based VaR model, the banks' models deliver comparable – or even more – VaR violations than the GARCH based model. This explains the somewhat surprising result of a lower equity exposure and draw-down, but a more negative minimum return of the Historical Simulation managed strategies. In total, the Historical Simulation based strategies are more conservative on average, but fail to correctly manage downside risk just when downside risk protection is most needed. This result again highlights the need of a fast adapting risk model when portfolio risk is managed. As before, a higher forecasting accuracy, as examined in Section 5.2, coincides with a superior performance in terms of higher minimum returns. However, even the dynamically managed strategies exhibit minimum daily returns that are similar in magnitude to the minimum return of the 60/40 portfolio. This highlights the fact that unpredictable negative price jumps cannot be completely avoided by risk targeting but, as supposed by [Longin \(2000\)](#), these jumps are best managed by models using EVT. By comparing the minimum return of the EVT, FHS and skewed t distribution based approaches that use the same volatility model, the minimum return of the EVT based models is always higher (less negative) than the minimum return of the other models. As expected, the results found for the minimum daily return reverse when the maximum daily return is compared. In this case, the volatility managed strategies exhibit higher maximum returns than the downside risk managed strategies. This indicates that downside risk timing seems superior in crash periods, but volatility timing is superior in bull markets. This motivates a strategy that switches between downside risk targeting in down-periods and volatility targeting in up-periods, which will be examined in Section 5.5.

The performance evaluation in Table V does not consider transaction costs. However, many studies demonstrate that volatility managing is still beneficial, even when realistic transaction costs are considered (see [Moreira and Muir \(2017\)](#), [Kirby and Ostdiek \(2012\)](#), [Fleming et al. \(2003\)](#), [Fleming et al. \(2001\)](#), [Marquering and Verbeek \(2004\)](#), [Harvey et al. \(2018\)](#) and [Bollerslev et al. \(2018\)](#)). In unreported results, we find that most downside risk managed strategies exhibit lower turnovers than the volatility managed strategies (see [Kirby and Ostdiek \(2012, p. 442\)](#) for a definition of the turnover). Hence, the superiority of downside risk managed

strategies, especially CVaR managed strategies, compared to the volatility managed strategies would be even more striking if realistic transaction costs were considered. However, the skewed t distribution based strategy produces a higher turnover compared to the FHS and EVT based approaches. Hence, the outperformance of the skewed t distribution based strategy over the FHS and EVT based strategies will be lower after transaction costs. Nevertheless, in Appendix D.3, we show that risk targeting is also beneficial when a rebalancing buffer is used. Using a rebalancing buffer lowers the strategies' turnover, which translates into lower transaction costs. Thus, risk targeting strategies can also be used in practice and should be superior to non-managed strategies, even when transaction costs are considered.

Table VI. Lowest and Highest DAX Returns

This table shows the returns for the days when the DAX exhibits the most extreme (positive or negative) returns. Panel A shows the five days with the lowest DAX returns as well as the corresponding returns of the target risk strategies and the 60/40 portfolio on these days. Panel B shows the five days with the highest DAX returns as well as the corresponding returns of the target risk strategies and the 60/40 portfolio on these days. All entries correspond to daily returns and are given in percent.

Model	Panel A: Low DAX Returns (%)					Panel B: High DAX Returns (%)				
Vola Hist	-4.125	-5.638	-2.751	-2.134	-1.119	1.425	1.952	1.774	2.180	3.204
Vola EWMA	-3.466	-5.279	-2.664	-1.858	-1.229	1.523	1.952	2.100	2.193	2.679
Vola GARCH	-3.114	-5.292	-2.622	-1.654	-1.292	1.690	1.912	2.444	2.264	2.301
VaR Hist	-3.211	-4.989	-4.108	-3.275	-2.690	2.548	2.680	3.456	4.467	5.291
VaR EWMA FHS	-3.157	-4.331	-2.192	-1.368	-0.903	1.126	1.976	1.553	1.625	1.985
VaR EWMA EVT	-3.038	-4.235	-2.001	-1.365	-0.901	1.125	1.742	1.547	1.622	1.976
VaR EWMA Stsk	-3.072	-4.764	-1.962	-1.610	-0.866	1.528	1.666	1.940	1.863	2.276
VaR GARCH FHS	-2.998	-4.668	-2.453	-1.399	-1.101	1.468	1.916	2.082	1.913	1.956
VaR GARCH EVT	-2.945	-4.633	-2.145	-1.330	-1.040	1.376	1.809	1.981	1.833	1.855
VaR GARCH Stsk	-2.799	-4.838	-2.016	-1.463	-1.076	1.554	1.625	2.170	1.740	1.919
CVaR Hist	-3.037	-5.123	-3.415	-2.757	-2.326	2.494	2.691	3.380	3.869	4.165
CVaR EWMA FHS	-2.987	-4.006	-1.771	-1.216	-0.802	1.005	1.731	1.385	1.449	1.770
CVaR EWMA EVT	-2.998	-3.895	-1.769	-1.214	-0.799	0.999	1.684	1.380	1.444	1.765
CVaR EWMA Stsk	-2.969	-4.666	-1.482	-1.558	-0.730	1.510	1.580	1.904	1.788	2.239
CVaR GARCH FHS	-2.964	-4.583	-1.986	-1.223	-0.963	1.270	1.836	1.835	1.697	1.718
CVaR GARCH EVT	-2.907	-4.407	-1.923	-1.197	-0.943	1.246	1.778	1.799	1.663	1.684
CVaR GARCH Stsk	-2.715	-4.749	-1.469	-1.412	-1.006	1.528	1.526	2.100	1.657	1.885
DAX	-8.492	-7.164	-7.073	-7.012	-6.838	7.632	7.845	10.344	11.277	11.402
60/40	-4.167	-4.098	-3.575	-3.243	-3.253	3.330	3.290	4.364	4.903	5.097

In order to better assess how risk targeting works in different market environments and if risk targeting mitigates extremely negative returns, we next consider the days when the underlying risky asset, i.e. the DAX, suffers the highest losses or obtains the highest gains. Table VI reports the five lowest and five highest daily DAX returns in conjunction with the corresponding returns of the target risk strategies and the 60/40 portfolio. On the days with the worst DAX returns, the target risk strategies and the 60/40 portfolio deliver significantly higher, i.e. less negative, daily

returns. The highest reduction of the negative returns is achieved by the risk targeting strategies, especially for the strategies that manage the portfolio's CVaR. In line with our earlier results, the highest reduction is again achieved by the dynamically managed strategies. In contrast, the reduction of the HS managed tail risk targeting strategies is significantly lower. Thus, in order to mitigate extreme crashes, a fast adapting risk model should be used to manage portfolio risk. Similarly, the returns of the 60/40 portfolio are typically more negative than the returns of the risk targeting strategies. One exception is the day with the second lowest DAX return, where the target risk strategies' losses are higher than the loss of the 60/40 portfolio. This day indicates a day with an unpredictable negative price jump in the DAX, as examined in [Liu et al. \(2003\)](#) and [Das and Uppal \(2004\)](#). In line with [Longin \(2000\)](#), the best results for this day within the risk targeting strategies is achieved by the EVT based strategies, i.e. (unpredictable) negative price jumps are best managed by EVT. In total, Panel A of Table VI show that extremely negative returns can be significantly reduced by targeting a constant level of portfolio risk, especially when CVaR is managed.

The aforementioned results reverse when the highest daily DAX returns are regarded. In this case, the DAX delivers higher returns than all the remaining models.⁷² The lowest returns are achieved by the risk targeting strategies, especially when CVaR is used as a risk measure. The low returns of the risk targeting strategies is line with the observation that the highest daily returns typically occur during extreme crash periods, so called “bear market rallies”. As a consequence, the exposure of the risk targeting strategies to the risky asset on these days is quite low and the risk targeting strategies fail to capture these extremely high returns. However, in order to achieve a high long-term performance, avoiding highly negative returns is more crucial than achieving highly positive returns.⁷³ Furthermore, the reduction of extremely negative returns also fits better to the preferences of investors who treat losses and gains asymmetrically and weight losses higher than gains. [Aït-Sahalia and Brandt \(2001\)](#) conjecture that loss aversion is highly related to downside risk managed portfolio strategies. This is confirmed by our

⁷²The results of Table VI are also influenced by the chosen volatility target. We have chosen a quite low volatility target. To better capture the upside potential of high DAX returns and to lower the underperformance on days with high DAX returns, risk-seeking or less risk-averse investors should use a higher risk target.

⁷³For example, a return of -5% has to be compensated by a return of 5.26% .

results, since CVaR targeting delivers the most convincing mitigation of extreme losses, which fits well to the preferences of loss-averse investors.

Table VII. Performance Results During and After the Financial Crisis

Panel A shows the performance of all strategies from 15.07.2008 to 15.07.2009, i.e. during the height of the financial crisis. Panel B shows the performance of all strategies from 16.07.2009 to 15.07.2011, i.e. the time following the financial crisis and before the European debt crisis. See Table V for a description of Return, Volatility, SR, MDD, Min, Max. - marks a negative Sharpe Ratio.

	Panel A: 15.07.2008 - 15.07.2009						Panel B: 16.07.2009 - 15.07.2011					
Model	Return	Volatility	SR	MDD	Min	Max	Return	Volatility	SR	MDD	Min	Max
Vola Hist	-3.51	13.16	-	16.93	-2.75	3.20	9.84	12.46	0.709	12.35	-3.03	3.16
Vola EWMA	-3.41	12.83	-	16.91	-2.66	2.93	10.49	12.21	0.776	11.91	-2.85	2.86
Vola GARCH	-4.07	13.56	-	17.71	-2.62	2.98	10.22	11.46	0.803	9.97	-2.35	2.45
VaR Hist	-8.53	17.20	-	22.38	-4.11	5.29	7.12	6.11	1.003	4.17	-1.10	1.76
VaR EWMA FHS	-1.22	10.29	-	12.94	-2.19	2.41	9.09	11.12	0.727	11.52	-2.78	2.60
VaR EWMA EVT	-1.53	9.63	-	12.36	-2.00	2.20	8.20	9.49	0.759	9.66	-2.29	2.21
VaR EWMA Stsk	-0.16	10.81	-	12.02	-1.96	2.42	8.48	10.10	0.740	10.68	-2.69	2.74
VaR GARCH FHS	-2.93	12.23	-	15.71	-2.45	2.76	9.46	10.98	0.770	9.95	-2.39	2.45
VaR GARCH EVT	-2.55	11.08	-	14.26	-2.15	2.42	8.70	9.59	0.803	8.48	-1.99	2.05
VaR GARCH Stsk	-0.08	11.49	-	12.52	-2.02	3.26	8.70	9.51	0.809	8.96	-2.28	2.47
CVaR Hist	-5.92	14.99	-	19.01	-3.42	4.17	6.99	5.98	1.003	4.09	-1.08	1.72
CVaR EWMA FHS	-0.96	8.56	-	10.79	-1.77	1.96	7.28	8.76	0.718	9.55	-2.28	2.02
CVaR EWMA EVT	-0.93	8.55	-	10.76	-1.77	1.96	7.46	8.62	0.751	8.95	-2.12	2.00
CVaR EWMA Stsk	0.67	9.72	-	10.73	-1.56	2.24	7.49	9.23	0.704	10.08	-2.60	2.70
CVaR GARCH FHS	-2.13	10.24	-	13.09	-1.99	2.25	8.09	9.12	0.778	8.52	-2.00	1.94
CVaR GARCH EVT	-1.95	10.01	-	12.74	-1.92	2.18	8.04	8.84	0.797	7.97	-1.87	1.88
CVaR GARCH Stsk	0.05	10.13	-	11.52	-1.75	2.10	8.00	8.70	0.805	8.45	-2.20	2.44
DAX	-18.36	41.48	-	44.53	-7.07	11.40	19.41	18.43	0.993	12.29	-3.33	5.30
60/40	-8.15	18.80	-	23.13	-3.58	5.10	9.89	9.48	0.936	6.86	-1.72	2.64

Based on the results of Table VI, we find that volatility and downside risk targeting behave quite differently in different market environments. In uptrending markets, volatility targeting delivers higher returns, whereas downside risk targeting is more convincing in crash periods. This again motivates a strategy that switches between CVaR and volatility targeting as examined later. To strengthen this observation, we next assess risk targeting in two subsamples, one high risk and one low risk period. Table VII shows the performance of the strategies in the period from 15.07.2008 to 15.07.2011, i.e. during the height of the financial crisis and the time following the financial crisis. We split this period in two subsamples. Splitting the sample in two subsamples is appealing, since [Dreyer and Hubrich \(2019, Fig. 2\)](#) and [Dopfel and Ramkumar \(2013\)](#) find that the profitability of risk targeting can be highly different for periods that include the global financial crisis or start after the global financial crisis. The first subperiod, given in Panel A, covers the financial crisis and ranges from 15.07.2008 to 15.07.2009. This period is marked by highly negative returns and high risk. During the financial crisis, the dynamically

managed risk targeting strategies have significantly higher (less negative) returns than the two benchmark portfolios. Furthermore, the higher returns of the risk targeting strategies are accompanied by lower risk measured by volatility, drawdown and minimum return. For example, the DAX and the 60/40 portfolio exhibit an (annualized) return of -18.36% and -8.15% as well as a volatility of 41.48% and 18.80% , respectively. In contrast, return and volatility of the EWMA managed target volatility strategy are -3.41% and 12.83% , respectively. However, the CVaR-EWMA-Stsk strategy is even more convincing and achieves a positive return of 0.67% with a volatility of only 9.72% . Hence, CVaR targeting significantly outperforms volatility targeting and the two benchmark portfolios, as long as CVaR is managed by a dynamic risk model. In contrast, the statically managed target VaR and target CVaR strategies (VaR-HS and CVaR-HS) exhibit significantly lower returns with higher risk than the dynamically managed strategies. Furthermore, there are significant differences between the EWMA and GARCH managed strategies. The EWMA managed strategies achieve higher returns with lower risk than the GARCH managed strategies. A possible explanation for the better results of the EWMA model could be the higher estimation risk for the GARCH model during highly volatile periods.

The second subperiod, ranging from 16.07.2009 to 15.07.2011, covers the time following the financial crisis, but excludes the European financial debt crisis. This period is marked by a continuously uptrending market with high returns and low risk. Results for the second subperiod are given in Panel B. For this period, the DAX clearly outperforms the remaining strategies. This is in line with [Dreyer and Hubrich \(2019, Fig. 2\)](#) who also find that volatility targeting is outperformed by the non-managed portfolio, once the sample starts after the global financial crisis. Further, in this period, the different target risk strategies perform significantly diverse. The unconditional models, VaR-HS and CVaR-HS, perform very well in this calm market and, by taking less risk, exhibit high Sharpe Ratios. Moreover, the volatility targeting strategies produce higher returns than the downside risk targeting strategies. This again motivates a strategy that switches between volatility and CVaR targeting, where volatility targeting is only used in uptrending markets. However, the higher return of volatility targeting is also accompanied by

higher risk. Consequently, the Sharpe Ratios of the dynamically managed target volatility, VaR and CVaR strategies are quite similar. In contrast, the Sharpe Ratios of the two benchmark portfolios are slightly higher than the Sharpe Ratios of the target risk strategies. However, the differences in the risk-adjusted performance are only small compared to the differences in Panel A. This demonstrates that risk targeting strategies are able to lower the downside risk in extreme crashes, but still (partly) capture the upside potential of the DAX. In particular, the downside risk managed strategies significantly outperform the remaining strategies in bear markets, but only slightly underperform the volatility targeting strategies in uptrending markets.

5.4 The Economic Value of Risk Targeting

In the previous section, we assessed the (risk-adjusted) performance and drawdown protection ability of risk targeting and found that CVaR targeting is superior to volatility targeting. However, our conclusions are so far based on *unconditional* risk-adjusted performance measures, like the Sharpe Ratio and Calmar Ratio. The performance evaluation based on unconditional risk measures has several disadvantages. First, these measures do not account for a time-varying volatility (Han, 2005, Marquering and Verbeek, 2004).⁷⁴ As a consequence, unconditional performance measures are not appropriate for strategies that time volatility (Boguth et al., 2011, Cederburg and O'Doherty, 2016). Second, the Sharpe Ratio does not account for higher moments. Hence, this measures is suboptimal for dynamic trading strategies that reduce a portfolio's tail risk, such as the strategies examined here (Dreyer and Hubrich, 2019, Sec. 5). In particular, unconditional performance measures do not incorporate skewness preferences of investors (Schneider et al., 2020). Third, when using an unconditional performance evaluation measure, performance evaluation can be biased since the rebalancing interval does not coincide with the evaluation period (Boguth et al., 2011, Footnote 6). Similarly, Benartzi and Thaler (1995) show that even long-term investors have quite short evaluation periods. Dreyer and Hubrich (2019) find that performance measurement is highly influenced by the performance

⁷⁴Marquering and Verbeek (2004, p. 419-421) write: "It is important to realize that the Sharpe ratio does not appropriately take into account time-varying volatility. The risk of the dynamic strategies is typically overestimated by the sample standard deviation, particularly in the presence of volatility timing, because the ex post (unconditional) standard deviation is an inappropriate measure for the (conditional) risk an investor was facing at each point in time. This indicates a potentially severe disadvantage of the use of Sharpe ratios to evaluate dynamic strategies."

evaluation frequency. Due to these disadvantages of the Sharpe Ratio, we next assess the economic value of volatility, VaR and CVaR timing, where we define the economic value as the annualized fee an investor is willing to pay to switch from a static portfolio allocation to a risk-managed portfolio. The economic value of volatility timing has been frequently shown in the literature.⁷⁵ These studies find huge improvements of volatility timing in terms of high utility gains for mean-variance investors. For example, [Fleming et al. \(2001\)](#), [Fleming et al. \(2003\)](#), [Kirby and Ostdiek \(2012\)](#), [Han \(2005\)](#) and [Taylor \(2014\)](#) examine utility gains in a multivariate framework using several asset classes, whereas [Marquering and Verbeek \(2004\)](#), [Moreira and Muir \(2017\)](#), [Moreira and Muir \(2019\)](#) and [Bollerslev et al. \(2018\)](#) work with only one risky asset. Further, [Jondeau and Rockinger \(2006\)](#) examine the economic value of portfolio strategies that incorporate higher moments and the authors find that the opportunity costs of ignoring higher moments can become very large when asset returns are non-normally distributed or investors are highly risk-averse. Similarly, [Jondeau and Rockinger \(2012\)](#) assess the economic value of dynamic timing strategies that also incorporate higher moments, like skewness and kurtosis, and find a higher economic value for these strategies compared to strategies that only time volatility. [Ghysels et al. \(2016\)](#) find that investors are willing to pay high fees to switch from mean-variance optimization to mean-variance-skewness optimization. These studies again demonstrate that portfolio allocations that incorporate information on higher moments are highly valuable for investors and increase the investors' utility.

In most studies, the economic value is defined as the maximum fee (in percent) a mean-variance investor is willing to pay to switch from one strategy to another strategy. However, as stated in Section 3.1, investors typically have preferences for moments higher than volatility and the mean-variance framework is not very realistic. For that reason, we further follow [Jondeau and Rockinger \(2012\)](#) and [Dreyer and Hubrich \(2019\)](#) and also calculate the economic value for an investor with constant relative risk aversion (CRRA). CRRA utility is frequently used

⁷⁵See [Fleming et al. \(2001\)](#), [Fleming et al. \(2003\)](#), [Kirby and Ostdiek \(2012\)](#), [Marquering and Verbeek \(2004\)](#), [Taylor \(2014\)](#), [Han \(2005\)](#), [Moreira and Muir \(2017\)](#) and [Bollerslev et al. \(2018\)](#) for studies on the economic value of volatility timing for mean-variance investors. Calculating the economic value, defined as the fee an investor is willing to pay to switch from one strategy to another strategy, is similar to calculating the certainty equivalent return as done by [Ang and Bekaert \(2002\)](#), [Dreyer and Hubrich \(2019\)](#), [Cederburg et al. \(2020\)](#), [Ghysels et al. \(2016\)](#), [Das and Uppal \(2004\)](#), [Guidolin and Timmermann \(2008\)](#), [DeMiguel et al. \(2009\)](#) and [Moreira and Muir \(2019\)](#).

in portfolio selection problems.⁷⁶ [Guidolin and Timmermann \(2008\)](#), [Jondeau and Rockinger \(2012\)](#) and [Bali et al. \(2009\)](#) show that, for reasonable levels of risk aversion, CRRA utility implies that investors prefer higher skewness and lower kurtosis, which is in line with the study of [Scott and Horvath \(1980\)](#).⁷⁷ In particular, “[t]he CRRA utility function implies that investors are tail-risk averse.” ([Dreyer and Hubrich, 2019](#), p. 47). For that reason, by calculating the economic value of risk targeting for an investor with CRRA utility, we explicitly take preferences for higher moments into account and consider the tail risk reduction of the strategies. Finally, in order to regard that investors weight losses higher than gains, we additionally calculate the economic value for loss-averse investors. The portfolio selection for loss-averse investors has been examined by [Benartzi and Thaler \(1995\)](#), [Aït-Sahalia and Brandt \(2001\)](#) and [Ang et al. \(2005\)](#). [Aït-Sahalia and Brandt \(2001\)](#) compare optimal portfolios for mean-variance investors, CRRA investors and loss-averse investors. The authors find that mean-variance and CRRA preferences produce only slightly different optimal portfolio selections (see also [Guidolin and Timmermann \(2008\)](#)), but loss aversion leads to a significantly different optimal portfolio. Similarly, [Ang et al. \(2005\)](#) examine the portfolio selection for CRRA investors and investors with disappointment aversion, who also treat gains and losses asymmetrically. The authors find more realistic asset allocations under disappointment aversion and that disappointment aversion can resemble portfolio allocations under CRRA preferences, whereas the opposite does not hold. Furthermore, several studies show that equity holdings of real investors are typically much lower than predicted for mean-variance or CRRA investors, whereas loss aversion successfully explains the low equity exposure of real investors (see [Benartzi and Thaler \(1995\)](#) and [Ang et al. \(2005\)](#)). Thus, the loss aversion framework seems to be more realistic than the mean-variance

⁷⁶See, for example, [Ang and Bekaert \(2002\)](#), [Liu et al. \(2003\)](#), [Das and Uppal \(2004\)](#), [Aït-Sahalia and Brandt \(2001\)](#), [Guidolin and Timmermann \(2008\)](#), [Ghysels et al. \(2016\)](#) for studies that examine the portfolio allocation for CRRA investors.

⁷⁷Several studies show that the CRRA utility framework is mainly driven by the preferences for mean, variance, skewness and kurtosis. For example, [Guidolin and Timmermann \(2008\)](#) compare the asset allocation under CRRA preferences with portfolio allocations under four moment preferences and find only minor differences between both approaches. Hence, portfolio selection under CRRA utility is mainly driven by preferences for the first four moments. A similar result also holds for investors with constant absolute risk aversion (CARA) as shown by [Bali et al. \(2009\)](#) and [Jondeau and Rockinger \(2006\)](#). [Jondeau and Rockinger \(2006\)](#) show that portfolio allocations under CARA utility is mainly driven by preferences for the first four moments and that portfolio allocations under CARA and CRRA utility produce similar results. For that reason, we do not calculate the economic value for CARA utility.

and CRRA frameworks. In total, we expect quite similar results for the mean-variance and CRRA investors, but quite different (and more realistic) results for the loss-averse investor.

As first method to calculate the economic value of risk targeting, we follow [Fleming et al. \(2001\)](#), [Fleming et al. \(2003\)](#), [Han \(2005\)](#) and [Kirby and Ostdiek \(2012\)](#) and assume that the investor's true utility function can be approximated by quadratic utility. For this investor, the realized day t utility is given by

$$U_{MV}(R_{t,a}) = W_{t-1}(1 + R_{t,a}) - \frac{1}{2}\gamma_{abs}W_{t-1}^2(1 + R_{t,a})^2, \quad (62)$$

where γ_{abs} is the investor's absolute risk aversion, W_{t-1} denotes the investor's wealth on day $t - 1$ and $R_{t,a}$ denotes the day t return of strategy a . We call an investor with preferences as in Equation (62) a mean-variance investor, since this approach is highly related to the mean-variance theory ([Fleming et al., 2001](#), p. 334). By assuming that this investor has a constant relative risk aversion γ , Equation (62) can be rewritten as

$$U_{MV}(R_{t,a}) = W_0 \left((1 + R_{t,a}) - \frac{\gamma}{2(1 + \gamma)}(1 + R_{t,a})^2 \right). \quad (63)$$

The economic value of a strategy a is then given by the percentage fee Δ_{MV} the investor with utility in Equation (62) is willing to pay to switch from the 60/40 portfolio to the strategy a . The fee Δ_{MV} is defined by equating the expected utilities

$$\mathbb{E}(U_{MV}(R_{t,a} - \Delta_{MV})) = \mathbb{E}(U_{MV}(R_{t,b})), \quad (64)$$

where $R_{t,b}$ denotes the return of the 60/40 portfolio. The expected utility in Equation (64) is then estimated by the average realized utility. Hence, the fee Δ_{MV} is calculated by solving

$$\overline{U}_{MV}(R_{1,a} - \Delta_{MV}, \dots, R_{T,a} - \Delta_{MV}) = \overline{U}_{MV}(R_{1,b}, \dots, R_{T,b}), \quad (65)$$

where $\overline{U}_{MV}(R_1, \dots, R_T) = \frac{1}{T} \sum_{t=1}^T (1 + R_t) - \frac{\gamma}{2(1+\gamma)}(1 + R_t)^2$. We calculate the fee Δ_{MV} for levels of risk aversion given by $\gamma = 2, 5, 10$ and 15 , which are in line with previous studies using this approach (see [Marquering and Verbeek \(2004\)](#), [Aït-Sahalia and Brandt \(2001\)](#), [Jondeau and Rockinger \(2006\)](#) and [Jondeau and Rockinger \(2012\)](#) for example).

In the case of the CRRA investor, the realized day t utility is given by

$$U_{CRRA}(R_{t,a}) = \begin{cases} \frac{(1+R_{t,a})^{(1-\gamma)}}{1-\gamma}, & \text{if } \gamma > 1 \\ \ln(1 + R_{t,a}), & \text{if } \gamma = 1. \end{cases} \quad (66)$$

Since we choose the same levels of γ as stated above, the investor's utility simplifies to the case $U_{CRRRA}(R_{t,a}) = \frac{(1+R_{t,a})^{(1-\gamma)}}{1-\gamma}$. Following [Jondeau and Rockinger \(2012\)](#), the economic value for an investor with CRRRA utility is defined by equating the expected utilities

$$\mathbb{E}(U_{CRRRA}(R_{t,a} - \Delta_{CRRRA})) = \mathbb{E}(U_{CRRRA}(R_{t,b})), \quad (67)$$

which is again estimated by the average realized utility. The percentage fee Δ_{CRRRA} is then calculated by solving

$$\bar{U}_{CRRRA}(R_{1,a} - \Delta_{CRRRA}, \dots, R_{T,a} - \Delta_{CRRRA}) = \bar{U}_{CRRRA}(R_{1,b}, \dots, R_{T,b}), \quad (68)$$

where $\bar{U}_{CRRRA}(R_1, \dots, R_T) = \frac{1}{T} \sum_{t=1}^T \frac{(1+R_t)^{(1-\gamma)}}{1-\gamma}$.

Lastly, to account for the loss aversion of investors, we use a utility function that gives more weight on negative returns. Following [Aït-Sahalia and Brandt \(2001\)](#) and [Benartzi and Thaler \(1995\)](#), we define the investor's day t utility by

$$U_{LA}(R_{t,a}) = \begin{cases} (R_{t,a})^b, & \text{if } R_{t,a} \geq 0 \\ -l(-R_{t,a})^b, & \text{if } R_{t,a} < 0, \end{cases} \quad (69)$$

where $l > 1$ determines the investor's loss aversion and b measures the degree of risk seeking for negative returns and risk aversion for positive returns (see [Aït-Sahalia and Brandt \(2001, p. 1314\)](#) or [Benartzi and Thaler \(1995, p. 79\)](#)).⁷⁸ Typical values of l and b are in the range of $l = 2.25$ and $b = 0.88$, which are motivated empirically. Similar to [Aït-Sahalia and Brandt \(2001\)](#), we choose the four combinations of $l = 2.0, 3.0$ and $b = 0.8, 1$. A loss aversion of $l = 2$ implies that the disutility of a loss is twice as great as the utility of a positive return of the same magnitude ([Benartzi and Thaler, 1995, p. 74](#)).⁷⁹ The economic value for a loss-averse investor is then given by equating the expected utilities

$$\mathbb{E}(U_{LA}(R_{t,a} - \Delta_{LA})) = \mathbb{E}(U_{LA}(R_{t,b})). \quad (70)$$

⁷⁸We also used the risk-free rate instead of a zero return to define the cut-off point which determines a loss or a gain. Results for the economic value were nearly identical for both choices and the risk-free rate based results are not reported here.

⁷⁹Another possibility to assess the economic value of an investor with unexpected utility would be to use preferences of an ambiguity-averse investor as in [Aït-Sahalia and Brandt \(2001\)](#) and [Jondeau and Rockinger \(2012\)](#) or preferences of a disappointment-averse investor as in [Ang et al. \(2005\)](#). See also [Jondeau and Rockinger \(2012, Footnote 17\)](#) for a list of studies that incorporate ambiguity aversion in asset allocation.

As above, we calculate Δ_{LA} by solving

$$\bar{U}_{LA}(R_{1,a} - \Delta_{LA}, \dots, R_{T,a} - \Delta_{LA}) = \bar{U}_{LA}(R_{1,b}, \dots, R_{T,b}), \quad (71)$$

where $\bar{U}_{LA}(R_1, \dots, R_T) = \frac{1}{T} \sum_{t=1}^T R_t^b \cdot \mathbb{1}_{\{R_t \geq 0\}} - l(-R_t)^b \cdot \mathbb{1}_{\{R_t < 0\}}$.

Table VIII shows the values Δ_i for the three utility functions, i.e. Δ_i is defined as the annualized percentage fee a mean-variance, CRRA or loss-averse investor is willing to pay to switch from the 60/40 strategy to one of the risk timing strategies. In addition, by choosing the same subperiods as in Table VII, we also examine the economic value of risk timing during and after the financial crisis. For these subperiods, we only report the results for a risk aversion of $\gamma = 5$ and $\gamma = 10$ in the case of the mean-variance and CRRA investor as well as $l = 2$ and $l = 3$ combined with $b = 0.8$ for the loss-averse investor.

Panel A of Table VIII shows the economic value for a mean-variance investor. The economic value over the whole sample is positive for almost all risk targeting strategies and levels of risk aversion. Further, we find that downside risk timing delivers a significantly higher economic value than volatility timing and that managing CVaR delivers the highest economic value. In other words, a mean-variance investor should manage CVaR instead of volatility. Interestingly, as in [Marquering and Verbeek \(2004\)](#), we find that the economic value of volatility timing is decreasing in the level of risk aversion γ , but this result reverses when the economic value of downside risk timing is assessed. In this case, the economic value is increasing in the risk aversion, i.e. for highly risk-averse mean-variance investors, CVaR timing becomes more important. For example, a mean-variance investor with a risk aversion of $\gamma = 15$ would pay an annualized fee of 0.756% to switch from the 60/40 portfolio to the GARCH managed strategy, but the same investor is not willing to pay a positive fee to switch to the HSD managed strategy. However, the same investor would even pay an annualized fee of 4.253% to switch from the 60/40 portfolio to the CVaR-GARCH-Stsk strategy. The differences between volatility and downside risk timing become even more striking in the crash period. During the financial crisis, an investor with a risk aversion of $\gamma = 5$ would pay an annualized fee of 9.184% to switch from the 60/40 portfolio to the EWMA managed strategy. However, the same investor is even willing to pay an annualized fee of 15.384% to switch to the CVaR-EWMA-Stsk strategy. Thus, in crash peri-

Table VIII. Economic Value of Risk Targeting

This table shows the economic value, given as annualized percentage fee Δ_i an investor is willing to pay to switch from the 60/40 portfolio to a risk timing strategy for a given utility function U_i , $i \in \{MV, CRRA, LA\}$. Panel A shows results for a mean-variance investor. Panel B shows results for an investor with CRRA utility. Panel C shows results for a loss-averse investor. γ determines the investor's risk aversion. l determines the investor's loss aversion and b measures the investor's degree of risk seeking for negative returns and risk aversion for positive returns.

	Whole Sample				Crash		Recovery	
Panel A: Δ_{MV}	$\gamma = 2$	$\gamma = 5$	$\gamma = 10$	$\gamma = 15$	$\gamma = 5$	$\gamma = 10$	$\gamma = 5$	$\gamma = 10$
Vola Hist	0.618	0.280	-0.283	-0.843	8.881	13.836	-1.343	-2.953
Vola EWMA	0.876	0.669	0.323	-0.021	9.184	14.394	-0.637	-2.100
Vola GARCH	0.799	0.789	0.772	0.756	8.018	12.641	-0.527	-1.556
VaR Hist	-0.049	0.682	1.912	3.158	0.731	2.172	-1.493	-0.186
VaR EWMA FHS	0.940	1.237	1.735	2.236	12.968	20.081	-1.396	-2.227
VaR EWMA EVT	1.009	1.521	2.381	3.248	12.906	20.407	-1.540	-1.542
VaR EWMA Stsk	1.168	1.552	2.194	2.841	13.930	20.780	-1.527	-1.828
VaR GARCH FHS	0.799	1.015	1.375	1.737	10.057	15.743	-0.995	-1.752
VaR GARCH EVT	0.898	1.346	2.098	2.856	11.070	17.580	-1.121	-1.171
VaR GARCH Stsk	1.126	1.653	2.537	3.430	13.672	20.057	-1.099	-1.114
CVaR Hist	0.347	1.206	2.653	4.121	5.073	8.469	-1.581	-0.236
CVaR EWMA FHS	1.031	1.629	2.634	3.648	14.001	22.157	-2.121	-1.798
CVaR EWMA EVT	1.066	1.733	2.856	3.992	14.038	22.200	-1.900	-1.513
CVaR EWMA Stsk	1.196	1.800	2.816	3.841	15.384	22.992	-2.093	-1.977
CVaR GARCH FHS	0.991	1.424	2.150	2.881	11.954	19.037	-1.503	-1.337
CVaR GARCH EVT	1.057	1.602	2.517	3.440	12.253	19.490	-1.453	-1.161
CVaR GARCH Stsk	1.068	1.794	3.016	4.253	14.491	21.799	-1.441	-1.088
Panel B: Δ_{CRRA}	$\gamma = 2$	$\gamma = 5$	$\gamma = 10$	$\gamma = 15$	$\gamma = 5$	$\gamma = 10$	$\gamma = 5$	$\gamma = 10$
Vola Hist	0.731	0.387	-0.197	-0.793	7.878	12.680	-1.022	-2.649
Vola EWMA	0.945	0.733	0.371	0.000	8.130	13.173	-0.345	-1.820
Vola GARCH	0.803	0.789	0.759	0.723	7.076	11.540	-0.322	-1.358
VaR Hist	-0.292	0.438	1.668	2.917	0.430	1.810	-1.753	-0.452
VaR EWMA FHS	0.841	1.135	1.625	2.117	11.550	18.457	-1.232	-2.074
VaR EWMA EVT	0.839	1.348	2.202	3.065	11.418	18.707	-1.542	-1.551
VaR EWMA Stsk	1.041	1.422	2.057	2.697	12.566	19.230	-1.469	-1.775
VaR GARCH FHS	0.728	0.940	1.292	1.645	8.910	14.410	-0.845	-1.610
VaR GARCH EVT	0.749	1.194	1.939	2.692	9.768	16.084	-1.112	-1.167
VaR GARCH Stsk	0.952	1.476	2.355	3.247	12.401	18.629	-1.097	-1.115
CVaR Hist	0.062	0.918	2.365	3.837	4.384	7.688	-1.849	-0.510
CVaR EWMA FHS	0.833	1.427	2.426	3.438	12.390	20.322	-2.187	-1.873
CVaR EWMA EVT	0.844	1.509	2.627	3.762	12.426	20.363	-1.979	-1.599
CVaR EWMA Stsk	0.995	1.596	2.606	3.630	13.877	21.285	-2.118	-2.006
CVaR GARCH FHS	0.847	1.277	1.996	2.721	10.543	17.423	-1.538	-1.377
CVaR GARCH EVT	0.876	1.418	2.327	3.248	10.814	17.844	-1.513	-1.226
CVaR GARCH Stsk	0.828	1.550	2.768	4.005	13.040	20.150	-1.512	-1.162
	$b = 0.8$		$b = 1$		$b = 0.8$		$b = 0.8$	
Panel C: Δ_{LA}	$l = 2$	$l = 3$	$l = 2$	$l = 3$	$l = 2$	$l = 3$	$l = 2$	$l = 3$
Vola Hist	-4.057	-6.599	-4.707	-7.490	19.566	28.406	-9.141	-13.852
Vola EWMA	-3.191	-5.377	-3.606	-5.932	20.476	29.792	-8.289	-12.770
Vola GARCH	-1.786	-3.157	-1.865	-3.224	17.193	25.036	-6.377	-9.761
VaR Hist	8.009	12.568	10.125	15.791	6.462	10.148	9.090	15.170
VaR EWMA FHS	1.423	1.749	1.978	2.617	31.116	45.090	-5.690	-8.333
VaR EWMA EVT	3.564	5.064	4.635	6.710	32.736	47.798	-1.133	-1.114
VaR EWMA Stsk	2.641	3.537	3.426	4.733	30.469	43.812	-2.489	-3.272
VaR GARCH FHS	0.251	0.013	0.613	0.591	22.955	33.377	-5.268	-7.836
VaR GARCH EVT	2.574	3.590	3.473	4.971	26.938	39.426	-1.381	-1.666
VaR GARCH Stsk	3.851	5.479	5.013	7.230	28.531	40.709	-0.666	-0.561
CVaR Hist	9.139	14.167	11.688	18.014	13.800	20.473	9.395	15.689
CVaR EWMA FHS	4.589	6.648	5.887	8.645	36.478	53.296	0.674	2.023
CVaR EWMA EVT	5.332	7.800	6.836	10.110	36.525	53.359	1.188	2.758
CVaR EWMA Stsk	5.171	7.479	6.541	9.562	34.719	50.002	-0.055	0.858
CVaR GARCH FHS	2.565	3.526	3.439	4.864	30.050	43.969	-0.239	0.328
CVaR GARCH EVT	3.752	5.352	4.918	7.129	30.919	45.186	0.620	1.669
CVaR GARCH Stsk	6.149	9.106	7.867	11.715	32.692	47.158	1.759	3.466

ods, mean-variance investors are willing to pay extremely high fees to switch to a risk targeting strategy, where the willingness to pay for CVaR managed strategies is significantly higher than the willingness to pay for volatility managed strategies. Hence, investors are willing to pay extremely high fees to mitigate crashes, which is best done by managing CVaR. This is in line with the results of [Bollerslev and Todorov \(2011\)](#) and [Chabi-Yo et al. \(2018\)](#) that investors are crash-averse. Furthermore, during the financial crisis, we find that the EWMA model is again superior to the GARCH model. This result is in line with the results of Table VII and could be explained by the fact that the EWMA model is not influenced by estimation risk. Further, during the crash period, investors are willing to pay much lower fees to switch to the static VaR-HS and CVaR-HS models. Hence, static models fail to achieve a good downside risk protection just when it is most needed. This is again in line with our earlier findings that accurate risk models are superior to static risk models, especially in crash periods. As expected, these results reverse when we regard the period that starts immediately after the financial crisis. In this period, all risk targeting strategies exhibit a negative economic value. This is again in line with our earlier findings. However, the utility gains in the crash period are significantly higher in magnitude than the utility losses in the calm period. Panel B shows the economic value of risk timing for an investor with CRRA utility. In this case, we also incorporate preferences for higher moments and consider that investors dislike left tail risk. However, results are quite similar to the results of the mean-variance investor shown in Panel A. For CRRA investors, managing CVaR is superior to managing volatility, especially in times of bear markets and for highly risk-averse investors. However, results reverse during the uptrending period. As mentioned above, the observation that results for the mean-variance and CRRA investors are quite similar is in line with the studies on the optimal portfolio allocation of mean-variance and CRRA investors. These studies find quite similar portfolios for both investors, whereas the optimal portfolio of loss-averse investors is highly different and more in line with portfolios in practice. Thus, we expect different and more realistic results for the economic value of the loss-averse investor.

Panel C shows the economic value of risk targeting for a loss-averse investor. The economic value for a loss-averse investor is significantly different to the economic value of a mean-

variance or CRRA investor. As stated above, this result is in line with [Aït-Sahalia and Brandt \(2001\)](#) who find similar results when mean-variance or CRRA preferences are used in portfolio selection problems, but vastly different allocations for loss-averse investors. Interestingly, over the whole period, the economic value of the target volatility strategies is negative, regardless of the level of loss aversion and volatility model. In other words, a loss-averse investor would pay a positive fee to switch away from a target volatility strategy to the 60/40 portfolio. In contrast, the economic value of downside risk targeting is always positive and typically very high. As before, we find that CVaR targeting produces the highest economic value, i.e. a loss-averse investor should time CVaR or at least VaR instead volatility. Somewhat surprising, we find a higher economic value for the unconditional models (VaR-HS and CVaR-HS). However, this finding can be explained by the lower average equity exposure of these strategies. Hence, these strategies are more conservative and should be more appealing for loss-averse investors who have a preference for conservative strategies. The extremely high economic value for loss-averse investors who manage downside risk can partly be explained by the daily evaluation period used in the economic value calculation. [Benartzi and Thaler \(1995\)](#) show that loss aversion is more pronounced for shorter evaluation periods, i.e. the shorter the evaluation period for a loss-averse investor the less attractive are investments with higher risk. Similar horizon effects have been found by [Aït-Sahalia and Brandt \(2001\)](#) for loss aversion, but not for mean-variance and CRRA preferences. The authors conclude that loss aversion implies that short-term investors are extremely risk-averse, whereas long-term investors become more risk-neutral.⁸⁰ During the financial crisis, the economic value of risk targeting becomes extremely high, i.e. a loss-averse investor is willing to pay extremely high fees for downside risk protection during crash periods. This again confirms the result of [Bollerslev and Todorov \(2011\)](#) and [Chabi-Yo et al. \(2018\)](#) that investors are crash-averse. For example, a loss-averse investor with

⁸⁰[Dreyer and Hubrich \(2019\)](#) also find that performance evaluation measures can be highly different when different evaluation periods are used. We also calculated the economic value for a loss-averse investor by first aggregating the daily returns to monthly returns. As expected, the economic value for a loss-averse investor calculated with monthly returns is smaller in magnitude than the economic value calculated with daily data. Nevertheless, the economic value of volatility targeting is still negative for all combinations and the economic value of downside risk targeting is still positive for all combinations, except for the VaR-GARCH-FHS model for $b = 0.8$. The highest economic value is again obtained by the strategies that manage CVaR. Furthermore, the economic value for a CRRA investor is nearly unchanged when using monthly returns instead of daily returns. This result is also found by [Aït-Sahalia and Brandt \(2001\)](#) for the optimal portfolio choice under CRRA preferences and loss aversion.

parameters $b = 0.8$ and $l = 3$ would pay an annualized fee of 29.792% to switch from the 60/40 portfolio to the EWMA managed target volatility strategy. However, the same investor would even pay a fee of 53.359% per year to switch to a CVaR managed strategy. Furthermore, during the crash period, we find a significantly higher economic value for dynamically managed strategies. This again indicates that more accurate risk models are more successful in managing extremely negative returns in crash periods. Interestingly, opposed to the results of the mean-variance and CRRA investor, we even find a positive economic value of almost all CVaR targeting strategies in the uptrending market, whereas the economic value of volatility targeting is negative and high in magnitude. In other words, a loss-averse investor should time CVaR instead of volatility, regardless of whether the market is in a bull or a bear regime.

The fees given in Table VIII are extremely high compared to the fees found by [Bollerslev et al. \(2018\)](#). The authors argue that even their fees, in the range of 0.5%, are extremely beneficial for investors. This highlights the advantage of risk targeting, especially CVaR targeting, found for our data set. However, there are several differences between our study and the study of [Bollerslev et al. \(2018\)](#), which explain the differences in the magnitude of the fees. First, [Bollerslev et al. \(2018\)](#) calculate utility gains of several volatility targeting strategies, relying on different volatility models, against a benchmark volatility targeting strategy. In other words, the authors choose a certain target volatility strategy as benchmark model, whereas we choose the 60/40 portfolio as benchmark, which is similar to [Marquering and Verbeek \(2004\)](#). Second, the authors only compare the differences between several volatility forecasting models, whereas we also compare the differences between volatility and downside risk targeting. In line with the results of [Bollerslev et al. \(2018\)](#), differences within the volatility targeting strategies are only small, whereas the differences between volatility and CVaR targeting are significantly higher. For example, a mean-variance investor with a risk aversion of $\gamma = 5$ would pay a fee of $0.789\% - 0.280\% = 0.509\%$ per year to switch from the HSD managed strategy to the GARCH managed strategy. This result is comparable to the finding of [Bollerslev et al. \(2018\)](#) and again demonstrates the positive relation between forecasting accuracy – or equivalently a more constant portfolio volatility – and utility gains. However, the same investor would even

pay $1.800\% - 0.280\% = 1.52\%$ per year to switch from the HSD managed strategy to the CVaR-EWMA-Stsk strategy. Third, [Bollerslev et al. \(2018\)](#) rebalance the weight of the volatility targeting strategy monthly, whereas we use daily rebalancing. Since the authors show that a higher accuracy typically coincides with higher utility gains, daily rebalancing should also produce a higher economic value. Since daily rebalancing also induces higher transaction costs, the extremely high fees of Table VIII should be lower in practice.⁸¹ Nevertheless, in Section D.3, we show that risk targeting is still beneficial when portfolio weights are only rebalanced when the DAX's risk changes dramatically. Fourth, additional to mean-variance preferences, we also calculate the economic value for loss-averse investors, who are willing to pay extremely high fees for a crash risk reduction that is best done by CVaR targeting.

Finally, we test if the economic value found in Table VIII is also statistically significant. [Bollerslev et al. \(2018\)](#) use the DM-test to statistically compare the utility benefit of several volatility models used in a volatility targeting strategy. [Taylor \(2014, Sec. 2.2\)](#) presents a conditional test, that extends the DM-test, to assess if advanced forecasting models produce higher utility gains than simple forecasting models. [Kirby and Ostdiek \(2012\)](#) use a bootstrap based test to assess the significance of utility gains. A similar approach is also used by [DeMiguel et al. \(2009\)](#) and [Cederburg et al. \(2020\)](#) to test for differences in the certainty equivalent return for mean-variance investors. We follow these approaches and apply the tests that were presented in Section 4.1 to test if a strategy produces a significantly higher utility. These tests are also frequently used to test for a superior (risk-adjusted) performance of technical trading rules or mutual funds (see [Sullivan et al. \(1999\)](#), [Hsu et al. \(2010\)](#), [Barras et al. \(2010\)](#), [Bajgrowicz and Scaillet \(2012\)](#) among others). Results of these tests are shown in Table IX, where we only show results for $\gamma = 10$ for the mean-variance and CRRA investor as well as $l = 2$ and $b = 1$ for the loss-averse investor. Whenever a benchmark model is needed, we choose the 60/40 portfolio as benchmark. Panel A shows results for the mean-variance investor. The DM-test indicates that almost all downside risk targeting strategies produce higher utilities than the 60/40 portfolio, whereas all target volatility strategies do not produce statistically higher utilities. The RC-test fails to reject any null hypothesis which again demonstrates the weak-

⁸¹Tail risk targeting for monthly rebalancing is examined in [Rickenberg \(2020a\)](#).

Table IX. Testing the Utility Gain of Risk Targeting

This table shows the results of the tests presented in Section 4.1 used to test the significance of the utility gains. Panel A shows results for a mean-variance investor with $\gamma = 10$. Panel B shows results for a CRRA investor with $\gamma = 10$. Panel C shows results for a loss-averse investor with $b = 1$ and $l = 2$. The description of the columns is given in Tables I and II.

Panel A: MV	DM-test	p^{RC}	p^{SPA}	p^{SQ}	Step-RC st	Step-SPA st	$FDR^+ = 10\%$
Vola Hist	-0.24	19.69	0.04	0.08	0	0	0
Vola EWMA	0.29	26.15	0.07	0.30	0	0	0
Vola GARCH	0.79	30.32	0.11	0.15	0	0	15
VaR Hist	2.35	55.93	1.75	21.36	1	1	3
VaR EWMA FHS	1.58	47.12	0.62	4.15	0	2	13
VaR EWMA EVT	2.16	64.79	0.97	21.36	1	1	9
VaR EWMA Stsk	1.96	58.15	0.77	16.17	1	1	12
VaR GARCH FHS	1.39	39.00	0.20	0.99	0	0	14
VaR GARCH EVT	2.09	56.18	0.55	7.74	1	1	10
VaR GARCH Stsk	2.46	76.40	3.02	36.14	1	1	5
CVaR Hist	3.16	74.85	42.98	92.67	1	1	1
CVaR EWMA FHS	2.29	75.56	1.18	39.16	1	1	8
CVaR EWMA EVT	2.49	86.18	55.07	92.67	1	1	4
CVaR EWMA Stsk	2.40	85.74	47.75	92.67	1	1	7
CVaR GARCH FHS	2.09	58.51	0.10	10.22	1	1	11
CVaR GARCH EVT	2.41	75.18	21.02	36.14	1	1	6
CVaR GARCH Stsk	2.73	100.00	100.00	100.00	1	1	2
Panel B: CRRA	DM-test	p^{RC}	p^{SPA}	p^{SQ}	Step-RC st	Step-SPA st	$FDR^+ = 10\%$
Vola Hist	-0.27	20.49	0.01	0.06	0	0	0
Vola EWMA	0.27	26.74	0.03	0.31	0	0	0
Vola GARCH	0.76	31.12	0.04	0.16	0	0	15
VaR Hist	2.34	56.40	1.87	22.34	1	1	7
VaR EWMA FHS	1.57	47.61	0.66	4.16	0	2	13
VaR EWMA EVT	2.15	65.11	0.70	22.34	1	1	9
VaR EWMA Stsk	1.95	58.54	0.95	16.36	2	1	12
VaR GARCH FHS	1.37	39.83	0.10	0.91	0	0	14
VaR GARCH EVT	2.08	56.38	0.26	7.42	1	1	10
VaR GARCH Stsk	2.45	76.84	3.40	36.21	1	1	6
CVaR Hist	3.14	74.35	42.39	92.73	1	1	1
CVaR EWMA FHS	2.29	76.22	1.20	40.30	1	1	8
CVaR EWMA EVT	2.48	86.93	55.26	92.73	1	1	3
CVaR EWMA Stsk	2.39	85.67	47.91	92.73	1	1	4
CVaR GARCH FHS	2.08	58.86	0.02	10.47	1	1	11
CVaR GARCH EVT	2.41	76.08	21.72	36.21	1	1	5
CVaR GARCH Stsk	2.73	100.00	100.00	100.00	1	1	2
Panel C: LA	DM-test	p^{RC}	p^{SPA}	p^{SQ}	Step-RC st	Step-SPA st	$FDR^+ = 10\%$
Vola Hist	-2.72	0.00	0.00	0.00	0	0	0
Vola EWMA	-2.25	0.00	0.00	0.00	0	0	0
Vola GARCH	-1.45	0.00	0.00	0.00	0	0	0
VaR Hist	9.09	52.46	0.00	0.18	1	1	1
VaR EWMA FHS	0.94	0.21	0.00	0.00	0	0	13
VaR EWMA EVT	2.35	2.54	0.00	0.00	1	1	9
VaR EWMA Stsk	1.79	0.81	0.00	0.00	1	1	12
VaR GARCH FHS	0.20	0.05	0.00	0.00	0	0	0
VaR GARCH EVT	1.96	0.64	0.00	0.00	1	1	10
VaR GARCH Stsk	2.96	2.69	0.00	0.00	1	1	6
CVaR Hist	10.68	100.00	100.00	100.00	1	1	2
CVaR EWMA FHS	2.91	6.05	0.00	0.00	1	1	7
CVaR EWMA EVT	3.39	10.13	0.05	0.18	1	1	5
CVaR EWMA Stsk	3.38	8.81	0.06	0.17	1	1	3
CVaR GARCH FHS	1.90	0.64	0.00	0.00	1	1	11
CVaR GARCH EVT	2.73	2.70	0.00	0.00	1	1	8
CVaR GARCH Stsk	4.49	15.07	0.03	0.18	1	1	4

nesses of the RC-test. In contrast, the SPA-test rejects the null-hypotheses of all target volatility and target VaR strategies, whereas the null-hypotheses for most target CVaR strategies cannot be rejected. Thus, the SPA-test indicates that the target CVaR strategies are significantly more valuable for mean-variance investors than volatility and VaR targeting. Results for the MCS are quite similar to the results of the DM-test. None of the target volatility strategies is contained in the MCS, which is also confirmed by the stepwise approaches, where we only show results for the studentized versions. The Step-SPA approach identifies almost all downside risk targeting strategies as superior, whereas the target volatility strategies are not identified as superior. As expected, the FDR approach produces the largest set of superior models and picks all models, except for the HSD and EWMA model. Panel B shows results for the CRRA investor. As in Table VIII, these results are again quite similar to the results of the mean-variance investor.

Panel C shows results for the loss-averse investor. The DM-test again indicates that almost all downside risk targeting strategies produce statistically higher utility gains than the 60/40 portfolio. In contrast, the volatility targeting strategies produce lower utilities, where the utilities of the HSD and EWMA based strategies are even statistically lower with a test statistic that is lower than -1.64 . Results for the RC-test, SPA-test and the MCS approach are very different to the findings of the DM-test. These tests indicate that the Historical Simulation based target CVaR strategy clearly outperforms the remaining models. This result is also in line with the high economic value of this strategy shown in Table VIII. In contrast, the stepwise approaches and the FDR approach produce large sets of optimal models and pick (almost) all downside risk targeting strategies, whereas none of the target volatility strategies is chosen. The FDR approach further shows that the target CVaR strategies are typically picked in the first steps. The differences between the results of the MCS and the stepwise approaches can be explained by their construction. The stepwise approaches identify superior models (compared to the 60/40 portfolio) and then test the remaining models in the next steps. In contrast, the MCS eliminates bad performing models and then tests all the remaining models in the next steps. Hence, in the MCS approach, a good performing model remains in the test set until the last step. Thus, if one model clearly outperforms the remaining models, all other models are identified as inferior to

this model. Due to the significantly higher economic value of the CVaR-HS model for a loss-averse investor, all other models are clearly eliminated in the first steps. In contrast, the stepwise approaches pick the HS based CVaR model and all other models that also produce a significantly higher utility than the 60/40 portfolio in the first steps. Thus, the stepwise approach picks all models that are superior to the 60/40 portfolio, whereas the MCS picks *the best* model(s) among all models. In total, results of Panel C demonstrate that loss-averse investors should time downside risk instead of volatility, where CVaR timing produces the best results. This again confirms the suggestion of [Aït-Sahalia and Brandt \(2001, p. 1315-1316\)](#) that loss aversion is highly related to CVaR based portfolio constructions, as examined by [Basak and Shapiro \(2001\)](#).

5.5 Switching Strategies

Results so far indicate that volatility targeting produces higher returns in uptrending markets, whereas CVaR targeting provides a better drawdown protection. However, in uptrending markets, the CVaR targeting approach is typically too conservative. For that reason, we next examine strategies that switch between volatility and CVaR targeting, based on indicators that indicate if the following day is an up- or down-day. Thus, we combine the superiority of the volatility targeting strategy in calm markets with the superiority of CVaR targeting in crash periods. Combining different portfolio strategies is frequently examined in the literature ([DeMiguel et al., 2009](#), [Garlappi et al., 2006](#), [Kan and Zhou, 2007](#), [Tu and Zhou, 2011](#)). Similarly, [Wang et al. \(2012\)](#) switch between different target risk levels, where a more conservative target is chosen if a crash regime is expected. This is similar to our approach of switching to a more conservative strategy when a down-market is expected. Furthermore, [Taylor \(2014\)](#) proposes to switch between several forecasting models using an estimate of the current market environment, which is again similar to switching between different target risk strategies. A combined strategy that manages portfolio risk by the portfolio's CVaR in times of bear markets, but switches to a volatility based strategy in bull markets, should be successful in capturing the upside potential while simultaneously drawdowns are mitigated. Another possibility would be to buy the risky asset, i.e. $w_t = 1$, in bull markets and use a CVaR based strategy in bear markets. However,

controlling an asset's risk can even be advantageous in bull markets ([Barroso and Santa-Clara, 2015](#), Table 6). Following [Tu and Zhou \(2011\)](#), we define the weight of day t as

$$w_t^{switch} = \delta_t \cdot w_t^{CVaR} + (1 - \delta_t) \cdot w_t^{vol}, \quad (72)$$

where $\delta_t \in \mathbb{R}$ is the weight placed on the target CVaR strategy, w_t^{CVaR} is the day t weight of the CVaR targeting strategy and w_t^{vol} is the day t weight of the volatility targeting strategy. Several possibilities to define the crash indicator δ_t are feasible. For example, a regime-switching process as in [Ang and Bekaert \(2002\)](#), [Guidolin and Timmermann \(2008\)](#) and [Wang et al. \(2012\)](#) could be used to determine bull and bear regimes. However, since risk targeting is also relevant for practical implementations, we will rely on simple models to determine δ_t . Our first two switching strategies use a crash indicator δ_t that equals one if a negative return on day t is expected and zero else, given information up to day $t - 1$. Hence, these approaches use either volatility or CVaR targeting. In order to determine $\delta_t \in \{0, 1\}$, we use methods from the literature on technical analysis ([Bajgrowicz and Scaillet, 2012](#), [Hsu et al., 2010](#), [Moskowitz et al., 2012](#), [Sullivan et al., 1999](#)), where we use the two most prominent methods, i.e. Moving Averages (MA) and Time Series Momentum (TSMOM). Based on the MA approach, the indicator δ_t is given by

$$\delta_t = \begin{cases} 1, & \text{if } S_{t-1} \leq \text{MA}_{t-1,n} \\ 0, & \text{if } S_{t-1} > \text{MA}_{t-1,n}, \end{cases} \quad (73)$$

where $\text{MA}_{t-1,n} = \frac{1}{n} \sum_{i=1}^n S_{t-i}$ denotes the Moving Average with a length of n days. Hence, if the risky asset is in an uptrend, given by $S_{t-1} > \text{MA}_{t-1,n}$, the portfolio is managed by the portfolio's volatility. In contrast, if the risky asset is in a downtrend, given by $S_{t-1} \leq \text{MA}_{t-1,n}$, the portfolio is managed by the more conservative CVaR targeting approach.

Based on the TSMOM approach of [Moskowitz et al. \(2012\)](#), the indicator δ_t is given by

$$\delta_t = \begin{cases} 1, & \text{if } S_{t-1} \leq S_{t-1-n} \\ 0, & \text{if } S_{t-1} > S_{t-1-n}. \end{cases} \quad (74)$$

Hence, the portfolio on day t is managed by volatility if the price of day $t - 1$ is higher than the price of day $t - 1 - n$, i.e. we use volatility targeting when the risky asset is in an uptrend. In contrast, during a downtrend, i.e. $S_{t-1} \leq S_{t-1-n}$, the more conservative CVaR targeting

approach is used. We use $n = 200$ days for the MA and TSMOM approach, which is the most common choice made by the literature and practitioners.⁸²

Table X. Performance Results of Risk Targeting: Switching Strategies

This table shows performance results of the strategies that switch between the EWMA volatility targeting strategy and the CVaR targeting strategies for three different indicators δ_t . Panel A shows results for the indicator δ_t based on a 200 day Time Series Momentum (TSMOM) rule. Panel B shows results for the indicator δ_t based on a 200 day Moving Average (MA) rule. Panel C shows results for the indicator δ_t based on the volatility forecast $\hat{\sigma}_t$ of the EWMA model. The description of the columns is given in Table V.

Panel A: TSMOM Indicator	Return	Vola	SR	z_{JK}	MDD	Calmar	VaR	CVaR	Min	Max
Vola Hist	3.12	12.84	0.102	-	41.23	0.032	1.38	1.82	-5.64	5.14
Index	2.73	23.46	0.040	-0.57	70.42	0.013	2.36	3.46	-8.49	11.40
60/40	2.37	11.94	0.048	-0.59	40.07	0.014	1.24	1.74	-4.32	5.10
EWMA/CVaR Hist	4.36	12.33	0.206	1.94	31.65	0.080	1.29	1.75	-5.12	4.98
EWMA/CVaR EWMA FHS	3.92	11.71	0.180	2.22	37.03	0.057	1.24	1.65	-4.01	4.98
EWMA/CVaR EWMA EVT	3.96	11.64	0.184	2.29	36.62	0.059	1.23	1.64	-3.91	4.98
EWMA/CVaR EWMA Stsk	4.24	11.73	0.206	3.00	34.96	0.069	1.24	1.65	-4.67	4.98
EWMA/CVaR GARCH FHS	3.83	12.07	0.167	2.02	38.81	0.052	1.28	1.68	-4.58	4.98
EWMA/CVaR GARCH EVT	3.93	11.93	0.177	2.26	37.68	0.056	1.26	1.67	-4.41	4.98
EWMA/CVaR GARCH Stsk	4.19	11.76	0.202	2.61	35.38	0.067	1.24	1.65	-4.75	4.98
Panel B: MA Indicator	Return	Vola	SR	z_{JK}	MDD	Calmar	VaR	CVaR	Min	Max
Vola Hist	3.12	12.84	0.102	-	41.23	0.032	1.38	1.82	-5.64	5.14
Index	2.73	23.46	0.040	-0.57	70.42	0.013	2.36	3.46	-8.49	11.40
60/40	2.37	11.94	0.048	-0.59	40.07	0.014	1.24	1.74	-4.32	5.10
EWMA/CVaR Hist	4.28	12.31	0.199	1.75	31.08	0.079	1.29	1.75	-5.12	4.98
EWMA/CVaR EWMA FHS	3.87	11.66	0.176	2.09	36.38	0.056	1.23	1.64	-4.01	4.98
EWMA/CVaR EWMA EVT	3.92	11.58	0.181	2.17	35.98	0.058	1.22	1.63	-3.91	4.98
EWMA/CVaR EWMA Stsk	4.07	11.72	0.192	2.62	34.86	0.064	1.24	1.66	-4.67	4.98
EWMA/CVaR GARCH FHS	3.85	12.00	0.169	2.02	37.97	0.053	1.28	1.67	-4.58	4.98
EWMA/CVaR GARCH EVT	3.94	11.86	0.179	2.23	36.82	0.058	1.26	1.65	-4.41	4.98
EWMA/CVaR GARCH Stsk	4.08	11.71	0.193	2.36	34.48	0.066	1.24	1.65	-4.75	4.98
Panel C: Volatility Indicator	Return	Vola	SR	z_{JK}	MDD	Calmar	VaR	CVaR	Min	Max
Vola Hist	3.12	12.84	0.102	-	41.23	0.032	1.38	1.82	-5.64	5.14
Index	2.73	23.46	0.040	-0.57	70.42	0.013	2.36	3.46	-8.49	11.40
60/40	2.37	11.94	0.048	-0.59	40.07	0.014	1.24	1.74	-4.32	5.10
EWMA/CVaR Hist	2.42	12.26	0.051	-0.28	36.28	0.017	0.94	1.85	-7.60	10.87
EWMA/CVaR EWMA FHS	3.17	9.03	0.152	0.57	30.55	0.045	0.95	1.33	-3.55	4.17
EWMA/CVaR EWMA EVT	3.14	8.68	0.154	0.58	29.17	0.046	0.91	1.28	-3.40	4.14
EWMA/CVaR EWMA Stsk	3.80	9.13	0.217	1.30	25.64	0.077	0.96	1.33	-4.45	4.11
EWMA/CVaR GARCH FHS	3.25	10.23	0.141	0.52	36.28	0.040	1.08	1.47	-4.33	3.81
EWMA/CVaR GARCH EVT	3.25	9.58	0.151	0.65	32.89	0.044	1.02	1.38	-4.10	3.77
EWMA/CVaR GARCH Stsk	3.36	8.99	0.173	0.68	28.06	0.055	0.93	1.28	-4.56	8.40

The two indicators defined above are dummy variables, taking a value of $\delta_t = 1$ if a negative return is likely and zero else. As a consequence, the weight of day t is either given by the volatility targeting strategy *or* the CVaR targeting strategy. We next define a third indicator, where the weight of day t is given as a combination of the volatility and CVaR targeting strate-

⁸²Moskowitz et al. (2012) find good results for the TSMOM strategy for periods between one and 36 months. This corresponds to a window size between 21 and 756 days. We also used other lengths and found good results for other choices of n . For example, choosing $n = 150$ produces even higher risk-adjusted returns compared to $n = 200$. However, since $n = 200$ is the most relevant length, we only show results for this choice.

gies. This is similar to the approach of combining different forecasting methods (Allen et al., 2012, Halbleib and Pohlmeier, 2012, Taylor, 2014). If market risk increases, measured by expected volatility of day t , we place more weight on the CVaR targeting strategy, whereas CVaR targeting becomes less important when market risk decreases.⁸³ More formally, we define δ_t as

$$\delta_t = \frac{\hat{\sigma}_t}{\sigma^{\text{target}}}, \quad (75)$$

where σ^{target} is the chosen volatility target and $\hat{\sigma}_t$ is the volatility forecast of one of the volatility models. Generally, market timing strategies based on expected volatility are frequently used by practitioners (Christoffersen and Diebold, 2006, Copeland and Copeland, 1999). Defining δ_t with respect to the chosen volatility target is appealing since more risk-averse investors choose lower levels of σ^{target} , which implies higher values of δ_t . This fits well to our earlier finding that more risk-averse investors obtain higher utility gains from CVaR targeting compared to volatility targeting. Hence, by choosing δ_t as a function of σ^{target} , more risk-averse investors place higher weights on CVaR targeting, whereas risk-seeking investors place higher weights on volatility targeting. Thus, the portfolio of a risk-averse investor is more conservative than the portfolio of a risk-seeking investor.

The weight of the switching strategy using the indicator given in Equation (75) can be rewritten as

$$w_t^{\text{switch}} = w_t^{\text{vol}} + \left(\frac{w_t^{\text{CVaR}}}{w_t^{\text{vol}}} - 1 \right). \quad (76)$$

Hence, this switching strategy is similar to the volatility targeting strategy with weight w_t^{vol} , but the switching strategy places more (less) weight on the risky asset when the weight of the CVaR targeting strategy is higher (lower) than the weight of the volatility targeting strategy. This strategy is similar to the approach of Packham et al. (2017) who examine tail risk hedging strategies based on the difference of VaR forecasts under a normality assumption and forecasting methods that take non-normalities into account. By definition, the CVaR takes non-normalities into account and w_t^{CVaR} should be higher (lower) than w_t^{vol} when the market is in an bull (bear) regime

⁸³We also used an indicator based on the two indicators defined above, given by $\delta_t = (\delta_t^{\text{MA}} + \delta_t^{\text{TSMOM}})/2$. Thus, this strategy uses either volatility targeting, CVaR targeting or an equally weighted combination of both strategies. However, results were quite similar to the previous approaches and are not reported.

with lower (higher) left tail risk. Thus, this switching strategy should be similar to the volatility targeting strategy, but should react more sensitive to changes in the market environment, where the weight is lowered in a down-market and increased in an up-market, governed by changes in higher moments like skewness and kurtosis.

Table XI. Economic Value of Risk Targeting: Switching Strategies

This table shows the economic value, given as annualized percentage fee Δ_i an investor is willing to pay to switch from the 60/40 portfolio to a strategy that switches between volatility and CVaR targeting, for a given utility function U_i , $i \in \{MV, CRRA, LA\}$. Panel A shows the economic value for a mean-variance investor. Panel B shows the economic value for an investor with CRRA utility. Panel C shows the economic value for a loss-averse investor. γ indicates the investor's risk aversion. l determines the investor's loss aversion and b measures the investor's degree of risk seeking for negative returns and risk aversion for positive returns.

	Whole Sample				Crash		Recovery	
Panel A: Δ_{MV}	$\gamma = 2$	$\gamma = 5$	$\gamma = 10$	$\gamma = 15$	$\gamma = 5$	$\gamma = 10$	$\gamma = 5$	$\gamma = 10$
EWMA/CVaR Hist	0.013	-0.105	-0.302	-0.498	-16.012	-26.657	-4.204	-2.298
EWMA/CVaR EWMA FHS	1.097	2.026	3.593	5.183	19.340	29.305	-4.034	-3.127
EWMA/CVaR EWMA EVT	1.095	2.119	3.849	5.606	19.426	29.379	-3.844	-2.866
EWMA/CVaR EWMA Stsk	1.697	2.605	4.136	5.689	29.427	39.157	-4.048	-3.429
EWMA/CVaR GARCH FHS	1.056	1.632	2.601	3.578	15.483	24.424	-3.070	-2.366
EWMA/CVaR GARCH EVT	1.120	1.894	3.198	4.517	16.074	25.207	-3.162	-2.312
EWMA/CVaR GARCH Stsk	1.283	2.224	3.812	5.422	25.367	32.485	-3.284	-2.399
Panel B: Δ_{CRRA}	$\gamma = 2$	$\gamma = 5$	$\gamma = 10$	$\gamma = 15$	$\gamma = 5$	$\gamma = 10$	$\gamma = 5$	$\gamma = 10$
EWMA/CVaR Hist	0.054	-0.045	-0.194	-0.381	-13.517	-24.043	-4.581	-2.689
EWMA/CVaR EWMA FHS	0.790	1.714	3.277	4.870	17.388	27.079	-4.216	-3.322
EWMA/CVaR EWMA EVT	0.756	1.776	3.502	5.266	17.476	27.155	-4.040	-3.074
EWMA/CVaR EWMA Stsk	1.396	2.301	3.831	5.391	27.527	37.061	-4.173	-3.560
EWMA/CVaR GARCH FHS	0.865	1.438	2.401	3.376	13.723	22.415	-3.212	-2.517
EWMA/CVaR GARCH EVT	0.864	1.635	2.934	4.256	14.279	23.158	-3.332	-2.491
EWMA/CVaR GARCH Stsk	0.972	1.917	3.530	5.192	24.114	31.687	-3.461	-2.582
	$b = 0.8$		$b = 1$		$b = 0.8$		$b = 0.8$	
Panel C: Δ_{LA}	$l = 2$	$l = 3$	$l = 2$	$l = 3$	$l = 2$	$l = 3$	$l = 2$	$l = 3$
EWMA/CVaR Hist	9.677	14.863	11.314	17.176	-10.945	-15.003	14.550	24.395
EWMA/CVaR EWMA FHS	9.891	14.798	12.402	18.670	49.383	71.451	4.782	9.462
EWMA/CVaR EWMA EVT	11.059	16.621	13.999	21.135	49.497	71.493	5.333	10.248
EWMA/CVaR EWMA Stsk	10.397	15.350	13.112	19.398	52.726	72.999	3.289	7.092
EWMA/CVaR GARCH FHS	5.107	7.432	6.407	9.413	42.252	62.125	3.018	6.225
EWMA/CVaR GARCH EVT	7.255	10.771	9.161	13.674	43.604	64.214	4.131	8.045
EWMA/CVaR GARCH Stsk	10.306	15.553	13.425	20.117	46.414	64.530	5.446	10.048

Results for the three indicators are given in Table X, where we only show results for the strategies that switch between the EWMA model and the CVaR targeting strategies. For a better comparison to our previous results, we also show results for the HSD based target volatility strategy, the DAX and the 60/40 portfolio. Panel A shows results for the indicator δ_t based on the TSMOM strategy. Compared to the HSD model, switching between the EWMA based strategy and the target CVaR strategies successfully heightens the return while simultaneously

the volatility is reduced. The switching strategies provide an enhanced risk-return profile, indicated by a higher Sharpe Ratio and Calmar Ratio, than the individual strategies given in Table V. For example, the strategy that switches between the EWMA model and the CVaR-EWMA-Stsk strategy increases the Sharpe Ratio of the HSD model by $0.206/0.102 - 1 = 101.96\%$. The high increase of the Sharpe Ratio can also be seen by the Sharpe Ratio test of [Jobson and Korkie \(1981\)](#). All switching strategies provide a statistically higher Sharpe Ratio than the HSD model, whereas only one model in Table V was able to provide a statistically higher Sharpe Ratio. Further, the switching strategies also provide a higher drawdown protection, indicated by the lower MDD and minimum return. Thus, the switching approach seems to capture the benefits of volatility and CVaR targeting and produces higher returns with lower risk. Panel B shows results for the indicator based on the 200 day Moving Average, which are quite similar to the TSMOM based results. As before, all switching strategies produce significantly higher Sharpe Ratios than the HSD model. Panel C shows results for the volatility based indicator δ_t . Interestingly, although some strategies based on this indicator produce the highest Sharpe Ratios in Table X, none of these strategies produces a significantly higher Sharpe Ratio with [Jobson and Korkie \(1981\)](#) test statistics lower than 1.64. However, this can be explained by a lower correlation of these switching strategies with the HSD strategy compared to the other indicators. Furthermore, the switching strategies based on the volatility based indicator δ_t exhibit the lowest drawdowns among the three indicators, which is in line with Equation (76) that this strategy is similar to the volatility targeting strategy, but more sensitive to up- and down-markets.

Table XI shows the economic value of the switching strategies that use the volatility based indicator δ_t for the mean-variance, CRRA and loss-averse investors. We only show results for the volatility based indicator δ_t , since Table X indicates that none of the switching strategies based on this indicator produces significant performance gains for the test of [Jobson and Korkie \(1981\)](#). Based on the results of Table XI, we will test in Table XII if these utility gains are statistically significant. Thus, our results in Table XII are conservative and results for the TSMOM and MA indicator are superior to the results shown here. The economic value is again calculated with respect to the 60/40 portfolio, i.e. the numbers in this table correspond to the

annual percentage fee an investor is willing to pay to switch away from the 60/40 portfolio to one of the switching strategies. Consequently, these numbers can be directly compared to the results of Table VIII. Results of Panels A and B in Table XI are similar to the results of Table VIII, but higher in magnitude. Thus, mean-variance and CRRA investors are willing to pay higher fees for the switching strategies over the whole sample and the crash period compared to the individual strategies. However, during the calm period, these investors prefer the 60/40 portfolio. Thus, a possible extension of our switching approach would be to switch between the CVaR managed strategy and a non-managed static portfolio. Panel C shows results for the loss-averse investor. Results are again similar to Table VIII, but higher in magnitude for the whole period and the period capturing the financial crisis. Interestingly, during the crisis period, the economic value of the strategy that switches to the Historical Simulation based target CVaR strategy is negative, although this strategy was quite convincing in Table VIII. However, this result is in line with our earlier findings that, especially on days with an extremely negative return, dynamically managed strategies are superior to a statically managed strategies. Since the switching strategy places higher weights on CVaR targeting when a crash is expected, the HS based switching strategy should perform worse during extreme crashes. During the calm period, the economic value of all switching strategies becomes positive and high in magnitude. This holds especially for the strategy that switches to the Historical Simulation based strategy. Thus, loss-averse investors are willing to pay high fees to have access to a strategy that switches between volatility and CVaR targeting, even when the market is trending upwards. In contrast, a loss-averse investor is not willing to pay a positive fee to use the EWMA based target volatility strategy as shown in Table VIII. In particular, Table XI again shows that conditional models produce higher utility gains than unconditional models. This is again in line with our earlier findings that there is a link between forecasting accuracy and utility gains for risk targeting investors.

To summarize Table XI, switching between volatility and CVaR targeting heightens utility gains for all three investors compared to the static 60/40 portfolio. Further, utility gains of the switching strategies are higher in magnitude compared to the economic value of the individual

Table XII. Testing the Utility Gain of Risk Targeting: Switching Strategies

This table shows the results of the tests presented in Section 4.1 used to test for the significance of the utility gains. Panel A shows results for a mean-variance investor with $\gamma = 10$. Panel B shows results for a CRRA investor with $\gamma = 10$. Panel C shows results for a loss-averse investor with $b = 1$ and $l = 2$. The description of the columns is given in Tables I and II.

Panel A: Mean-Variance	DM-test	p^{RC}	p^{SPA}	p^{SQ}	Step-RC st	Step-SPA st	$FDR^+ = 10\%$
Vola Hist	-0.23	3.66	0.30	0.80	0	0	0
Vola EWMA	0.28	5.91	0.57	1.64	0	0	0
Vola GARCH	0.77	7.51	1.21	2.81	0	0	0
EWMA/CVaR Hist	-0.14	8.43	11.47	11.39	0	0	0
EWMA/CVaR EWMA FHS	2.17	74.39	9.12	48.84	1	1	5
EWMA/CVaR EWMA EVT	2.27	82.19	45.80	84.39	1	1	2
EWMA/CVaR EWMA Stsk	2.23	100.00	100.00	100.00	1	1	1
EWMA/CVaR GARCH FHS	1.82	41.73	0.69	14.90	1	1	6
EWMA/CVaR GARCH EVT	2.16	64.85	27.44	46.11	1	1	3
EWMA/CVaR GARCH Stsk	2.25	79.42	43.03	84.39	1	1	4
Panel B: CRRA	DM-test	p^{RC}	p^{SPA}	p^{SQ}	Step-RC st	Step-SPA st	$FDR^+ = 10\%$
Vola Hist	-0.26	3.97	0.24	0.76	0	0	0
Vola EWMA	0.26	5.99	0.60	1.45	0	0	0
Vola GARCH	0.75	7.57	1.13	2.57	0	0	0
EWMA/CVaR Hist	-0.11	9.01	12.07	11.47	0	0	0
EWMA/CVaR EWMA FHS	2.17	74.66	9.26	48.31	1	1	5
EWMA/CVaR EWMA EVT	2.28	82.64	47.38	87.53	1	1	2
EWMA/CVaR EWMA Stsk	2.23	100.00	100.00	100.00	1	1	1
EWMA/CVaR GARCH FHS	1.82	42.04	0.60	14.00	1	1	6
EWMA/CVaR GARCH EVT	2.16	66.10	28.55	46.16	1	1	3
EWMA/CVaR GARCH Stsk	2.23	80.82	46.72	87.53	1	1	4
Panel C: Loss Aversion	DM-test	p^{RC}	p^{SPA}	p^{SQ}	Step-RC st	Step-SPA st	$FDR^+ = 10\%$
Vola Hist	-2.63	0.00	0.00	0.00	0	0	0
Vola EWMA	-2.16	0.00	0.00	0.00	0	0	0
Vola GARCH	-1.40	0.00	0.00	0.00	0	0	0
EWMA/CVaR Hist	5.03	43.80	36.99	69.53	1	1	1
EWMA/CVaR EWMA FHS	4.33	43.64	0.00	0.22	1	1	2
EWMA/CVaR EWMA EVT	4.76	100.00	100.00	100.00	1	1	3
EWMA/CVaR EWMA Stsk	4.64	64.23	32.40	69.53	1	1	4
EWMA/CVaR GARCH FHS	2.71	0.03	0.00	0.00	1	1	7
EWMA/CVaR GARCH EVT	3.70	2.36	0.00	0.02	1	1	6
EWMA/CVaR GARCH Stsk	5.32	66.24	41.12	69.53	1	1	5

risk targeting strategies as shown in Table VIII. This confirms the earlier finding of Table X that the switching approach further enhances the risk-return profile. We will next test if these utility increases are also statistically significant. Table XII shows results for the tests that test for statistically significant utility gains of the switching strategies. For a better comparison, we also include the three volatility targeting strategies. Table XII shows that the switching strategies produce statistically significant utility increases, whereas the volatility targeting strategies do not. For all three investors, most switching strategies produce significantly higher utilities compared to the 60/40 portfolio, whereas the volatility targeting strategies do not exhibit statistically significant utility increases. In line with our earlier findings, the most convincing

results are again found for the strategies that switch to a conditionally managed CVaR strategy. Consequently, results of Table XII show that switching between volatility and CVaR targeting produces statistically significant higher utilities for all three investors, whereas the test of [Jobson and Korkie \(1981\)](#) does not indicate that the switching strategies based on the volatility indicator δ_t exhibit statistically higher Sharpe Ratios. This again highlights that the Sharpe Ratio is a suboptimal performance measure for dynamic trading strategies ([Han, 2005](#), [Marquering and Verbeek, 2004](#)).

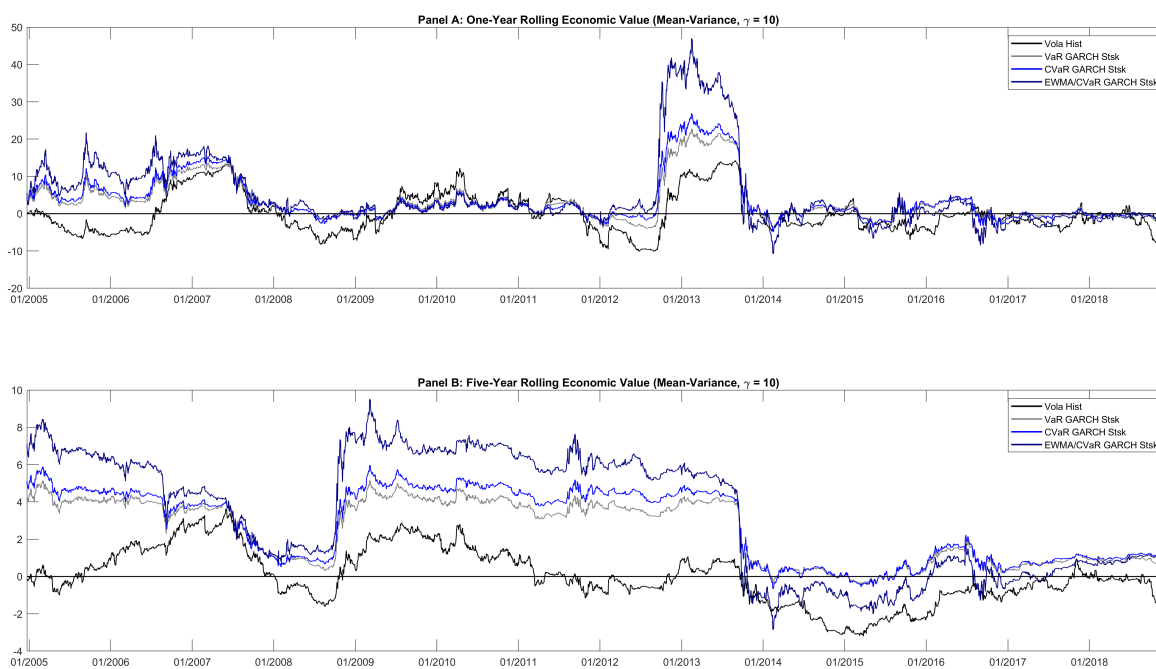


Figure I. One and Five Year Rolling Economic Value for a Mean-Variance Investor. This figure plots the one and five year rolling economic value measured by Δ_{MV} with respect to the 60/40 portfolio for a mean-variance investor with a risk aversion of $\gamma = 10$. Panel A shows the economic value for an investor with an investment horizon of one year, whereas Panel B shows the economic value for an investor with an investment horizon of five years. The dates on the x axes correspond to the end dates of the one or five year investment horizon.

We have so far focused our analysis on the economic value calculated over the whole sample. However, most investors typically have short evaluation periods ([Benartzi and Thaler, 1995](#)). Further, timing short-term risk is also beneficial for long-term investors ([Moreira and Muir, 2019](#)), i.e. even long-term investors should be concerned about short-term utility gains. For that reason, similar to Figure 2 of [Marquering and Verbeek \(2004\)](#), we next plot in Figure I the rolling one and five year economic value for a mean-variance investor who uses our risk

targeting and switching strategies. Thus, this figure plots the rolling annualized fee a mean-variance investor, with an investment horizon of one or five years, is willing to pay to switch from the 60/40 portfolio to the risk targeting strategies. Panel A shows the rolling annualized percentage fee for an investor with a risk aversion of $\gamma = 10$ and an investment horizon of one year, whereas Panel B shows the rolling economic value for an investor with an investment horizon of five years. Figure I demonstrates that a mean-variance investor with an investment horizon of one year is almost always willing to pay a positive fee to have access to risk targeting. This holds especially for the strategies that target a constant level of downside risk and the switching strategy. In line with our earlier findings, the utility gains in the crises are substantially higher than the utility losses in the low risk periods. Hence, investors are willing to pay very high fees to avoid crash periods, whereas their utility loss of lower returns in uptrending markets is significantly lower. In particular, during crash periods, the economic value of downside risk timing and the switching strategy is significantly higher than the economic value of volatility timing, whereas the economic value of volatility targeting, downside risk timing and the switching strategy is comparable in calm periods. Interestingly, during the crises periods, we find that the switching strategy outperforms all other strategies. Thus, even during crises, switching between volatility and CVaR targeting outperforms downside risk targeting. Results in Panel B are similar to the results in Panel A, but the differences between volatility targeting, downside risk targeting and the switching strategy become even larger. Thus, for investors with longer investment horizons, downside risk targeting becomes far more important than volatility targeting. This again holds especially for the strategy that switches between volatility and CVaR targeting. In particular, we find that, on average, the switching strategy produces the highest economic value, followed by CVaR targeting. In contrast, volatility targeting exhibits the lowest (average) economic value. Results for the CRRA investor are again similar to the results of Figure I and are not shown here.

Figure II shows the rolling one and five year economic value for a loss-averse investor with parameters $b = 0.8$ and $l = 3$. Panel A shows results for an investor with an investment horizon of one year. The economic value of downside risk targeting for a loss-averse investor is always

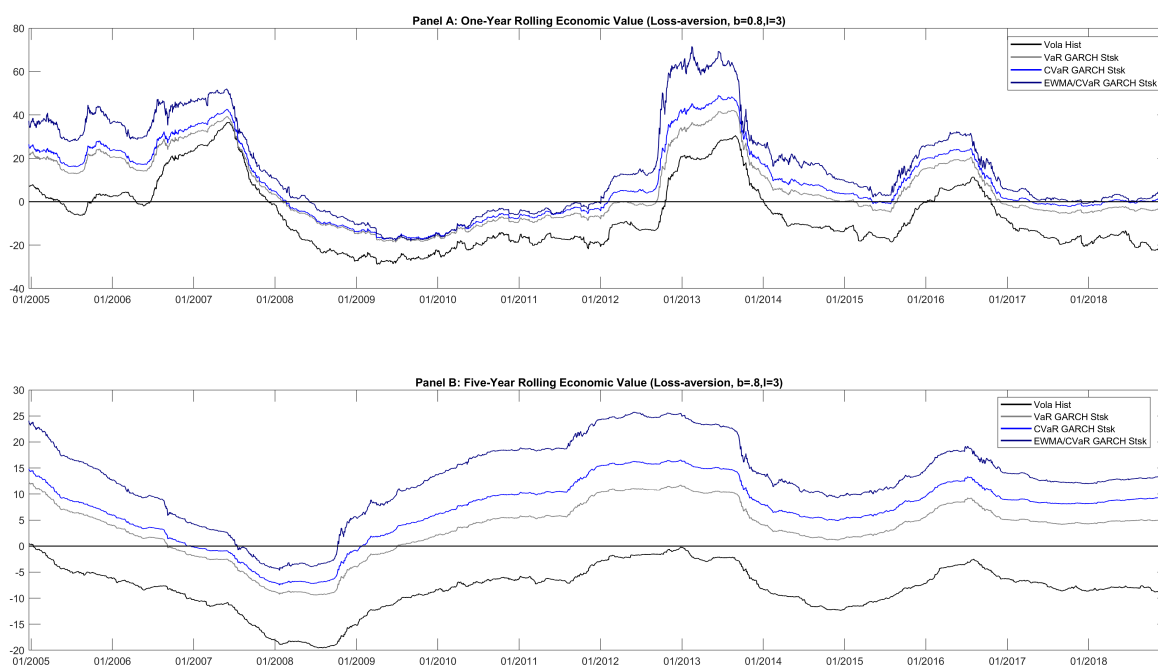


Figure II. One and Five Year Rolling Economic Value for a Loss-Averse Investor. This figure plots the one and five year rolling economic value measured by Δ_{LA} with respect to the 60/40 portfolio for a loss-averse investor with parameters $b = 0.8$ and $l = 3$. Panel A shows the economic value for an investor with an investment horizon of one year, whereas Panel B shows the economic value for an investor with an investment horizon of five years. The dates on the x axes correspond to the end dates of the one or five year investment horizon.

higher than the economic value of volatility targeting. As in Figure I, the economic value significantly increases for the periods that contain a crisis. Switching between volatility and CVaR targeting again outperforms all the remaining strategies, especially during crises. Panel B shows results for a loss-averse investor with an investment horizon of five years. Interestingly, the economic value of volatility targeting is always negative, i.e. a loss-averse investor with an investment horizon of five years is never willing to pay a positive fee to switch from the 60/40 portfolio to the HSD managed strategy. This is opposed to the finding of [Moreira and Muir \(2019\)](#) that even long-term investors should time short-term volatility.⁸⁴ In contrast, the economic value of downside risk targeting and the switching strategy is only negative for a short period. Thus, a loss-averse investor with an investment horizon of five years should almost always target downside risk or, even more advantageous, this investor should switch between volatility and CVaR targeting.

⁸⁴In contrast to our examination in Figure II, [Moreira and Muir \(2019\)](#) do not assess the economic value for loss-averse investors.

Finally, we compare the performance of the different risk targeting strategies by running time-series regressions of the returns from one strategy on the returns of the other strategies. Following [Moreira and Muir \(2017, Table I\)](#), we first run regressions of the risk-managed portfolios on the non-managed portfolio. Further, following [Daniel and Moskowitz \(2016, Table 8\)](#), we additionally regress the risk-managed strategies on the other risk-managed strategies. We follow the authors and rescale all strategies to the same level of volatility before running the regressions. Table XIII reports the annualized percentage alphas with the corresponding t -statistics.⁸⁵ We calculate the alphas for the switching strategies that use the same indicators as in Table X.

Table XIII. Spanning Tests: Portfolio Alphas

This table shows results of spanning test for the DAX and the risk targeting strategies. We run time-series regressions of each portfolio on the remaining strategies, where we use the DAX, the volatility targeting strategy, the CVaR targeting strategy and the strategy that switches between volatility and CVaR targeting. Panel A uses the TSMOM based indicator δ_t for the switching strategy, Panel B uses the MA based indicator and Panel C uses the volatility based indicator. We report annualized percentage alphas with corresponding t -statistics in parentheses. Bold numbers mark alphas that are significantly positive at the 10%, whereas red numbers mark alphas that are significantly negative at the 10%.

	DAX		Volatility		CVaR		Switching	
	α	t -stat	α	t -stat	α	t -stat	α	t -stat
Panel A: TSMOM Indicator								
DAX	-	-	-2.005	(-0.397)	-5.704	(-1.227)	-6.495	(-1.197)
Volatility	5.511	(0.990)	-	-	-3.421	(-1.446)	-5.331	(-2.699)
CVaR	9.399	(1.782)	4.285	(1.772)	-	-	-1.539	(-0.690)
Switch	12.063	(1.921)	6.187	(2.924)	2.253	(0.991)	-	-
Panel B: MA Indicator								
DAX	-	-	-2.005	(-0.397)	-5.704	(-1.227)	-6.139	(-1.134)
Volatility	5.511	(0.990)	-	-	-3.421	(-1.446)	-4.900	(-2.407)
CVaR	9.399	(1.782)	4.285	(1.772)	-	-	-1.090	(-0.479)
Switch	11.545	(1.860)	5.710	(2.646)	1.796	(0.780)	-	-
Panel C: Volatility Indicator								
DAX	-	-	-2.005	(-0.397)	-5.704	(-1.227)	-4.298	(-0.586)
Volatility	5.511	(0.990)	-	-	-3.421	(-1.446)	-3.629	(-0.713)
CVaR	9.399	(1.782)	4.285	(1.772)	-	-	-0.750	(-0.188)
Switch	13.941	(1.578)	8.033	(1.383)	3.464	(0.746)	-	-

Results in Table XIII show that the DAX always has a negative alpha when it is controlled for the performance of the risk targeting strategies. In contrast, all risk targeting strategies produce

⁸⁵[Cederburg et al. \(2020\)](#) state that the approach of [Moreira and Muir \(2017\)](#) has several disadvantages. Moreover, [Boguth et al. \(2011\)](#) and [Cederburg and O'Doherty \(2016\)](#) state that unconditional alphas are not appropriate to assess the performance of strategies that time volatility. Furthermore, [Schneider et al. \(2020\)](#) show that the unconditional alpha does not incorporate skewness preferences of investors. However, these regression results can be seen as a further robustness check of our economic value based findings. The economic value approach corrects for several disadvantage of the unconditional alpha and is a more powerful tool to compare the performance of dynamic trading strategies.

positive and economically large alphas when it is controlled for the performance of the DAX. In this case, the volatility targeting strategy produces an economically high alpha of 5.511%, which is in line with [Moreira and Muir \(2017, Table I\)](#) who find an alpha of 4.86% for the US market. However, the alpha is not statistically significant with a t -statistic of 0.99.⁸⁶ In contrast, the performance of CVaR targeting and the switching strategies cannot be explained by the DAX with an alpha of 9.399% for the CVaR targeting strategy and alphas between 11.545% and 13.941% for the switching strategies. These alphas are statistically significant at the 10% level for the TSMOM and MA based switching strategies with t -statistics of 1.921 and 1.860. Interestingly, among the switching strategies, the switching strategy using the volatility based indicator δ_t produces the highest alpha but the lowest t -statistic. This can be explained by the lower correlation of this strategy with the DAX. When we control for the volatility targeting strategy, the CVaR targeting and switching strategies produce economically high and mostly significant alphas. For example, with an alpha of 4.285% and a t -statistic of 1.772, CVaR targeting cannot be explained by volatility targeting. In contrast, when we control for the performance of the CVaR targeting strategy, the alpha of the volatility targeting strategy becomes negative. Furthermore, when we control for the switching approach, the volatility targeting strategy's alpha is even significantly negative in most cases and economically high in magnitude with values between -3.629% and -5.331%. Thus, not switching away from volatility targeting in periods when a negative return is expected, produces a significantly inferior performance. Similarly, the CVaR targeting strategy's alpha becomes negative, once we control for the switching approach.

In total, when we control for the switching strategy, all other strategies produce negative alphas. In contrast, the switching strategy's performance cannot be explained by volatility targeting, i.e. the switching approach produces economically high and statistically significant alphas in this case. The alphas of the switching strategies with respect to the CVaR targeting strategy are also positive, but not statistically significant. Thus, switching away from CVaR targeting in periods when the market is expected to be in an uptrend further improves the strategy's performance, but a huge part of the good performance of the switching strategy can be explained by CVaR targeting. Altogether, results in Table XIII confirm our earlier findings that risk targeting

⁸⁶We show in Appendix D.6 that the alphas are highly significant when a longer data set is used.

outperforms the non-managed portfolio. Furthermore, strategies that account for the DAX's tail risk outperform the volatility targeting strategy. However, the strategy that switches between volatility and CVaR targeting performs the best and produces economically high alphas with respect to the remaining strategies.

6 Conclusion

This paper studies dynamic trading strategies that target a predefined level of risk measured by volatility, Value at Risk (VaR) or Conditional Value at Risk (CVaR). We derive weights for these trading strategies and present several methods to estimate volatility, VaR and CVaR. Based on a data set for the German stock market, we find that risk targeting offers an enhanced risk-return profile, better drawdown protection and significant utility gains compared to a buy-and-hold equity investment and a static portfolio consisting of equities and bonds. Most convincing results are found for the strategies that target a constant level of portfolio CVaR over time. In particular, we find that mean-variance investors, CRRA investors and loss-averse investors should time downside risk, measured by CVaR, instead of volatility. This result especially holds for highly risk-averse or loss-averse investors and during crises. In particular, a loss-averse investor is not willing to pay a positive fee for volatility targeting, but the same investor would pay extremely high fees to have access to CVaR targeting. Generally, we find that risk should be managed by conditional risk models instead of unconditional models, as done by [Barroso and Santa-Clara \(2015\)](#), [Barroso and Maio \(2017\)](#) and [Moreira and Muir \(2017\)](#). This is in line with the result of [Bollerslev et al. \(2018\)](#) that a higher forecasting accuracy, and hence a more constant portfolio risk of risk targeting, typically coincides with higher performance benefits compared to static and less accurate forecasting models.

The risk-return profile and utility gains of risk targeting can further be improved by switching between volatility and CVaR targeting, where CVaR targeting is only used when a negative market return is expected. Based on three different crash indicators, we show that these switching strategies produce higher returns with lower risk compared to the volatility targeting strategies. Further, the mean-variance, CRRA and loss-averse investors are willing to pay high

fees to have access to these switching strategies. When the volatility targeting and switching strategies are simultaneously compared to the utility of a static portfolio allocation, utility gains of the switching strategies are statistically significant, whereas the utility gains of volatility targeting are insignificant. In particular, the returns of the switching strategy cannot be explained by volatility targeting, whereas volatility targeting becomes unprofitable, once we control for CVaR targeting or the switching strategies. Furthermore, the superiority of both strategies, the CVaR managed and switching strategies, holds in the long run and during the recent corona crisis.

Appendices

A Advantages of Volatility Targeting

This section summarizes several reasons why investors should target a constant level of volatility. See also [Dreyer and Hubrich \(2019\)](#) and references therein for further advantages of volatility targeting. Most of the advantages that will be summarized here also hold for the target VaR and target CVaR strategies presented in Section 3.

First, using the weight of Equation (3) implies that the weight of the risky asset is decreased in times of high volatility and increased in low volatile times. Since volatility is often associated with risk and investors are typically risk-averse ([Scott and Horvath, 1980](#)), targeting a constant level of volatility fits well to these investors' preferences. By choosing an adequate volatility target σ^{target} , investors can choose an investment strategy that fits well to their preferences and risk aversion ([Bollerslev et al., 2018](#)). [Zakamulin \(2015\)](#) and [Moreira and Muir \(2017\)](#) show that mean-variance investors should, under some assumptions, optimally choose the weight of the risky asset as $w_t = (\sigma^{\text{target}}/\sigma_t)^2$ (see also [Dopfel and Ramkumar \(2013\)](#)).⁸⁷ Based on this result, [Kirby and Ostdiek \(2012\)](#) suggest that investors should additionally decrease the sensitivity of w_t to volatility changes, which leads to the weight given in Equation (3).

⁸⁷The mean-variance framework is only suitable for elliptical distributions. Since asset returns usually do not follow an elliptical distribution, this weighting is not optimal for realistic return distributions ([Szegö, 2002](#), p.1254). We will revisit this issue in Section 3, where we present a similar weighting scheme based on risk measures that account for non-normalities in the asset return distribution.

Decreasing the sensitivity of w_t to volatility changes has the advantage that transaction costs are lowered and this strategy is more profitable in practice.⁸⁸

Second, especially during bear markets, which are associated with increases in volatility and correlations, investors seek for risk reduction methods (Ang and Bekaert, 2002).⁸⁹ The last financial crises were all accompanied by higher than normal volatilities (Liu et al., 2003, Moreira and Muir, 2017). In particular, times of an extremely high volatility typically coincide with times of a downward moving market (Campbell and Hentschel, 1992, French et al., 1987). Similarly, Moreira and Muir (2017) find that the probability of a recession is higher in times of high a market volatility. This is confirmed by Muir (2017) who shows that asset prices decline and stock market volatility increases in financial crises and recessions, but these effects reverse subsequently. Thus, times of a significantly higher volatility coincide with declining asset prices and these times should be avoided by investors. In contrast, the times following a crisis, which are marked by a declining volatility, offer an appealing risk-return profile for investors (Dopfel and Ramkumar, 2013, Moreira and Muir, 2019). Further, since volatilities and correlations between different equity markets increase simultaneously during bear markets, drawdowns in crises cannot simply be managed by diversification (Ang and Bekaert, 2002, Ang and Chen, 2002, Butler and Joaquin, 2002, Karolyi and Stulz, 1996, Longin and Solnik, 2001, Patton, 2004). For example, Chabi-Yo et al. (2018) find that extremely negative returns of stocks are more related than extremely positive returns, i.e. stocks tend to crash simultaneously. In particular, the authors show that the correlation of extremely negative returns among different stocks

⁸⁸Since volatility can only be estimated with an estimation error, Kirby and Ost diek (2012) suggest to scale the weight $w_t = (\sigma^{\text{target}}/\sigma_t)^2$ by a parameter η , called tuning parameter, that determines how aggressively the weight w_t reacts to changes in σ_t . This approach lowers the portfolio's turnover and, as a consequence, reduces transaction costs. By choosing $\eta = 0.5$, we obtain the weight of the target volatility strategy (see also Zakamulin (2015, p. 91)). Moreira and Muir (2017) compare the weight of the volatility and variance managed strategies. The authors find similar results, but less extreme weights and lower transaction costs for the volatility managed strategy. Zakamulin (2015, Exhibit 2) also find better results for the volatility managed strategy compared to the variance managed strategy.

⁸⁹Liu et al. (2003) find that most events with extremely negative returns are accompanied with high increases in volatility. Guidolin and Timmermann (2008) find a bear regime with low returns, negative alphas, high volatility and highly correlated assets and a bull regime with higher returns, positive alphas, lower volatility and less correlated returns (see also Wang et al. (2012, p. 27) and Hocquard et al. (2013)). Similarly, Ang and Bekaert (2002, p. 1139) find "a normal regime with low correlations, low volatilities, and a bear regime with higher correlations, higher volatilities, and lower conditional means." The bear states occurred during financial crises and/or global recessions that were accompanied with high market volatilities, indicating that periods of market distress are associated with high volatilities and low returns (Muir, 2017).

increases in crash periods ([Chabi-Yo et al., 2018](#), Figure 2). Similarly, [Bollerslev et al. \(2018\)](#) find co-movements, spillover effects and simultaneous spikes of volatilities between equities, bonds, commodities and currencies. Thus, risk characteristics between assets and asset classes are quite similar, especially in crash periods. Furthermore, [Jondeau and Rockinger \(2003\)](#) find that also higher moments, like (negative) skewness and kurtosis, increase simultaneously between markets during bear regimes. This indicates that the probability of an occurrence of large (negative) returns cannot be reduced by simply combining several risky assets. As stated above, these crash periods typically coincide with periods of an increased volatility. As a consequence, during high risk periods, the portfolio should be managed by simultaneously decreasing the exposure to a portfolio of risky assets and increasing the exposure to the riskless asset, as done by the target volatility strategy.⁹⁰ This also solves the problem identified by [Ang and Chen \(2002\)](#) and [Longin and Solnik \(2001\)](#) that investors incorrectly assess the benefits of diversification, and thus typically hold too much equities in bear markets, whereas they are underinvested in bull markets. However, by managing volatility, an investor is not protected against unpredictable tail events, marked by periods with extreme jumps in asset prices.⁹¹

Third, an often proclaimed justification of volatility targeting is the relation between volatility and future return. Although classical finance models, like the CAPM, indicate that higher risk should be compensated by higher expected returns (see [Merton \(1980\)](#) for example), many empirical studies find a negative relation between volatility and returns, i.e. a higher volatility coincides with lower or negative future returns ([Glosten et al., 1993](#)).⁹² A possible explanation

⁹⁰This result is also found by [Ang and Bekaert \(2002\)](#) who examine the optimal portfolio allocation when assets' correlations increase in crash regimes. By incorporating a risk-free asset, they find that, in the normal regime, the risky asset should be leveraged by being short in the risk-free asset, whereas money should be shifted to the risk-free asset in the bear regime. Furthermore, the authors find significant drawbacks of ignoring information about the regime, once the possibility of shifting money to the risk-free asset is introduced (see also [Patton \(2004\)](#)).

⁹¹See [Liu et al. \(2003\)](#) for a study on how jump risk, in both equity prices and volatility, affects the dynamic asset allocation between a risky and a riskless asset. In order to face jump risk, investors should avoid leveraged positions in the risky asset. Hence, an equity cap of 100% or a low volatility target should be used (see also [Das and Uppal \(2004\)](#)). Alternatively, investors should manage extreme losses, measured by downside risk, instead of return deviations. [Jarrow and Zhao \(2006\)](#) show that managing volatility differs from managing downside risk when asset returns exhibit jump risk.

⁹²A similar observation has also been found in cross-sectional analyses. See, for example, [Frazzini and Pedersen \(2014\)](#) who show that buying low beta assets and selling high beta assets produces high returns. Similarly, [Ang et al. \(2006b\)](#) and [Ang et al. \(2009\)](#) show that assets with a high past sensitivity to volatility changes, high idiosyncratic volatility or high total volatility have significantly lower returns than assets with a low past sensitivity to volatility changes, low idiosyncratic volatility or low total volatility, respectively. [Moreira and Muir \(2017, Sec. II.D\)](#) show that this effect is different to the time-series volatility effect examined here. [Haugen and Heins](#)

for the negative volatility-return relation is the *volatility feedback* effect, which is sometimes also called time-varying risk premium effect and is opposed to the well-known leverage effect (see [Glosten et al. \(1993, p. 1786\)](#) for an explanation of the leverage effect).⁹³ Based on this observation, an increase in volatility induces an immediate stock price decline. In other words, if tomorrow's volatility σ_{t+1} is expected to be higher than today's volatility σ_t , then tomorrow's weight w_{t+1} should be lower than today's weight w_t .⁹⁴ However, results in the academic literature on the relation between volatility and future returns are very mixed and a relation between volatility and returns is hard to confirm. [Lundblad \(2007\)](#) shows that very long data sets are needed, when the relation of volatility and future returns is examined. The author, using a data set ranging from 1836 to 2003, finds a positive relation between volatility and return. [Bali and Peng \(2006\)](#) using high-frequency-data based volatility measures find a significant and positive relation,⁹⁵ whereas [Bollerslev et al. \(2006\)](#), using similar volatility measures, find an insignificant or even negative relation. [Bollerslev and Zhou \(2006, p. 124-125\)](#) state that the risk-return relation in empirical investigations strongly depends on the volatility measure used in this investigation, which partly explains the inconsistent results in the academic literature (see also [Glosten et al. \(1993\)](#), [Ghysels et al. \(2005\)](#) and [Bollerslev et al. \(2013\)](#)). Furthermore, [Adrian](#)

[\(1975\)](#) find that the risk-return relation strongly depends on the sample period and whether the sample period is dominated by a bull or bear regime. Using a long data set, the authors find that "over the long run, stock portfolios with lesser variance in monthly returns have experienced greater average returns than their "riskier" counterparts" ([Haugen and Heins, 1975, p. 782](#)).

⁹³[Bollerslev et al. \(2006, p. 354\)](#) describe the volatility feedback effect as: "If volatility is priced, an anticipated increase in volatility would raise the required rate of return, in turn necessitating an immediate stock-price decline to allow for higher returns. Therefore, the causality underlying the volatility feedback effect runs from volatility to prices, as opposed to the leverage effect that hinges on the reverse causal relationship" (see also [Campbell and Hentschel \(1992\)](#), [Bekaert and Wu \(2000\)](#) and [Glosten et al. \(1993\)](#) for an explanation of the leverage and volatility feedback effect). For additional studies on the relation between volatility and return, see also [French et al. \(1987\)](#), [Bali and Peng \(2006\)](#), [Bollerslev et al. \(2006\)](#), [Bollerslev and Zhou \(2006\)](#), [Ghysels et al. \(2005\)](#), [Lundblad \(2007\)](#), [Bollerslev et al. \(1992\)](#) among others. See [Muir \(2017\)](#) for an examination why risk premiums or expected returns vary over time, rise modestly in recessions and spike in financial crises. See [Glosten et al. \(1993\)](#) and [Bekaert and Wu \(2000\)](#) on how the leverage effect influences results on the volatility feedback effect. [Bekaert and Wu \(2000, p. 7\)](#) find that "the leverage effects reinforces the volatility feedback effect" and both effects interact. Both effects and their interaction are also nicely visualized in [Bekaert and Wu \(2000, Figure 1\)](#).

⁹⁴The volatility feedback effect is reflected by the construction of the target volatility weighting given in Equation (3). More formally, from $\sigma_{t+1} > \sigma_t$ it follows $w_{t+1} = \sigma^{\text{target}}/\sigma_{t+1} < \sigma^{\text{target}}/\sigma_t = w_t$, i.e. an increase in volatility induces a decrease in the weight of the risky asset. Due to this relation, [Harvey et al. \(2018\)](#) find that, under the leverage effect, volatility targeting induces momentum, i.e. negative returns induce higher future volatilities and lower future weights of the risky asset. The authors find that this observations explains a part of the increase of the Sharpe Ratio of the volatility targeting strategy.

⁹⁵Similarly, [Ghysels et al. \(2005\)](#) using daily data to measure monthly volatility by advanced volatility measures find a positive and significant relation.

and Rosenberg (2008) show that the risk-return relation strongly depends on the examined time frequency of volatility. In line with the volatility feedback effect, the authors find a negative volatility-return relation for short-term volatility, but a positive relation for long-term volatility.

To summarize results in the academic literature, the relation between volatility and future returns is hard to identify and results in the literature are too mixed to draw a distinct conclusion (see also Harvey and Siddique (1999) and references therein). Backus and Gregory (1993) theoretically confirm this observation. Similarly, Glosten et al. (1993) also argue that a positive and a negative relation would be consistent with theory. However, in order to achieve an enhanced risk-return profile by volatility targeting, a negative relation between volatility and future return is not needed. Moreira and Muir (2017) show that the relation between volatility and future *risk-adjusted* returns should be of main interest for volatility timing strategies instead of the risk-return relation (see also Dopfel and Ramkumar (2013)). Moreira and Muir (2017) theoretically show that an alpha of zero is obtained if movements of expected returns and volatility coincide. In particular, the authors find that the positive alpha of the volatility managed strategy is mainly driven by the negative relation between volatility and volatility-adjusted returns. In other words, volatility targeting produces positive alphas, since an increase in volatility is not compensated by an adequate increase in expected return. This is also empirically confirmed by the authors: although the authors cannot confirm a negative volatility-return relation, they find that volatility timing increases the Sharpe Ratio. The reason for the increasing Sharpe Ratio is that “changes in volatility are not offset by proportional changes in expected returns” (Moreira and Muir, 2017, p. 1611). In total, a high volatility in $t - 1$ is related to a low Sharpe Ratio in t . Hence, high volatility periods exhibit an unattractive risk-return profile and should be avoided by investors. Similarly, Dachraoui (2018, Eq. (2)) shows that the Sharpe Ratio of the target volatility strategy is given by the Sharpe Ratio of the risky asset and the correlation between the volatility and the risk-adjusted return of the risky asset. In particular, if volatility and risk-adjusted returns of the risky asset are negatively correlated, the Sharpe Ratio of the target volatility strategy is higher than the Sharpe Ratio of the risky asset. A sufficient condition for this negative correlation is that volatility and return are negatively correlated or uncorrelated.

This is confirmed by [Barroso and Maio \(2017\)](#) who find that volatility targeting works well since risk and future returns are nearly uncorrelated and risk is highly forecastable due to its persistent nature. Similarly, [Harvey et al. \(2018\)](#) find no clear pattern between the volatility of day $t - 1$ and the return of day t . However, due to the persistence of volatility, they find that a high volatility in $t - 1$ is related to a high volatility in t , which translates into a low Sharpe Ratio in t . Hence, an investor should be higher invested in the risky asset if the risky asset's volatility is low and vice versa, i.e. the investor should time volatility. Interestingly, even when volatility and return are weakly positively correlated, volatility targeting can still be advantageous. For example, [Moreira and Muir \(2019\)](#) find that an increase of volatility coincides with higher expected returns, but that the increase in expected return is much more persistent than the increase in volatility. Thus, investors should reduce the weight of the risky asset if short-term volatility increases, but the investors should then subsequently increase the exposure to the risky asset when volatility begins to decline.

The finding that investors should time (short-term) volatility does not fundamentally contradict the assumption that higher risk is compensated by higher expected returns. In the long run, assets with higher volatility typically earn higher risk premiums, but risk premiums typically fluctuate over time ([Lempérière et al., 2017](#), [Muir, 2017](#)). Thus, long-term investors benefit from investing in riskier assets like equities, whereas increases of short-term volatility of the same asset are related to low or negative returns ([Adrian and Rosenberg, 2008](#)). Hence, in the long run, stocks typically have higher long-term returns than bonds and investors with a long investment horizon should participate in the stock market. Similarly, [Blitz et al. \(2019\)](#) state that “[t]he relation between risk and return only seems to be positive across entire asset classes, since stocks have higher returns than bonds, and corporate bond returns are higher than government bond returns, in the long run”. However, most investors fail to capture the long-term potential of stocks, due to too short evaluation periods and the higher volatility of stocks ([Benartzi and Thaler, 1995](#)). [Benartzi and Thaler \(1995\)](#) show that the evaluation period of long-term investors is typically much shorter than their investment horizon. In other words, investors with an investment horizon of years act like investors with a horizon of several months.

This hinders these investor to fully participate from the long-term performance potential of stocks, since investors with a short evaluation period are more sensitive to changes in market volatility ([Moreira and Muir, 2019](#)). For these long-term investors, timing short-term volatility can also be beneficial to capture the long-term potential of stocks and simultaneously manage short-term risk. [Ang and Bekaert \(2002\)](#) show that even for investors with longer horizons it is possible to act myopically, as done by risk targeting, instead of solving complex long-term portfolio problems. Consequently, risk targeting is an easy method to make the long-term potential of equity investments available for investors with short evaluation periods. In particular, even highly risk-averse investors can participate from the huge long-term return potential of risky assets, where investors can choose the risk level they are willing to accept. Further, besides making stock market investments available for all investors, volatility targeting can even enhance the risk-adjusted performance for long-term investors by dynamically timing the risky asset's short-term risk ([Moreira and Muir, 2019](#)).

Fourth, many studies have shown that volatility timing can add substantial economic value in terms of higher risk-adjusted returns and high utility gains. [Fleming et al. \(2001\)](#), [Fleming et al. \(2003\)](#), [Han \(2005\)](#), [Kirby and Ostdiek \(2012\)](#) and [Taylor \(2014\)](#) examine the economic value of volatility timing strategies, i.e. strategies that rely on estimates of the covariance matrix solely, in a multivariate setting and the authors find that these strategies are superior to non-managed portfolios, even after transaction costs.⁹⁶ Although these studies use multivariate data sets, these studies demonstrate that volatility timing is typically related to higher risk-adjusted returns and high utility gains for mean-variance investors. Similarly, in a univariate setting, [Marquering and Verbeek \(2004\)](#) find substantial increases in the Sharpe Ratio and utility if volatility timing is added to return timing. [Moreira and Muir \(2019\)](#) confirm this finding for investors with a long investment-horizon. [Moreira and Muir \(2017\)](#) and [Bollerslev et al. \(2018\)](#), using a similar framework as in our paper, also demonstrate the vast utility gains and Sharpe Ratio increases of volatility targeting. Furthermore, [Busse \(1999\)](#) examines volatility timing used by mutual

⁹⁶These studies examine the economic value of volatility timing for long-only portfolios. Volatility timing can also be applied to long-short strategies. For example, [Moskowitz et al. \(2012\)](#) and [Kim et al. \(2016\)](#) use volatility timing to manage the risk of the Time Series Momentum (TSMOM) strategy. Similarly, [Asness et al. \(2013\)](#) and [Goyal and Jegadeesh \(2017\)](#) use volatility timing to weight the assets of the cross-sectional momentum strategy.

funds and finds higher Sharpe Ratios for funds using volatility timing. Additionally, [Daniel and Moskowitz \(2016\)](#), [Cederburg et al. \(2020\)](#), [Barroso and Santa-Clara \(2015\)](#), [Moreira and Muir \(2017\)](#) and [Barroso and Maio \(2017\)](#) demonstrate the vast potential of volatility timing overlayed on several portfolio strategies, especially in terms of a drawdown reduction and an improvement of risk-adjusted returns. This especially holds for portfolios with a high left tail risk, such as momentum or betting against beta. Generally, volatility targeting works best for assets that deviate strongly from normality ([Perchet et al., 2016](#), Exhibit 11). This is shown by the authors in a Monte Carlo Study and the authors find that the gains of volatility targeting are higher for assets that strongly deviate from normally distributed returns, exhibit high levels of volatility clustering or have fatter tails. Similarly, [Harvey et al. \(2018\)](#) find that volatility targeting works well for risky assets, like equities or portfolios that contain equities, but not for assets with lower risk, like bonds. Applying volatility targeting to equities usually produces “benchmark-comparable levels of return with lower risk” ([Benson et al., 2014](#), p. 89).

Fifth, the target volatility strategy focuses on the risk, measured by volatility of the risky asset solely and ignores the information about future returns ([Bollerslev et al., 2018](#)). Furthermore, a forecast of the whole return distribution is not needed, which is a tenuous task ([Aït-Sahalia and Brandt, 2001](#)). This is appealing, since future volatility can be estimated much more precisely than the whole distribution or future returns, which minimizes the estimation risk of this approach ([Merton, 1980](#)). [Kirby and Ostdiek \(2012\)](#) show that portfolio allocations that rely on estimates of returns and volatilities exhibit very high estimation risk, whereas estimation risk is only small for volatility based allocations.⁹⁷ For that reason, portfolio strategies that are based on a forecast of the asset’s return typically perform bad in real-time. For example, [Moreira and Muir \(2017\)](#) find higher utility gains for volatility timing strategies than for expected return timing strategies. [Marquering and Verbeek \(2004\)](#) examine both, return and volatility timing, and find that timing returns and volatility is superior to strategies that only time returns. Similarly, [Moreira and Muir \(2019\)](#) find that return timing strategies are highly influenced by estimation risk, whereas volatility timing strategies are quite robust against estimation risk. As

⁹⁷Due to the high estimation risk, especially for mean returns, portfolio allocations under estimation risk are frequently examined in the financial literature ([DeMiguel et al., 2009](#), [Garlappi et al., 2006](#), [Kan and Zhou, 2007](#), [Tu and Zhou, 2011](#)).

a consequence, volatility timing outperforms return timing in practice.

Sixth, investors are typically crash-averse and dislike periods of highly negative returns (Bollerslev and Todorov, 2011, Chabi-Yo et al., 2018, van Oordt and Zhou, 2016). Volatility timing has proven to be a good and easy method for drawdown protection, which makes this approach appealing for investors who typically dislike huge drawdowns (Barroso and Maio, 2017, Barroso and Santa-Clara, 2015, Benson et al., 2014, Moreira and Muir, 2017, Perchet et al., 2016). Generally, volatility targeting is a simple but effective tail risk hedging instrument and significantly reduces left tail risk and fits well to investors with higher order preferences (Dreyer and Hubrich, 2019). Similarly, Harvey et al. (2018) find that volatility targeting successfully reduces the likelihood of extremely negative returns. Thus, volatility targeting also fits well to the loss-aversion of most investors (Aït-Sahalia and Brandt, 2001, Benartzi and Thaler, 1995). In particular, risk targeting is an easy way to manage portfolio risk dynamically. Cuoco et al. (2008) highlight the importance of managing portfolio risk dynamically, i.e. reevaluating portfolio weights frequently.

Seventh, risk-averse investors typically want to hedge against changes in volatility (see Ang et al. (2006b), Adrian and Rosenberg (2008) and references therein).⁹⁸ Adrian and Rosenberg (2008) find that investors are willing to pay for methods that protect them from changes in volatility. Similarly, in a cross-sectional setting, Baltussen et al. (2018) find that assets with a high volatility of volatility (vol-of-vol) underperform assets with a more constant volatility. Thus, assets with a more constant volatility produce higher returns than assets that exhibit higher volatility changes. Furthermore, assets with higher volatility changes also exhibit higher downside risk. Similarly, a highly fluctuating portfolio volatility is a driver of the portfolio's tail risk and a stabilization of the portfolio volatility is rewarded by a lower crash risk (Dreyer and Hubrich, 2019). Thus, investors strongly benefit from a stabilization of portfolio volatility. The investors' demand to hedge against these changes in volatility has led to the introduction of many new financial instruments, like variance swaps (Bollerslev and Todorov, 2011, Footnote 11). Volatility targeting is an easy way to hedge against this volatility risk without using

⁹⁸Generally, the variance risk premium, which measures the "compensation for the risk associated with temporal changes in the variation of the price level" is examined by Bollerslev and Todorov (2011) and Bollerslev et al. (2015).

any financial derivatives.

Eighth, liabilities of institutional investors, like insurance companies or pension funds, are often less volatile than investments in risky assets. Targeting a constant level of volatility can help to match the volatility of the investments with the volatility of the liabilities. For example, [Banerjee et al. \(2016, p. 1\)](#) write that “a risk control strategy may provide a smoother path of asset returns and could more closely align the performance of the institution’s assets to the characteristics of its liabilities.”

Ninth, a skill of a portfolio manager can typically be separated into his ability to time the market and the ability to pick the right stocks (see [Agarwal and Naik \(2004\)](#) and references therein). Risk targeting can be used separately as market timing tool, which is independent of the asset selection process ([Zakamulin, 2015](#)). In practice, market timing and volatility timing are fundamentally related as documented by [Christoffersen and Diebold \(2006\)](#) and [Copeland and Copeland \(1999\)](#). The authors state that market timing strategies based on measures of volatility are frequently used by practitioners. Hence, portfolio managers can focus on picking the right assets without accounting for the current market environment, which is separately managed by a risk targeting strategy. Similarly, [Cederburg et al. \(2020\)](#), [Daniel and Moskowitz \(2016\)](#), [Barroso and Santa-Clara \(2015\)](#), [Barroso and Maio \(2017\)](#), [Rickenberg \(2020a\)](#) and [Moreira and Muir \(2017\)](#) use volatility targeting for several portfolio strategies, where the asset allocation is determined in the first step and these portfolios are then managed by volatility targeting. [Moreira and Muir \(2017, Sec. I.E\)](#), [Rickenberg \(2020a\)](#) and [Zakamulin \(2015\)](#) also examine volatility targeting applied to volatility weighted portfolios, where the portfolio allocation is chosen first based on the assets’ volatility.

B Portfolio Risk

B.1 Portfolio Value at Risk

In this section, we derive the Value at Risk (VaR) for the portfolio loss given in Equation (10). We denote the conditional cumulative distribution function (cdf) of the risky asset’s loss L_t , based on the information $\mathcal{F}_{t-1}$ available at time $t - 1$, by $F_{L_t|\mathcal{F}_{t-1}}$. Moreover, we assume that

$F_{L_t|\mathcal{F}_{t-1}}$ is continuous and strictly increasing and we denote the corresponding $(1 - \alpha)$ -quantile by $F_{L_t|\mathcal{F}_{t-1}}^{-1}(1 - \alpha)$. For a positive weight w_t , the day t VaR of the portfolio loss, denoted by $\text{VaR}_\alpha^{P,t}$, is given by

$$\begin{aligned}
& \mathbb{P}(L_t^P \leq \text{VaR}_\alpha^{P,t} \mid \mathcal{F}_{t-1}) = 1 - \alpha \\
& \Leftrightarrow \mathbb{P}(w_t \cdot L_t - (1 - w_t) \cdot R_t^f \leq \text{VaR}_\alpha^{P,t} \mid \mathcal{F}_{t-1}) = 1 - \alpha \\
& \Leftrightarrow \mathbb{P}(L_t \leq (\text{VaR}_\alpha^{P,t} + (1 - w_t) \cdot R_t^f)/w_t \mid \mathcal{F}_{t-1}) = 1 - \alpha \\
& \Leftrightarrow F_{L_t|\mathcal{F}_{t-1}}((\text{VaR}_\alpha^{P,t} + (1 - w_t) \cdot R_t^f)/w_t) = 1 - \alpha \\
& \Leftrightarrow (\text{VaR}_\alpha^{P,t} + (1 - w_t) \cdot R_t^f)/w_t = F_{L_t|\mathcal{F}_{t-1}}^{-1}(1 - \alpha) \\
& \Leftrightarrow \text{VaR}_\alpha^{P,t} = w_t \cdot F_{L_t|\mathcal{F}_{t-1}}^{-1}(1 - \alpha) - (1 - w_t) \cdot R_t^f.
\end{aligned} \tag{77}$$

Since the VaR of the risky asset, denoted by VaR_α^t , is given by the $(1 - \alpha)$ -quantile of the risky asset's (conditional) loss distribution, i.e. $\text{VaR}_\alpha^t = F_{L_t|\mathcal{F}_{t-1}}^{-1}(1 - \alpha)$, the VaR of the portfolio is given by

$$\text{VaR}_\alpha^{P,t} = w_t \cdot \text{VaR}_\alpha^t - (1 - w_t) \cdot R_t^f. \tag{78}$$

B.2 Portfolio Conditional Value at Risk

In this section, we derive the Conditional Value at Risk (CVaR) for the portfolio loss given in Equation (10). The CVaR of the portfolio loss L_t^P , denoted by $\text{CVaR}_\alpha^{P,t}$, is given by

$$\text{CVaR}_\alpha^{P,t} = \mathbb{E}(L_t^P \mid L_t^P \geq \text{VaR}_\alpha^{P,t}, \mathcal{F}_{t-1}) \tag{79}$$

$$= \mathbb{E}(w_t \cdot L_t - (1 - w_t) \cdot R_t^f \mid w_t \cdot L_t - (1 - w_t) \cdot R_t^f \geq \text{VaR}_\alpha^{P,t}, \mathcal{F}_{t-1}) \tag{80}$$

From Equation (78), i.e. $\text{VaR}_\alpha^{P,t} = w_t \cdot \text{VaR}_\alpha^t - (1 - w_t) \cdot R_t^f$, and since the weight w_t and the riskless return R_t^f are $\mathcal{F}_{t-1}$ -measurable, it follows

$$\begin{aligned}
\text{CVaR}_\alpha^{P,t} &= \mathbb{E}(w_t \cdot L_t - (1 - w_t) \cdot R_t^f \mid L_t \geq \text{VaR}_\alpha^t, \mathcal{F}_{t-1}) \\
&= w_t \cdot \mathbb{E}(L_t \mid L_t \geq \text{VaR}_\alpha^t, \mathcal{F}_{t-1}) - (1 - w_t) \cdot R_t^f \\
&= w_t \cdot \text{CVaR}_\alpha^t - (1 - w_t) \cdot R_t^f,
\end{aligned} \tag{81}$$

where $\text{CVaR}_\alpha^t := \mathbb{E}(L_t \mid L_t \geq \text{VaR}_\alpha^t, \mathcal{F}_{t-1})$ denotes the CVaR of the risky asset.

C Backtesting Target Risk Strategies

C.1 Backtesting Target VaR Strategies

By definition, the variable H_t^P is equal to one if $L_t^P - \text{VaR}_\alpha^{\text{target}} > 0$ and zero else. From Equation (10) it follows that the portfolio loss is given by

$$L_t^P = w_t \cdot L_t - (1 - w_t) \cdot R_t^f. \quad (82)$$

Moreover, given the weight w_t , the portfolio VaR equals the predefined VaR level $\text{VaR}_\alpha^{\text{target}}$, and hence from Equation (78) we obtain

$$\text{VaR}_\alpha^{\text{target}} = \text{VaR}_\alpha^{P,t} = w_t \cdot \text{VaR}_\alpha^t - (1 - w_t) \cdot R_t^f. \quad (83)$$

Consequently, we have

$$L_t^P - \text{VaR}_\alpha^{\text{target}} = w_t \cdot L_t - w_t \cdot \text{VaR}_\alpha^t = w_t \cdot (L_t - \text{VaR}_\alpha^t). \quad (84)$$

Since the weight w_t is strictly positive, it follows

$$L_t^P - \text{VaR}_\alpha^{\text{target}} > 0 \Leftrightarrow L_t - \text{VaR}_\alpha^t > 0. \quad (85)$$

Therefore, the variable H_t^P is equal to H_t .

C.2 Backtesting Target CVaR Strategies

Given the weight w_t , the target CVaR equals the portfolio CVaR, and hence from Equation (81) it follows

$$\text{CVaR}_\alpha^{\text{target}} = \text{CVaR}_\alpha^{t,P} = w_t \cdot \text{CVaR}_\alpha^t - (1 - w_t) \cdot R_t^f. \quad (86)$$

Therefore, the difference between the portfolio loss and the target CVaR is given by

$$L_t^P - \text{CVaR}_\alpha^{\text{target}} = w_t \cdot L_t - w_t \cdot \text{CVaR}_\alpha^t = w_t \cdot (L_t - \text{CVaR}_\alpha^t). \quad (87)$$

Moreover, from Equation (2) and since w_t and R_t^f are $\mathcal{F}_{t-1}$ -measurable, we obtain

$$\sqrt{\text{var}(R_t^P \mid \mathcal{F}_{t-1})} = \sqrt{\text{var}(w_t \cdot R_t \mid \mathcal{F}_{t-1})} = w_t \cdot \sqrt{\text{var}(R_t \mid \mathcal{F}_{t-1})} = w_t \cdot \sigma_t. \quad (88)$$

Consequently, from Equations (87), (88) and (37) it follows

$$\frac{L_t^P - \text{CVaR}_\alpha^{\text{target}}}{\sqrt{\text{var}(R_t^P \mid \mathcal{F}_{t-1})}} = \frac{w_t \cdot (L_t - \text{CVaR}_\alpha^t)}{w_t \cdot \sigma_t} = \frac{L_t - \text{CVaR}_\alpha^t}{\sigma_t} = L_t^* - \text{CVaR}_\alpha^{t,*}. \quad (89)$$

D Additional Results

This section shows additional results for the risk targeting strategies. In Section D.1, we demonstrate that tail risk targeting can be used as an alternative to absolute return strategies. In Section D.2, we show results for other significance levels α . Section D.3 examines the profitability of risk targeting when a rebalancing buffer is used. In Section D.4, we examine risk targeting for leverage constrained investors. Section D.5 shows results for the US and small caps, proxied by the S&P 500 and the SDAX. In Section D.6, we examine the profitability of risk targeting in the long run using almost 100 years of data. Finally, in Section D.7, we conduct an out-of-sample study and demonstrate the benefits of risk targeting during the recent corona crisis. Throughout this section, we only show results for the volatility and CVaR targeting strategies as well as the strategies that switch between the EWMA model and the CVaR targeting strategies based on the TSMOM indicator. Further, we only show results of the strategies' return, volatility, Sharpe Ratio, maximum drawdown and the economic value Δ_{MV} of a mean-variance investor with a moderate risk aversion of $\gamma = 5$. The economic value Δ_{MV} measures the annualized percentage fee an investor is willing to pay to switch from the 60/40 portfolio to a risk targeting strategy. Additionally, we use the [Jobson and Korkie \(1981\)](#) and [Diebold and Mariano \(1995\)](#) tests to test for the significance of the Sharpe Ratio and utility increases.⁹⁹

D.1 Risk Targeting as Absolute Return Strategy

This section demonstrates that downside risk targeting can be used as an alternative to absolute return and other hedge fund strategies as examined in [Fung and Hsieh \(1997\)](#) and [Agarwal and Naik \(2004\)](#). In Figure III, we plot a target VaR strategy with a VaR target of $\text{VaR}_\alpha^{\text{target}} = 0.5\%$ and a significance level of $\alpha = 0.4\%$. Hence, a daily return lower than -0.5% should occur only once a year. Days with a return lower than -0.5% are marked with a red cross. As can be seen from the figure, the strategy is successful in mitigating extremely negative returns

⁹⁹We use the HSD managed strategy as benchmark for both tests. By doing this, we can assess how different characteristics affect the profitability of risk targeting using volatility or CVaR as risk measure. Thus, the tests show if a risk targeting strategy outperforms the HSD based volatility targeting strategy. This is opposed to Tables IX and XII, where we used the 60/40 portfolio as benchmark for the [Diebold and Mariano \(1995\)](#) test. Our conclusions were similar to the case when the DM-test is calculated with respect to the 60/40 portfolio.

and produces positive or moderately negative returns with a very high probability. The target VaR strategy's long-term return is as high as the return of the 60/40 portfolio and only slightly lower than the return of the DAX. However, the target risk strategy also takes much less risk with significantly lower drawdowns. Hence, this strategy is appealing for highly risk-averse or loss-averse investors who want to benefit from the long-term potential of equity markets. We only show results for the VaR-EWMA-FHS strategy in Figure III since this strategy is easy to estimate and implement, and hence could be interesting for practitioners. Other risk targeting strategies produce similar or even superior performance charts.

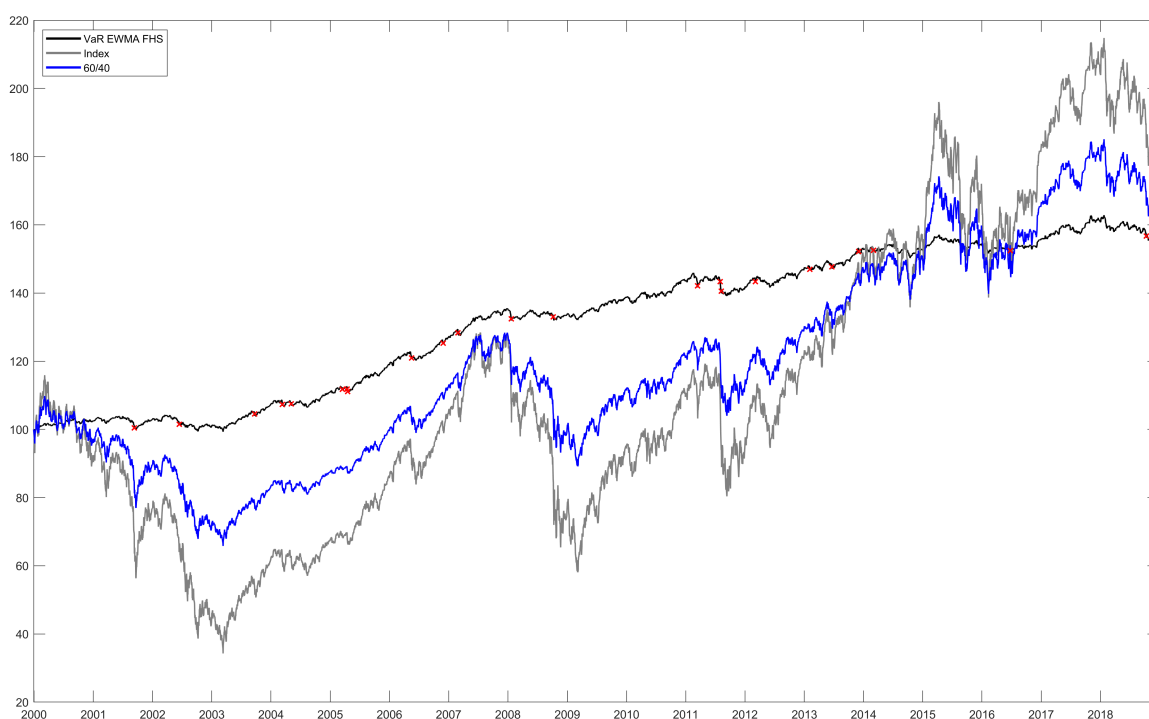


Figure III. Cumulative Return of VaR Targeting. This figure plots the cumulative return of the DAX, the 60/40 portfolio and a target VaR strategy. The target VaR strategy uses the EWMA volatility model combined with Filtered Historical Simulation (FHS), a VaR target of $\text{VaR}_\alpha^{\text{target}} = 0.5\%$ and a significance level of $\alpha = 0.4\%$. Days when the portfolio return is lower than -0.5% are marked with a red cross.

D.2 Tail Risk Targeting for Different Significance Levels

This section examines performance results of the tail risk targeting strategies for different significance levels α . In Table XIV, we show additional performance results of volatility and CVaR targeting for significance levels of 1%, 2.5% and 5%. The chosen significance levels

are frequently used in the literature on VaR and CVaR forecasting (see [Bali et al. \(2008\)](#) for example). The target CVaR level is recalculated to match the chosen volatility target by using Equation (46). In line with the results of Tables V and VIII, we find that downside risk targeting is superior to volatility targeting in terms of higher Sharpe Ratios, lower drawdowns and a higher economic value. Further, also in line with the previous results, we find that managing risk by conditional models outperforms the strategies based on HSD or Historical Simulation. This result is most pronounced for high significance levels. Generally, downside risk targeting becomes less attractive if a higher significance level α is chosen. This result is in line with [Ghysels et al. \(2016\)](#) and [Happersberger et al. \(2019\)](#). We find that tail risk targeting resembles volatility targeting when higher significance levels are chosen. This finding is quite intuitive, since a lower α means that extreme losses in the far left tail are managed. In contrast, using a higher α means that also moderate losses are taken into account. In total, results of Table XIV highlight that investors should manage extreme losses by using CVaR combined with a low value of α . A similar results has also been found by [Basak and Shapiro \(2001\)](#). However, even for a significance level of $\alpha = 5\%$, CVaR targeting is typically superior to volatility targeting. Interestingly, the strategies based on the skewed t distribution are very stable in terms of the Sharpe Ratio for different levels of α . However, the drawdown and the economic value indicate that low levels of α are also superior for these models.

In Table XV, we show results for the strategies that switch between volatility and CVaR targeting for the same significance levels. Throughout the section, we use the EWMA volatility model and the TSMOM based indicator δ_t . In line with our main results, we find that the strategies that switch between volatility and CVaR targeting outperform the non-managed strategy or the strategies that use either volatility or CVaR targeting. However, in line with Table XIV, we find better results for the strategies that use lower significance levels α . This is again quite intuitive, since the switching approach typically uses CVaR targeting only in adverse market periods. These crash periods are best managed by tail risk measures that use low significance levels and manage *extreme* losses instead of moderate losses. Moreover, we again find that conditional models produce the most convincing results. All switching strategies based on con-

Table XIV. Performance Results for Different Significance Levels

This table shows additional performance results for the volatility and CVaR targeting strategies using the DAX and three different significance levels α . Panel A shows results for $\alpha = 1\%$, Panel B shows results for $\alpha = 2.5\%$ and Panel C shows results for $\alpha = 5\%$. Return and Volatility denote the annualized return and volatility in percent. SR denotes the annualized Sharpe Ratio. z_{JK} denotes the test statistic of the [Jobson and Korkie \(1981\)](#) test. MDD and Δ MDD denote the maximum drawdown and the reduction of the maximum drawdown in relation to the drawdown of the risky asset. $\Delta_{MV}^{\gamma=5}$ denotes the economic value for a mean-variance investor with risk aversion $\gamma = 5$. DM-test denotes the [Diebold and Mariano \(1995\)](#) test that tests if a strategy produces a significantly higher utility for a mean-variance investor with risk aversion $\gamma = 5$, where we use the HSD strategy as benchmark. Return, Volatility, MDD, Δ MDD and $\Delta_{MV}^{\gamma=5}$ are given in percent. Values of z_{JK} and DM-test that are higher than 1.645 are given in bold.

Panel A: Significance Level of $\alpha = 1\%$								
Model	Return	Volatility	SR	z_{JK}	MDD	Δ MDD	$\Delta_{MV}^{\gamma=5}$	DM-test
Vola Hist	3.115	12.843	0.102	-	41.228	41.458	0.280	-
Vola EWMA	3.336	12.499	0.122	0.965	40.522	42.459	0.671	1.720
Vola GARCH	3.191	11.965	0.116	0.389	39.233	44.290	0.792	1.123
CVaR Hist	2.316	9.550	0.055	-0.462	30.298	56.978	0.982	0.493
CVaR EWMA FHS	3.193	10.541	0.132	1.006	35.754	49.231	1.443	1.820
CVaR EWMA EVT	3.230	10.232	0.139	1.311	34.685	50.749	1.609	1.891
CVaR EWMA Stsk	3.408	10.529	0.152	1.612	34.268	51.340	1.659	2.110
CVaR GARCH FHS	3.174	10.875	0.126	0.616	37.396	46.898	1.279	1.509
CVaR GARCH EVT	3.191	10.516	0.132	0.789	36.157	48.659	1.452	1.624
CVaR GARCH Stsk	3.273	10.100	0.145	1.015	34.017	51.697	1.707	1.754
Panel B: Significance Level of $\alpha = 2.5\%$								
Model	Return	Volatility	SR	z_{JK}	MDD	Δ MDD	$\Delta_{MV}^{\gamma=5}$	DM-test
Vola Hist	3.115	12.843	0.102	-	41.228	41.458	0.280	-
Vola EWMA	3.336	12.499	0.122	0.965	40.522	42.459	0.671	1.720
Vola GARCH	3.191	11.965	0.116	0.389	39.233	44.290	0.792	1.123
CVaR Hist	2.061	10.223	0.027	-0.714	33.217	52.833	0.462	0.130
CVaR EWMA FHS	3.227	10.975	0.129	1.062	37.056	47.382	1.287	1.927
CVaR EWMA EVT	3.230	10.694	0.133	1.239	36.002	48.878	1.413	1.955
CVaR EWMA Stsk	3.438	11.075	0.147	1.671	35.830	49.123	1.449	2.261
CVaR GARCH FHS	3.137	11.082	0.120	0.488	37.919	46.156	1.151	1.437
CVaR GARCH EVT	3.114	10.830	0.121	0.519	37.086	47.338	1.240	1.490
CVaR GARCH Stsk	3.354	10.614	0.146	1.079	35.353	49.800	1.570	1.840
Panel C: Significance Level of $\alpha = 5\%$								
Model	Return	Volatility	SR	z_{JK}	MDD	Δ MDD	$\Delta_{MV}^{\gamma=5}$	DM-test
Vola Hist	3.115	12.843	0.102	-	41.228	41.458	0.280	-
Vola EWMA	3.336	12.499	0.122	0.965	40.522	42.459	0.671	1.720
Vola GARCH	3.191	11.965	0.116	0.389	39.233	44.290	0.792	1.123
CVaR Hist	2.027	10.958	0.022	-0.745	35.590	49.464	0.116	-0.119
CVaR EWMA FHS	3.217	11.194	0.126	1.023	37.421	46.864	1.179	1.960
CVaR EWMA EVT	3.231	11.084	0.129	1.136	37.018	47.436	1.242	1.995
CVaR EWMA Stsk	3.461	11.515	0.143	1.622	37.056	47.382	1.270	2.368
CVaR GARCH FHS	3.101	11.237	0.115	0.370	38.026	46.004	1.046	1.350
CVaR GARCH EVT	3.081	11.121	0.115	0.357	37.773	46.363	1.079	1.368
CVaR GARCH Stsk	3.414	11.032	0.145	1.082	36.386	48.332	1.445	1.874

ditional risk models produce statistically higher Sharpe Ratios for significance levels of 1% and 2.5%. In contrast, none of the volatility targeting or HS based strategies produces a significantly higher Sharpe Ratio. Similarly, all switching strategies that are based on a conditional CVaR model produce economically high utility gains for the mean-variance investor for all

three significance levels, whereas the HS based switching strategy only slightly heightens the investor's utility. Furthermore, the utility increases of the conditional models are all statistically significant for all three significance levels.

Table XV. Performance Results for Different Significance Levels: Switching Strategies

This table shows additional performance results for the strategies that switch between volatility and CVaR targeting using the DAX and three different significance levels α . Panel A shows results for $\alpha = 1\%$, Panel B shows results for $\alpha = 2.5\%$ and Panel C shows results for $\alpha = 5\%$. The description of the columns is given in Table XIV.

Panel A: Significance Level of $\alpha = 1\%$								
Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta \frac{\gamma=5}{MV}$	DM-test
Vola Hist	3.115	12.843	0.102	-	41.228	41.458	0.280	-
DAX	2.726	23.457	0.040	-0.569	70.424	-	-7.523	-2.548
EWMA/CVaR Hist	4.197	12.490	0.190	1.641	32.334	54.086	1.514	1.634
EWMA/CVaR EWMA FHS	3.805	11.842	0.168	2.109	37.938	46.130	1.452	2.720
EWMA/CVaR EWMA EVT	3.875	11.728	0.175	2.233	37.051	47.389	1.575	2.787
EWMA/CVaR EWMA Stsk	4.110	11.845	0.193	2.932	35.823	49.133	1.749	3.160
EWMA/CVaR GARCH FHS	3.741	12.158	0.158	1.844	39.070	44.521	1.236	2.250
EWMA/CVaR GARCH EVT	3.827	12.005	0.167	2.082	38.000	46.042	1.395	2.469
EWMA/CVaR GARCH Stsk	4.104	11.882	0.192	2.535	36.104	48.733	1.724	2.724
Panel B: Significance Level of $\alpha = 2.5\%$								
Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta \frac{\gamma=5}{MV}$	DM-test
Vola Hist	3.115	12.843	0.102	-	41.228	41.458	0.280	-
DAX	2.726	23.457	0.040	-0.569	70.424	-	-7.523	-2.548
EWMA/CVaR Hist	3.865	12.888	0.159	0.985	34.401	51.151	0.986	0.903
EWMA/CVaR EWMA FHS	3.692	11.974	0.157	1.994	38.565	45.239	1.278	2.711
EWMA/CVaR EWMA EVT	3.746	11.870	0.163	2.096	37.696	46.473	1.381	2.755
EWMA/CVaR EWMA Stsk	3.930	12.022	0.176	2.705	37.008	47.450	1.487	3.083
EWMA/CVaR GARCH FHS	3.669	12.233	0.152	1.691	39.224	44.303	1.129	2.133
EWMA/CVaR GARCH EVT	3.692	12.131	0.155	1.789	38.460	45.387	1.201	2.254
EWMA/CVaR GARCH Stsk	3.979	12.065	0.179	2.315	37.075	47.355	1.513	2.601
Panel C: Significance Level of $\alpha = 5\%$								
Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta \frac{\gamma=5}{MV}$	DM-test
Vola Hist	3.115	12.843	0.102	-	41.228	41.458	0.280	-
DAX	2.726	23.457	0.040	-0.569	70.424	-	-7.523	-2.548
EWMA/CVaR Hist	3.671	13.301	0.140	0.601	36.618	48.004	0.580	0.361
EWMA/CVaR EWMA FHS	3.607	12.034	0.149	1.839	38.528	45.292	1.166	2.634
EWMA/CVaR EWMA EVT	3.642	11.991	0.152	1.929	38.207	45.747	1.221	2.692
EWMA/CVaR EWMA Stsk	3.789	12.171	0.162	2.355	37.942	46.124	1.276	2.921
EWMA/CVaR GARCH FHS	3.589	12.284	0.145	1.497	39.044	44.559	1.026	1.971
EWMA/CVaR GARCH EVT	3.599	12.240	0.146	1.543	38.787	44.924	1.057	2.040
EWMA/CVaR GARCH Stsk	3.878	12.222	0.169	2.035	37.820	46.297	1.338	2.433

In total, this section confirms our earlier results for other significance levels α . In particular, switching between volatility and CVaR targeting produces the most convincing risk-return profile for all significance levels. This holds especially when risk is estimated with a conditional risk model. However, results for lower significance levels are advantageous compared to higher levels of α . Thus, investors should manage extreme losses instead of moderate losses or return

deviations.

D.3 Tail Risk Targeting for Different Rebalancing Buffers

This section examines the performance of the risk targeting strategies when a rebalancing buffer η is used. Our main results rely on strategies where the weight of the risky asset is readjusted every day. In practice, readjusting the weight of the risky asset every day can lead to high transaction costs, which makes risk targeting less profitable. However, transaction costs can be significantly reduced by readjusting the weights less frequently or only in periods when risk changes dramatically. [Bollerslev et al. \(2018\)](#) find that reallocating portfolio weights less frequently reduces transaction costs without producing an inferior performance. Similarly, [Perchet et al. \(2016, Exhibit 5\)](#) find that daily, weekly and monthly rebalancing produce similar results. Further, [Perchet et al. \(2016, p. 36\)](#) suggest that a “reduction of turnover can be achieved with daily monitoring of volatility and rebalancing only when the volatility changes significantly”. For that reason, we show in Table XVI results for the volatility and CVaR targeting strategies for rebalancing buffers η of 5%, 10% and 15%, i.e. we define a corridor of 10%, 20% and 30% around the current weight where no rebalancing is done. Using a rebalancing buffer means that portfolio weights are only readjusted when the optimal portfolio weight w_t , under a given risk forecast, lies outside this corridor around the current weight. Thus, portfolio weights are only readjusted if portfolio risk changes substantially. This has the advantage that portfolio risk is monitored every day, but transaction costs are significantly lowered by only reacting to extreme changes. We denote the day t weight under a rebalancing buffer η by w_t^η and the day t weight *before* rebalancing on day t by $\tilde{w}_t^\eta$. Thus, as in [Kirby and Ostdiek \(2012, Eq. \(3\)\)](#), $\tilde{w}_t^\eta$ is defined by

$$\tilde{w}_t^\eta := \frac{w_{t-1}^\eta(1 + R_{t-1})}{w_{t-1}^\eta(1 + R_{t-1}) + (1 - w_{t-1}^\eta)(1 + R_{t-1}^f)}. \quad (90)$$

The day t weight using a rebalancing buffer of η is then given by

$$w_t^\eta = \begin{cases} \tilde{w}_t^\eta, & \text{if } (1 - \eta)\tilde{w}_t^\eta \leq w_t \leq (1 + \eta)\tilde{w}_t^\eta, \\ w_t, & \text{else} \end{cases}, \quad (91)$$

where w_t is the optimal weight for the target volatility, target VaR or target CVaR strategy. The approach of Equation (91) is similar to the approach suggested by [Perchet et al. \(2016, p. 33\)](#).

Table XVI. Performance Results for Different Rebalancing Buffers

This table shows additional performance results for the volatility and CVaR targeting strategies using the DAX and rebalancing buffers as in Equation (91). Panel A uses a rebalancing buffer of $\eta = 5\%$, Panel B uses a rebalancing buffer of $\eta = 10\%$ and Panel C uses a rebalancing buffer of $\eta = 15\%$. The description of the columns is given in Table XIV.

Panel A: Rebalancing Buffer of $\eta = 5\%$								
Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta_{MV}^{\gamma=5}$	DM-test
Vola Hist	3.116	12.849	0.102	-	41.268	41.401	0.278	-
Vola EWMA	3.272	12.394	0.118	0.750	39.915	43.322	0.661	1.565
Vola GARCH	3.073	11.870	0.107	0.138	39.069	44.523	0.723	0.963
CVaR Hist	2.385	9.275	0.064	-0.377	28.870	59.005	1.154	0.607
CVaR EWMA FHS	3.196	10.072	0.138	1.087	33.834	51.957	1.642	1.776
CVaR EWMA EVT	3.179	9.844	0.139	1.180	33.289	52.731	1.717	1.778
CVaR EWMA Stsk	3.376	10.140	0.154	1.476	32.926	53.246	1.791	1.973
CVaR GARCH FHS	3.123	10.595	0.124	0.554	37.003	47.457	1.351	1.478
CVaR GARCH EVT	3.157	10.248	0.132	0.754	35.400	49.733	1.531	1.596
CVaR GARCH Stsk	3.218	9.729	0.145	0.941	32.881	53.309	1.801	1.675
Panel B: Rebalancing Buffer of $\eta = 10\%$								
Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta_{MV}^{\gamma=5}$	DM-test
Vola Hist	3.158	12.968	0.104	-	41.597	40.933	0.258	-
Vola EWMA	3.177	12.378	0.111	0.274	39.791	43.498	0.576	1.099
Vola GARCH	3.170	11.813	0.115	0.295	37.597	46.613	0.845	1.142
CVaR Hist	2.321	9.293	0.057	-0.467	28.628	59.349	1.085	0.559
CVaR EWMA FHS	3.089	10.036	0.128	0.689	33.869	51.907	1.552	1.614
CVaR EWMA EVT	3.065	9.810	0.129	0.732	33.409	52.560	1.619	1.606
CVaR EWMA Stsk	3.312	10.152	0.148	1.169	33.051	53.069	1.724	1.849
CVaR GARCH FHS	3.262	10.516	0.138	0.799	35.435	49.684	1.522	1.604
CVaR GARCH EVT	3.304	10.178	0.147	1.018	33.863	51.916	1.705	1.705
CVaR GARCH Stsk	3.178	9.750	0.141	0.756	33.303	52.711	1.754	1.582
Panel C: Rebalancing Buffer of $\eta = 15\%$								
Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta_{MV}^{\gamma=5}$	DM-test
Vola Hist	3.029	13.034	0.094	-	41.222	41.466	0.098	-
Vola EWMA	3.135	12.436	0.107	0.514	40.977	41.814	0.507	1.352
Vola GARCH	3.032	11.787	0.104	0.259	38.728	45.008	0.722	1.163
CVaR Hist	2.380	9.482	0.062	-0.319	29.599	57.971	1.071	0.665
CVaR EWMA FHS	3.141	10.064	0.133	1.117	34.126	51.542	1.591	1.854
CVaR EWMA EVT	3.176	9.857	0.139	1.331	33.570	52.332	1.710	1.886
CVaR EWMA Stsk	3.276	10.101	0.145	1.349	33.537	52.379	1.709	1.913
CVaR GARCH FHS	3.125	10.523	0.125	0.723	35.662	49.361	1.383	1.625
CVaR GARCH EVT	3.143	10.180	0.131	0.870	34.336	51.243	1.546	1.677
CVaR GARCH Stsk	2.964	9.722	0.120	0.527	33.145	52.935	1.554	1.509

Results for the target volatility and target CVaR strategies for the three rebalancing buffers are shown in Table XVI. These results are similar to our main results that rebalance the weight on a daily basis. The Sharpe Ratios for the strategies using a rebalancing buffer of $\eta = 5\%$ and $\eta = 10\%$ are similar to the Sharpe Ratios with daily rebalancing as shown in Table V. This finding is in line with [Perchet et al. \(2016, Exhibit 5\)](#) who also find nearly identical results for daily and weekly rebalancing. Similarly, [Bollerslev et al. \(2018\)](#) conclude that it may be better to not trade every change in the optimal weight. They find better utility gains for the

strategies that adjust portfolio weights less frequently, especially when transaction costs are high and/or when models induce high day-to-day changes in the optimal weight. The authors combine the target volatility strategy with an approach that adjusts the weight only partially. Kirby and Ostdiek (2012) also find good results of models that are less sensitive to volatility changes. However, for a high rebalancing buffer of $\eta = 15\%$, we find that Sharpe Ratios (before transaction costs) are significantly lower. Nevertheless, risk targeting, especially CVaR targeting, still provides a convincing risk-return profile compared to the non-managed strategies.

In total, as in Table V, we again find the best results for the CVaR targeting strategies, regardless of the used rebalancing buffer. This holds especially when risk is managed by a conditional risk model. Thus, risk targeting, especially CVaR targeting, is still advantageous even when portfolio weights are readjusted less frequently. However, as expected, the strategies using a rebalancing buffer are slightly less profitable than the non-constrained strategies, at least before transaction costs.

In Table XVII, we show results for the switching strategies using the three rebalancing buffers.¹⁰⁰ In line with our previous results, switching between volatility and CVaR targeting significantly outperforms the non-managed and volatility managed strategies. Nearly all switching strategies exhibit significantly higher Sharpe Ratios and utilities. The best results are again found for the strategies that use conditional risk models. All switching strategies that use a conditional CVaR model produce statistically significant utility increases, even for high rebalancing buffers.

In total, this section shows that risk targeting is still advantageous to non-managed strategies when a rebalancing buffer is used. This is important since the rebalancing buffer can reduce transaction costs dramatically. This shows that risk targeting is an appealing tail risk hedging tool for practitioners. This holds particularly for the strategies that switch between volatility CVaR targeting, especially when conditional risk models are used.

¹⁰⁰We use rebalancing buffers that only determine the weights of the volatility and CVaR targeting strategies, whereas the switching approach of Equation (72) is not affected by the rebalancing buffer. Thus, we reallocate the portfolio weight every time when the indicator δ_t changes, even when the portfolio weight changes are small. More formally, the weight of the switching strategy with a rebalancing buffer η is given by $w_t^{switch,\eta} = \delta_t \cdot w_t^{CVaR,\eta} + (1 - \delta_t) \cdot w_t^{vol,\eta}$, where $w_t^{vol,\eta}$ and $w_t^{CVaR,\eta}$ are given by Equation (91). An alternative would be to apply the rebalancing buffer directly to the weights of the switching strategy.

Table XVII. Performance Results for Different Rebalancing Buffers: Switching Strategies

This table shows additional performance results for the strategies that switch between volatility and CVaR targeting using the DAX and rebalancing buffers as in Equation (91). Panel A uses a rebalancing buffer of $\eta = 5\%$, Panel B uses a rebalancing buffer of $\eta = 10\%$ and Panel C uses a rebalancing buffer of $\eta = 15\%$. The description of the columns is given in Table XIV.

Panel A: Rebalancing Buffer of $\eta = 5\%$								
Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta_{MV}^{\gamma=5}$	DM-test
Vola Hist	3.116	12.849	0.102	-	41.268	41.401	0.278	-
DAX	2.726	23.457	0.040	-0.570	70.424	-	-7.523	-2.557
EWMA/CVaR Hist	4.273	12.263	0.200	1.806	31.468	55.316	1.703	1.863
EWMA/CVaR EWMA FHS	3.867	11.613	0.176	2.112	36.333	48.409	1.621	2.617
EWMA/CVaR EWMA EVT	3.898	11.538	0.180	2.162	35.903	49.019	1.687	2.655
EWMA/CVaR EWMA Stsk	4.188	11.652	0.203	2.917	34.565	50.919	1.918	3.083
EWMA/CVaR GARCH FHS	3.758	11.969	0.162	1.851	38.535	45.281	1.345	2.308
EWMA/CVaR GARCH EVT	3.859	11.829	0.173	2.103	37.309	47.022	1.511	2.505
EWMA/CVaR GARCH Stsk	4.152	11.680	0.199	2.548	35.066	50.207	1.868	2.724
Panel B: Rebalancing Buffer of $\eta = 10\%$								
Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta_{MV}^{\gamma=5}$	DM-test
Vola Hist	3.158	12.968	0.104	-	41.597	40.933	0.258	-
DAX	2.726	23.457	0.040	-0.587	70.424	-	-7.523	-2.564
EWMA/CVaR Hist	4.066	12.237	0.183	1.414	31.378	55.444	1.513	1.495
EWMA/CVaR EWMA FHS	3.740	11.589	0.166	1.698	36.290	48.469	1.509	2.232
EWMA/CVaR EWMA EVT	3.765	11.511	0.169	1.745	35.953	48.949	1.570	2.259
EWMA/CVaR EWMA Stsk	4.026	11.628	0.190	2.398	34.472	51.050	1.770	2.665
EWMA/CVaR GARCH FHS	3.778	11.905	0.165	1.746	36.920	47.574	1.395	2.096
EWMA/CVaR GARCH EVT	3.863	11.766	0.174	1.951	35.751	49.235	1.544	2.235
EWMA/CVaR GARCH Stsk	3.968	11.667	0.184	2.010	35.211	50.001	1.695	2.274
Panel C: Rebalancing Buffer of $\eta = 15\%$								
Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta_{MV}^{\gamma=5}$	DM-test
Vola Hist	3.029	13.034	0.094	-	41.222	41.466	0.098	-
DAX	2.726	23.457	0.040	-0.495	70.424	-	-7.523	-2.524
EWMA/CVaR Hist	4.144	12.373	0.188	1.685	31.792	54.856	1.521	1.780
EWMA/CVaR EWMA FHS	3.703	11.621	0.162	1.836	37.148	47.252	1.457	2.429
EWMA/CVaR EWMA EVT	3.752	11.556	0.168	1.940	36.627	47.991	1.536	2.463
EWMA/CVaR EWMA Stsk	3.952	11.666	0.183	2.431	35.993	48.890	1.680	2.787
EWMA/CVaR GARCH FHS	3.569	11.954	0.147	1.477	38.752	44.974	1.167	2.157
EWMA/CVaR GARCH EVT	3.666	11.819	0.157	1.698	37.517	46.728	1.327	2.273
EWMA/CVaR GARCH Stsk	3.843	11.690	0.173	1.952	36.231	48.554	1.562	2.321

D.4 Tail Risk Targeting for Different Leverage Constraints

This section examines the profitability of risk targeting for investors who are leverage constrained. Our main results are based on unconstrained portfolio weights. However, in practice, many investors have short-sale and leverage constraints (see [Frazzini and Pedersen \(2014\)](#) and references therein). The portfolio weights w_t are positive by construction, but risk targeting investors could be forced to use leverage in periods with low market risk. Following [Perchet et al. \(2016, Exhibit 5\)](#) and [Moreira and Muir \(2017, Table IV\)](#), we additionally assess the impact of leverage constraints on the profitability of risk targeting. The day t weight w_t^c under a leverage

constraint is given by

$$w_t^c = \min(w_t, c), \quad (92)$$

where w_t is the optimal weight without leverage constraint. We choose values c of 1, 1.5 and 2, i.e. no leverage, a maximum leverage of 50% and a maximum leverage of 100%. This is in line with the choices of [Perchet et al. \(2016\)](#) who use a cap of $c = 1$ and $c = 2$ as well as [Moreira and Muir \(2017, Table IV-V\)](#) who use $c = 1$ and $c = 1.5$. [Perchet et al. \(2016, Exhibit 5\)](#) find that a cap of $c = 2$ produces nearly identical results compared to the unconstrained strategy, whereas a cap of $c = 1$ reduces the return, volatility and drawdown and produces a slightly lower Sharpe Ratio. Similarly, [Moreira and Muir \(2017, Table V\)](#) find similar Sharpe Ratios for the non-constrained and leverage constrained strategies. Results for the leverage constrained target volatility and target CVaR strategies are shown in Table XVIII. In line with [Perchet et al. \(2016\)](#) and [Moreira and Muir \(2017\)](#), we find that risk targeting is still advantageous when a maximum equity exposure is used. Interestingly, we find that CVaR targeting is much less influenced by the leverage constraints. In particular, CVaR targeting produces identical results for equity caps of $c = 1.5$ and $c = 2$ that are in line with the results of the unconstrained strategies in Table V. Thus, CVaR targeting needs much lower levels of leverage than the volatility targeting strategies. As a consequence, the advantages of CVaR targeting are even more pronounced for leverage constrained investors. Furthermore, as expected, the drawdown reduction ability of all risk targeting strategies is not influenced by the maximum equity exposure.

In Table XIX, we show results for the switching strategies under the three leverage constraints. In line with Table XVIII, we find that the switching approach is less influenced by the maximum equity exposure than the HSD managed strategy. All switching strategies produce significantly higher Sharpe Ratios and utilities for all three equity caps and significantly outperform the non-managed and HSD managed strategies, i.e. all switching strategies produce statistically significant Sharpe Ratio and utility increases for all three equity caps. Furthermore, the drawdown reduction ability is again not influenced by leverage constraints.

In total, results in this section show that investors also benefit from risk targeting when they have (tight) leverage constraints. This holds especially for the strategies that target a constant

Table XVIII. Performance Results for Different Leverage Constraints

This table shows additional performance results for the volatility and CVaR targeting strategies using the DAX and leverage constraints as in Equation (92). Panel A uses a maximum equity exposure of $c = 100\%$, Panel B uses a maximum equity exposure of $c = 150\%$ and Panel C uses a maximum equity exposure of $c = 200\%$. The description of the columns is given in Table XIV.

	Panel A: Maximum Equity Exposure of $c = 100\%$							
Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta_{MV}^{\gamma=5}$	DM-test
Vola Hist	2.954	12.409	0.093	-	41.228	41.458	0.343	-
Vola EWMA	3.102	12.198	0.106	0.761	40.522	42.459	0.592	1.340
Vola GARCH	3.136	11.908	0.112	0.634	39.233	44.290	0.766	1.079
CVaR Hist	2.431	9.260	0.069	-0.254	28.509	59.519	1.206	0.639
CVaR EWMA FHS	3.149	10.080	0.133	1.216	34.351	51.223	1.593	1.806
CVaR EWMA EVT	3.158	9.867	0.137	1.359	33.801	52.004	1.688	1.832
CVaR EWMA Stsk	3.307	10.083	0.149	1.597	33.121	52.970	1.747	2.017
CVaR GARCH FHS	3.237	10.670	0.134	1.160	36.998	47.465	1.430	1.700
CVaR GARCH EVT	3.265	10.324	0.141	1.382	35.505	49.583	1.605	1.806
CVaR GARCH Stsk	3.230	9.733	0.146	1.308	33.020	53.112	1.813	1.783
	Panel B: Maximum Equity Exposure of $c = 150\%$							
Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta_{MV}^{\gamma=5}$	DM-test
Vola Hist	2.989	12.785	0.093	-	41.228	41.458	0.188	-
Vola EWMA	3.318	12.491	0.121	1.420	40.522	42.459	0.658	2.099
Vola GARCH	3.191	11.965	0.116	0.688	39.233	44.290	0.792	1.366
CVaR Hist	2.431	9.260	0.069	-0.238	28.509	59.519	1.206	0.711
CVaR EWMA FHS	3.219	10.159	0.139	1.447	34.351	51.223	1.629	1.956
CVaR EWMA EVT	3.231	9.932	0.143	1.646	33.801	52.004	1.733	1.980
CVaR EWMA Stsk	3.382	10.141	0.155	1.823	33.121	52.970	1.797	2.146
CVaR GARCH FHS	3.234	10.678	0.134	1.069	36.998	47.465	1.424	1.757
CVaR GARCH EVT	3.263	10.329	0.141	1.289	35.505	49.583	1.602	1.852
CVaR GARCH Stsk	3.214	9.738	0.145	1.190	33.020	53.112	1.794	1.811
	Panel C: Maximum Equity Exposure of $c = 200\%$							
Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta_{MV}^{\gamma=5}$	DM-test
Vola Hist	3.085	12.830	0.100	-	41.228	41.458	0.258	-
Vola EWMA	3.336	12.499	0.122	1.091	40.522	42.459	0.671	1.846
Vola GARCH	3.191	11.965	0.116	0.458	39.233	44.290	0.792	1.185
CVaR Hist	2.431	9.260	0.069	-0.305	28.509	59.519	1.206	0.657
CVaR EWMA FHS	3.219	10.159	0.139	1.212	34.351	51.223	1.629	1.849
CVaR EWMA EVT	3.231	9.932	0.143	1.402	33.801	52.004	1.733	1.878
CVaR EWMA Stsk	3.385	10.141	0.155	1.603	33.121	52.970	1.800	2.046
CVaR GARCH FHS	3.234	10.678	0.134	0.855	36.998	47.465	1.424	1.636
CVaR GARCH EVT	3.263	10.329	0.141	1.063	35.505	49.583	1.602	1.740
CVaR GARCH Stsk	3.214	9.738	0.145	1.000	33.020	53.112	1.794	1.716

level of tail risk or the strategies that switch between volatility and CVaR targeting. In contrast, the strategy that is based on the HSD model is much more influenced by a maximum equity exposure. Thus, the outperformance of the CVaR targeting and switching strategies is even more pronounced for investors with leverage constraints.

Table XIX. Performance Results for Different Leverage Constraints: Switching Strategies

This table shows additional performance results for the strategies that switch between volatility and CVaR targeting using the DAX and leverage constraints as in Equation (92). Panel A uses a maximum equity exposure of $c = 100\%$, Panel B uses a maximum equity exposure of $c = 150\%$ and Panel C uses a maximum equity exposure of $c = 200\%$. The description of the columns is given in Table XIV.

Panel A: Maximum Equity Exposure of $c = 100\%$								
Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta_{MV}^{\gamma=5}$	DM-test
Vola Hist	2.954	12.409	0.093	-	41.228	41.458	0.343	-
DAX	2.726	23.457	0.040	-0.522	70.424	-	-7.523	-2.579
EWMA/CVaR Hist	4.121	12.030	0.191	1.835	31.650	55.058	1.669	1.795
EWMA/CVaR EWMA FHS	3.686	11.394	0.164	2.063	37.035	47.412	1.546	2.544
EWMA/CVaR EWMA EVT	3.723	11.318	0.169	2.123	36.619	48.002	1.618	2.600
EWMA/CVaR EWMA Stsk	3.999	11.409	0.191	2.870	34.956	50.363	1.846	3.004
EWMA/CVaR GARCH FHS	3.589	11.758	0.151	1.870	38.808	44.894	1.281	2.172
EWMA/CVaR GARCH EVT	3.687	11.613	0.161	2.113	37.680	46.495	1.446	2.393
EWMA/CVaR GARCH Stsk	3.952	11.443	0.186	2.476	35.379	49.763	1.785	2.608
Panel B: Maximum Equity Exposure of $c = 150\%$								
Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta_{MV}^{\gamma=5}$	DM-test
Vola Hist	2.989	12.785	0.093	-	41.228	41.458	0.188	-
DAX	2.726	23.457	0.040	-0.491	70.424	-	-7.523	-2.520
EWMA/CVaR Hist	4.345	12.323	0.204	2.095	31.650	55.058	1.742	2.056
EWMA/CVaR EWMA FHS	3.905	11.704	0.178	2.466	37.035	47.412	1.615	2.919
EWMA/CVaR EWMA EVT	3.946	11.629	0.183	2.525	36.619	48.002	1.691	2.962
EWMA/CVaR EWMA Stsk	4.222	11.718	0.205	3.252	34.956	50.363	1.919	3.341
EWMA/CVaR GARCH FHS	3.812	12.058	0.165	2.296	38.808	44.894	1.354	2.570
EWMA/CVaR GARCH EVT	3.910	11.917	0.176	2.524	37.680	46.495	1.518	2.761
EWMA/CVaR GARCH Stsk	4.176	11.750	0.200	2.836	35.379	49.763	1.858	2.924
Panel C: Maximum Equity Exposure of $c = 200\%$								
Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta_{MV}^{\gamma=5}$	DM-test
Vola Hist	3.085	12.830	0.100	-	41.228	41.458	0.258	-
DAX	2.726	23.457	0.040	-0.551	70.424	-	-7.523	-2.542
EWMA/CVaR Hist	4.362	12.331	0.206	1.981	31.650	55.058	1.755	1.978
EWMA/CVaR EWMA FHS	3.923	11.713	0.180	2.295	37.035	47.412	1.629	2.798
EWMA/CVaR EWMA EVT	3.964	11.638	0.184	2.361	36.619	48.002	1.705	2.845
EWMA/CVaR EWMA Stsk	4.240	11.726	0.206	3.076	34.956	50.363	1.932	3.224
EWMA/CVaR GARCH FHS	3.830	12.066	0.167	2.098	38.808	44.894	1.367	2.442
EWMA/CVaR GARCH EVT	3.928	11.925	0.177	2.335	37.680	46.495	1.532	2.639
EWMA/CVaR GARCH Stsk	4.194	11.759	0.202	2.678	35.379	49.763	1.872	2.819

D.5 Tail Risk Targeting for US Data and Small Caps

Table XX shows additional performance results for US data and small caps, proxied by the S&P 500 and the German small cap index SDAX, respectively. The data are also obtained from Datastream. Following [Marquering and Verbeek \(2004\)](#), we use the three month treasury bill rate as risk-free rate for the US data. Panel A contains results for the S&P 500, which are mainly in line with the results of Tables V and VIII for the DAX. The dynamically managed target risk strategies exhibit higher returns than the 60/40 portfolio with comparable risk. Further, returns of the dynamically managed strategies are also comparable to the return of the

S&P 500, but with only about half of the volatility. The Historical Simulation based strategy performs again significantly worse than the dynamically managed strategies. This can also be seen by the Sharpe Ratios of the strategies. The highest Sharpe Ratios are obtained by the dynamically managed CVaR targeting strategies, followed by the volatility managed strategies. The lowest Sharpe Ratios are obtained by the HS managed strategy, the 60/40 portfolio and the S&P 500. However, the Sharpe Ratio test of [Jobson and Korkie \(1981\)](#) indicates that only the strategy based on the EWMA model combined with the skewed t distribution produces a statistically higher Sharpe Ratio. Similar results also hold for the maximum drawdown. The highest drawdown reduction, given by ΔMDD , is obtained by the dynamically managed CVaR strategies, whereas statically or volatility managed strategies are less successful in reducing the drawdown. Results for the economic value are also in line with the findings for the Sharpe Ratio. The economic value for the volatility managed strategies is low or even negative, whereas the dynamically managed target CVaR strategies produce high economic values.

In Panel B, we show results for the German small cap index SDAX. Interestingly, the dynamically managed CVaR strategies exhibit higher returns with lower risk than the 60/40 portfolio and the SDAX. The volatility managed strategies produce even higher levels of return but also exhibit significantly higher risk. Thus, as before, the highest Sharpe Ratios are obtained by the dynamically managed target CVaR strategies. The [Jobson and Korkie \(1981\)](#) test shows that four target CVaR strategies, but none of the target volatility strategies, exhibit statistically significant higher Sharpe Ratios than the HSD managed strategy. Similarly, the drawdown reduction is again the highest for the CVaR managed strategies, whereas the drawdown reduction of the volatility managed strategies is only small. Interestingly, the drawdown of the HSD managed strategy is even higher than the drawdown of the SDAX. The economic value of risk targeting is again the highest for the CVaR managed strategies. However, the increase in the investor's utility is not statistically significant. Interestingly, we find that the skewed t distribution does not work well for the SDAX, although this strategy works well for the DAX and the S&P 500. Hence, different estimation methods can perform quite differently when different assets are used. A possibility to obtain more robust results for different assets would be to combine

Table XX. Performance Results for the S&P 500 and SDAX

This table shows additional performance results of the volatility and CVaR targeting strategies for the S&P 500 and the SDAX for the period 01.01.2000 to 31.12.2018. Panel A shows results for the S&P 500, whereas Panel B shows results for the SDAX. The description of the columns is given in Table XIV.

Panel A: Results for the S&P 500								
Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta_{MV}^{\gamma=5}$	DM-test
Vola Hist	4.760	13.091	0.236	-	36.587	36.766	-0.059	-
Vola EWMA	4.920	12.652	0.257	0.894	34.976	39.550	0.321	1.399
Vola GARCH	4.359	11.948	0.226	-0.210	33.524	42.059	0.129	0.325
CVaR Hist	3.083	9.103	0.159	-0.652	33.242	42.546	0.097	0.103
CVaR EWMA FHS	4.471	9.591	0.293	1.225	23.061	60.144	1.260	1.448
CVaR EWMA EVT	4.369	9.248	0.293	1.134	22.515	61.087	1.292	1.357
CVaR EWMA Stsk	5.123	9.934	0.347	2.525	25.619	55.722	1.755	2.150
CVaR GARCH FHS	4.527	10.050	0.285	0.924	26.064	54.953	1.132	1.336
CVaR GARCH EVT	4.404	9.736	0.282	0.863	24.509	57.640	1.138	1.289
CVaR GARCH Stsk	4.550	9.383	0.308	1.222	27.602	52.294	1.417	1.502
S&P 500	4.338	19.220	0.139	-0.751	57.859	-	-4.335	-1.585
60/40	3.395	10.139	0.173	-0.570	33.295	42.455	-	0.045
Panel B: Results for the SDAX								
Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta_{MV}^{\gamma=5}$	DM-test
Vola Hist	8.475	13.524	0.486	-	69.219	-2.524	1.658	-
Vola EWMA	8.389	12.655	0.513	1.060	64.101	5.057	2.040	0.593
Vola GARCH	8.572	12.109	0.551	1.190	57.488	14.851	2.490	0.648
CVaR Hist	4.727	8.172	0.354	-1.296	38.189	43.436	0.451	-0.438
CVaR EWMA FHS	6.809	8.667	0.570	2.278	46.291	31.436	2.277	0.299
CVaR EWMA EVT	6.673	8.389	0.573	2.270	45.036	33.296	2.245	0.271
CVaR EWMA Stsk	5.822	9.139	0.434	-1.112	47.413	29.774	1.162	-0.250
CVaR GARCH FHS	7.671	9.204	0.628	2.554	43.768	35.173	2.905	0.588
CVaR GARCH EVT	7.487	8.901	0.630	2.531	42.341	37.286	2.843	0.536
CVaR GARCH Stsk	5.503	8.861	0.413	-1.118	42.852	36.529	0.958	-0.308
SDAX	6.489	15.792	0.293	-1.707	67.515	-	-1.523	-1.190
60/40	5.033	10.190	0.313	-1.535	47.612	29.479	-	-0.612

several forecasting methods (Allen et al., 2012, Halbleib and Pohlmeier, 2012, Taylor, 2014).

In summary, the additional results for the S&P 500 and the SDAX confirm our earlier results for the DAX, i.e. portfolio risk is best managed by the portfolio's CVaR and portfolio risk should be estimated conditionally.

We next assess if our switching approach also works well for the S&P 500 and the SDAX. Results of the switching strategies are shown in Table XXI. The switching strategies are again successful in producing higher returns compared to the individual strategies with lower levels of volatility and drawdown. As a consequence, these strategies provide an enhanced risk-return profile with high Sharpe Ratio increases. The increase of the Sharpe Ratio is statistically significant for all dynamically managed strategies for the S&P 500. For the SDAX, all switching strategies exhibit significantly higher Sharpe Ratios. This finding is confirmed by the economic

Table XXI. Performance Results for the S&P 500 and SDAX: Switching Strategies

This table shows additional performance results for the strategies that switch between volatility and CVaR targeting using the S&P 500 and the SDAX. Panel A shows results for the S&P 500, whereas Panel B shows results for the SDAX. The description of the columns is given in Table XX.

Panel A: Results for the S&P 500								
Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta \frac{\gamma=5}{MV}$	DM-test
Vola Hist	4.760	13.091	0.236	-	36.587	36.766	-0.059	-
S&P 500	4.338	19.220	0.139	-0.751	57.859	-	-4.335	-1.694
EWMA/CVaR Hist	5.189	12.757	0.276	0.774	32.293	44.187	0.524	0.780
EWMA/CVaR EWMA FHS	5.424	11.787	0.318	1.952	26.331	54.492	1.230	2.186
EWMA/CVaR EWMA EVT	5.483	11.731	0.324	1.980	25.862	55.301	1.313	2.189
EWMA/CVaR EWMA Stsk	5.690	12.056	0.332	2.674	27.650	52.212	1.355	2.627
EWMA/CVaR GARCH FHS	5.506	12.123	0.316	2.112	27.504	52.465	1.145	2.249
EWMA/CVaR GARCH EVT	5.508	12.071	0.317	2.114	27.753	52.034	1.173	2.280
EWMA/CVaR GARCH Stsk	5.531	12.146	0.317	2.150	29.165	49.593	1.159	2.155
Panel B: Results for the SDAX								
Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta \frac{\gamma=5}{MV}$	DM-test
Vola Hist	8.475	13.524	0.486	-	69.219	-2.524	1.658	-
SDAX	6.489	15.792	0.293	-1.707	67.515	-	-1.523	-1.176
EWMA/CVaR Hist	10.282	11.870	0.703	3.616	46.276	31.458	4.224	1.538
EWMA/CVaR EWMA FHS	10.610	11.550	0.751	5.068	51.583	23.598	4.691	1.996
EWMA/CVaR EWMA EVT	10.700	11.501	0.762	5.061	50.807	24.746	4.800	1.997
EWMA/CVaR EWMA Stsk	9.910	11.734	0.680	4.116	51.845	23.210	3.939	1.606
EWMA/CVaR GARCH FHS	10.896	11.802	0.759	5.322	50.495	25.209	4.838	2.104
EWMA/CVaR GARCH EVT	11.004	11.736	0.772	5.381	49.798	26.241	4.972	2.120
EWMA/CVaR GARCH Stsk	10.340	11.772	0.714	4.388	49.516	26.660	4.327	1.723

value. All switching strategies produce economically high economic values and most utility increases are statistically significant. Hence, our simple switching approach does not only work well for the DAX, but also for the S&P 500 and the SDAX.

In total, results in this section show that risk targeting does not only work for the DAX, but also for US data and small caps. This holds especially for strategies that manage a portfolio's risk by dynamic risk models, where the best results are found for the strategies that manage a portfolio's tail risk. The portfolio's performance can further be improved by switching between volatility and CVaR targeting. In particular, mean-variance investors are willing to pay economically high and statistically significant fees to have access to these switching strategies.

D.6 Tail Risk Targeting in the Long Run

We have so far only examined a period of 18 years, which was marked by several crises. To assess if risk targeting is also beneficial in the long run, we use data for the US market from 1929 to 2018. Data for the US market and the risk-free rate are obtained from the website of Kenneth

French.¹⁰¹ This data set is also used by [Moreira and Muir \(2017\)](#) and [Dreyer and Hubrich \(2019\)](#). Results for the long sample are shown in Table XXII, where we only show results for the switching strategies. As argued above, although CVaR targeting is superior to volatility targeting for our main data set, this does not necessarily hold in the long run, since CVaR targeting could be too conservative in uptrending periods. Furthermore, as before, we only show results for the switching strategies based on the EWMA model and the TSMOM crash indicator. Results for the GARCH(1,1) model and the other crash indicators were again quite similar, but slightly less profitable. In line with [Moreira and Muir \(2017\)](#) and our earlier findings, we find that volatility targeting significantly enhances the risk-return profile by producing a higher Sharpe Ratio with lower drawdowns. This finding is also in line with [Moreira and Muir \(2019\)](#) who show that long-term investors should time short-term volatility. However, in line with our main results, we find that the performance of risk targeting can further be enhanced by switching between volatility and CVaR targeting. All switching strategies increase the portfolio's Sharpe Ratio and these increases are all statistically significant. Similarly, mean-variance investors are willing to pay high fees to switch away from volatility targeting to the switching approach. This improved risk-return profile is also accompanied by significantly lower drawdowns of the switching strategies. Interestingly, although the skewed t distribution works well for the S&P 500 in the short sample, the same approach does not work well for the long US sample. As mentioned above, a further extension could be to use an average of several CVaR forecasting methods ([Allen et al., 2012](#)).

Similar to [Moreira and Muir \(2017, Figure 3\)](#), Figure IV shows the cumulative return of the US market, the 60/40 portfolio, the HSD managed strategy and the strategy that switches between volatility and CVaR targeting for a 100\$ investment. For a better comparison, we rescale all strategies to the volatility of the US market. We only show results for the switching strategy that switches between the EWMA volatility model and the CVaR-EWMA-FHS model for the TSMOM indicator. This approach is easy to implement and does not need any estimated parameters. Thus, this approach could be appealing for practical implementations. Figure IV shows the clear outperformance of the risk targeting strategies. Both risk targeting strategies, the HSD

¹⁰¹http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

Table XXII. Performance Results for the US Market in the Long Run: Switching Strategies

This table shows additional performance results for the strategies that switch between volatility and CVaR targeting using data of the US market for the period November 1929 to December 2018. The description of the columns is given in Table XIV.

Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta \gamma_{MV}^5$	DM-test
Vola Hist	9.779	13.283	0.481	-	55.886	30.248	1.241	-
US Market	9.215	16.839	0.347	-2.218	80.121	-	-1.424	-2.425
60/40	8.620	13.597	0.387	-1.634	54.299	32.229	-	-1.465
EWMA/CVaR Hist	10.617	12.259	0.588	3.876	42.971	46.368	2.549	2.871
EWMA/CVaR EWMA FHS	10.580	11.801	0.607	5.918	39.002	51.321	2.740	3.890
EWMA/CVaR EWMA EVT	10.678	11.751	0.618	6.183	38.721	51.672	2.855	4.051
EWMA/CVaR EWMA Stk	10.046	12.027	0.553	3.785	45.484	43.232	2.133	2.560
EWMA/CVaR GARCH FHS	10.239	12.084	0.566	4.152	44.918	43.938	2.285	2.957
EWMA/CVaR GARCH EVT	10.345	12.023	0.577	4.578	43.646	45.524	2.413	3.215
EWMA/CVaR GARCH Stk	9.817	12.035	0.534	2.463	50.572	36.881	1.917	1.759

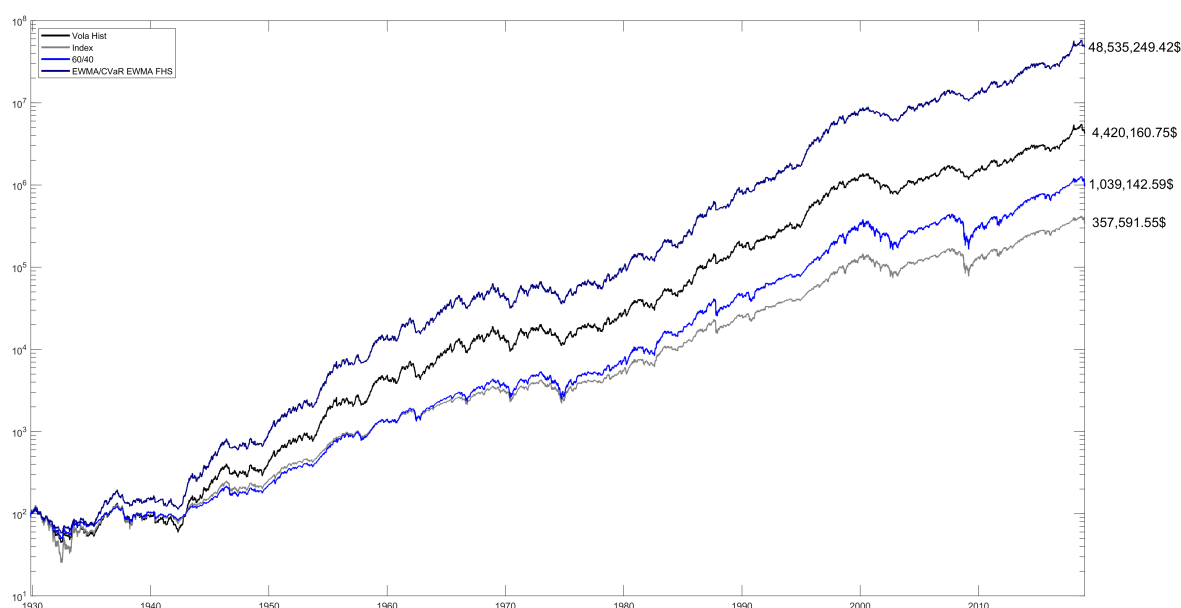


Figure IV. Cumulative Return of Risk Targeting. This figure plots the cumulative return of the US market, the 60/40 portfolio, the HSD target volatility strategy and a strategy that switches between volatility and CVaR targeting for the period 1929 to 2018. All strategies are rescaled to the same level of volatility.

managed strategy and the switching strategy, successfully capture the upside potential of the market while downside risk is limited. However, the strategy that switches between volatility and CVaR targeting clearly outperforms the HSD managed strategy. A 100\$ investment in the market portfolio would result in a terminal wealth of 357,591.55\$. Invested in the target volatility strategy, the terminal wealth would increase to 4,420,160.75\$. However, the strategy that switches between volatility and CVaR targeting produces a final wealth of even 48,535,249.42\$.

This is in line with the results of [Moreira and Muir \(2019\)](#) that even long-term investors should time short-term risk. In particular, the outperformance of the switching strategy is quite steady over time, i.e. switching between volatility and CVaR targeting is advantageous in bull and bear regimes. [Dreyer and Hubrich \(2019\)](#) also find that risk targeting applied to the long US data set works well in several sub periods.

Table XXIII. Spanning Tests: Portfolio Alphas

This table shows results of spanning test of the US market for the risk targeting strategies. We run time-series regressions of each portfolio on the remaining strategies, where we use the US market, the HSD based volatility targeting strategy, the CVaR targeting strategy and the strategy that switches between volatility and CVaR targeting based on the TSMOM indicator δ_t . We report annualized percentage alphas with corresponding t -statistics in parentheses. Alphas that are significantly positive at the 10% level are given in bold. Alphas that are significantly negative at the 10% level are given in red.

	US Market		Volatility		CVaR		Switching	
TSMOM indicator	α	t -stat	α	t -stat	α	t -stat	α	t -stat
US Market	-	-	-2.204	(-0.697)	-3.808	(-1.196)	-6.922	(-2.001)
Volatility	15.177	(3.984)	-	-	-1.046	(-0.664)	-7.122	(-5.503)
CVaR	18.132	(4.611)	3.571	(2.198)	-	-	-4.586	(-2.899)
Switching	27.088	(6.100)	9.737	(6.822)	7.639	(4.668)	-	-

Finally, following [Daniel and Moskowitz \(2016\)](#) and [Moreira and Muir \(2017\)](#), we next repeat the examination of Table XIII and run time-series regressions for the non-managed and risk-managed portfolios. Results for these spanning tests are shown in Table XXIII. In line with our main results, we find that the market's alpha is always negative, once we control for the returns of the risk targeting strategies. When we control for the switching strategy, the alpha is even negative and statistically significant with a t -statistic of -2.001 . In contrast, all risk targeting strategies have positive and statistically significant alphas with respect to the market.¹⁰² The highest alpha of 27.088% with a t -statistic of 6.100 is obtained by the switching strategy. The alpha of the volatility targeting strategy becomes negative when we control for the returns of the CVaR targeting or switching strategy. In particular, when we control for the switching approach, the alpha is even significantly negative with a t -statistic of -5.503 . Furthermore, CVaR targeting cannot be explained by volatility targeting, but exhibits a significantly negative alpha with respect to the switching approach. In contrast, the performance of the switching strategy

¹⁰²The alpha for the volatility targeting strategy is higher than the alpha found by [Moreira and Muir \(2017\)](#). A possible explanation for this finding could be that we use daily rebalancing, whereas [Moreira and Muir \(2017\)](#) use monthly rebalancing. As shown in Section D.3, increasing the rebalancing interval reduces the profitability of volatility targeting.

cannot be explained by any other strategy and the switching strategy's alphas are economically high with t -statistics between 4.668 and 6.822. In total, all strategies exhibit significantly negative alphas when we control for the switching strategy, whereas the switching strategy has significantly positive alphas in all cases.

D.7 Out-Of-Sample Study: Corona Crisis

Finally, this section examines the performance of risk targeting during the recent corona crisis. This crisis is marked by one of the fastest drawdowns in history. Financial markets all over the world crashed over 30% within several weeks. For example, between 19.02.2020 and 18.03.2020, the DAX exhibits a maximum drawdown of 38.78%. This fast and severe crash was not only limited to equities, but nearly all asset classes crashed abruptly and simultaneously. This high co-crash behavior of nearly all asset classes made an adequate tail risk protection very challenging during this period. Most widely-used portfolio risk management approaches that aim to mitigate a portfolio's crash risk, such as the risk parity approach, failed as portfolio risk management tools. For example, the "S&P Risk Parity Index - 12% Target Volatility (TR)", a risk parity index that also targets a volatility of 12%, exhibits a maximum drawdown of 29.14% during the corona crisis.¹⁰³ Thus, the recent corona crisis challenges currently used tail risk hedging strategies and demonstrates the importance of a fast adapting crash risk mitigation tool. Consequently, assessing the performance of risk targeting during this challenging time gives further insights on the ability of risk targeting to mitigate extreme crashes. Moreover, assessing the performance of risk targeting during the recent corona crisis is particularly interesting, since this examination can be seen as an out-of-sample test of our earlier results. The first version of this paper was written in the years 2016 to 2018. Thus, applying the same methods as examined in the main part to data from the recent corona crisis demonstrates the profitability of risk targeting, without the potential of data mining. Since our main results are based on data that end in December 2018, we use data for the period that ranges from January

¹⁰³Further details and data on this index can be obtained from <https://us.spindices.com/indices/strategy/sp-risk-parity-index-12-target-volatility-tr>. The aim of the index is described as: "The S&P Risk Parity Index – 12% Target Volatility seeks to measure the performance of a multi-asset risk parity strategy that allocates risk equally among equity, fixed income, and commodities futures contracts, while targeting a volatility level of 12%".

2019 to April 2020. This period is marked by an uptrending market in 2019 and the beginning of 2020. However, this calm and uptrending market ended abruptly by the severe crash in February and March 2020, initiated by the spread of the corona virus in Europe. Thus, this period is quite challenging for portfolio risk management approaches, since a good portfolio risk management tool has to adapt very fast to changing market environments.

Table XXIV. Performance Results for the DAX in the Corona Crisis

This table shows additional performance results for the DAX, the 60/40 portfolio, the volatility targeting strategies and the CVaR targeting strategies. The data range from 01.01.2019 to 30.04.2020. The description of the columns is given in Table XIV.

Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta_{MV}^{\gamma=5}$	DM-test
Vola Hist	-0.182	14.206	0.013	-	19.983	48.469	-0.598	-
Vola EWMA	2.066	13.422	0.181	1.774	18.673	51.847	2.078	2.155
Vola GARCH	3.507	12.908	0.301	1.908	18.036	53.492	3.799	1.986
CVaR Hist	-0.878	11.173	-0.046	-0.166	20.867	46.191	0.234	0.193
CVaR EWMA FHS	2.670	10.726	0.283	2.075	13.742	64.564	4.020	0.904
CVaR EWMA EVT	2.312	10.426	0.257	2.010	13.713	64.638	3.789	0.838
CVaR EWMA Stsk	2.911	9.673	0.339	1.867	13.107	66.202	4.711	0.769
CVaR GARCH FHS	3.578	10.560	0.374	2.301	14.088	63.671	5.014	1.086
CVaR GARCH EVT	3.450	10.200	0.375	2.339	13.684	64.712	5.041	1.008
CVaR GARCH Stsk	5.472	9.588	0.611	1.991	12.506	67.751	7.352	0.937
DAX	2.149	25.640	0.098	0.172	38.779	-	-7.093	-0.403
60/40	1.148	15.434	0.098	0.185	25.721	33.675	-	0.140

Table XXIV shows performance results for the DAX, the 60/40 portfolio, the volatility targeting strategies and the CVaR targeting strategies for the period starting in January 2019 and ending in April 2020. The DAX produces a slightly positive return with a high volatility and a drawdown of 38.779%. The bad risk-return profile during this period can also be seen by the low Sharpe Ratio of 0.098. The 60/40 portfolio produces a lower return with a lower volatility. This results in an equally high Sharpe Ratio of 0.098. Interestingly, the HSD managed strategy exhibits a negative return with a very low Sharpe Ratio.¹⁰⁴ In contrast, the EWMA and GARCH(1,1) managed strategies produce higher returns with lower levels of volatility. Both strategies exhibit significantly higher Sharpe Ratios than the HSD managed strategy. Similarly, the conditionally managed CVaR targeting strategies also produce statistically significant Sharpe Ratio increases. However, the HS managed CVaR strategy produces a negative return and a negative Sharpe Ratio. Thus, as before, more accurate risk models are rewarded with

¹⁰⁴The Sharpe Ratio of this strategy is positive since the annualized return of -0.182% is less negative than the return of the risk-free asset in this period.

higher performance and utility gains. All risk targeting strategies are successful in reducing the portfolio's drawdown. For example, by using a CVaR targeting strategy, the DAX's drawdown of 38.779% can be reduced to only 12.506%. This high drawdown reduction of CVaR targeting is accompanied by a higher return. Furthermore, mean-variance investors are willing to pay high fees to switch to a conditionally managed target risk strategy. The highest utility gains are again found for the CVaR targeting strategies. In total, results for the corona crisis indicate that risk targeting can enhance a portfolio's return while simultaneously crash risk is significantly reduced. In line with our earlier results, conditionally managed strategies outperform strategies that manage portfolio risk based on unconditional risk models. In particular, the HSD based target volatility strategy fails to enhance the risk-return profile during the corona crisis and underperforms the non-managed portfolio.

Table XXV. Performance Results for the DAX in the Corona Crisis: Switching Strategies

This table shows additional performance results for the strategies that switch between volatility and CVaR targeting using the DAX. The switching strategies use the EWMA volatility model and the TSMOM based indicator δ_t . The data range from 01.01.2019 to 30.04.2020. The description of the columns is given in Table XIV.

Model	Return	Volatility	SR	z_{JK}	MDD	ΔMDD	$\Delta \frac{\gamma=5}{MV}$	DM-test
Vola Hist	-0.182	14.206	0.013	-	19.983	48.469	-0.598	-
DAX	2.149	25.640	0.098	0.172	38.779	-	-7.093	-0.459
60/40	1.148	15.434	0.098	0.185	25.721	33.675	-	0.212
EWMA/CVaR Hist	-1.535	13.504	-0.087	-0.419	21.756	43.897	-1.564	-0.384
EWMA/CVaR EWMA FHS	1.553	12.290	0.156	0.915	16.376	57.771	2.155	1.102
EWMA/CVaR EWMA EVT	1.170	12.136	0.126	0.704	16.337	57.872	1.845	0.936
EWMA/CVaR EWMA Stsk	2.399	12.002	0.231	1.089	15.875	59.065	3.149	0.847
EWMA/CVaR GARCH FHS	2.328	12.326	0.219	1.242	16.319	57.918	2.916	1.151
EWMA/CVaR GARCH EVT	2.038	12.170	0.198	1.066	16.192	58.246	2.703	1.025
EWMA/CVaR GARCH Stsk	4.307	12.151	0.385	1.358	15.441	60.183	4.995	0.955

In Table XXV, we show results for the strategies that switch between the EWMA model and the CVaR targeting strategies based on the TSMOM indicator δ_t . All strategies that switch to a conditionally managed CVaR strategy clearly outperform the non-managed and HSD managed strategies. Furthermore, the switching strategies typically have higher Sharpe Ratios than the EWMA managed strategy. Thus, switching away from volatility targeting when a crash is expected produces a superior risk-return profile. Nevertheless, the switching strategies have slightly lower Sharpe Ratios than the CVaR managed strategies.¹⁰⁵ This finding is quite intu-

¹⁰⁵Results for the GARCH based switching strategies or the strategies that use the volatility based indicator δ_t

itive, since CVaR targeting is expected to perform well in an extreme crash period as examined here. In contrast, as shown in the previous section, the switching approach is beneficial in the long run, since using CVaR targeting on every day would be too conservative. Thus, results of Tables XXIV and XXV are in line with our earlier findings.

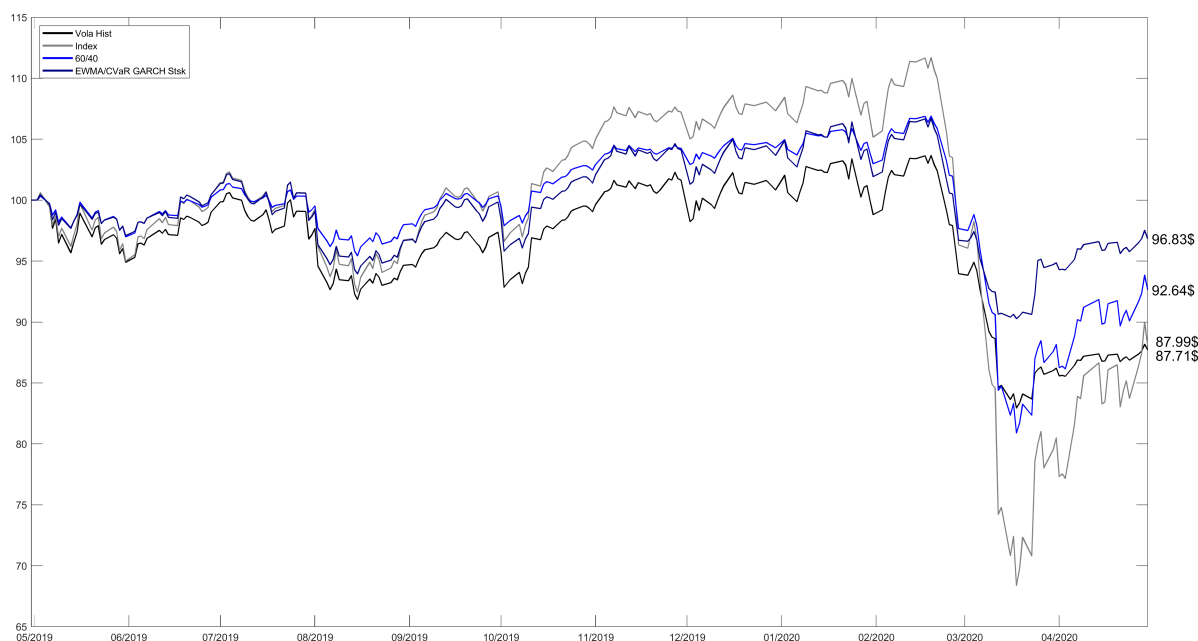


Figure V. Cumulative Return of Risk Targeting: Corona Crisis. This figure plots the cumulative return of the DAX, the 60/40 portfolio, the HSD based target volatility strategy and a strategy that switches between volatility and CVaR targeting for the one year period that ranges from May 2019 to April 2020.

In order to further demonstrate the profitability of the different strategies during the corona crisis, we show in Figure V the cumulative return of the DAX, the 60/40 portfolio, the HSD managed strategy and a switching strategy for the one year period from 01.05.2019 to 30.04.2020. This figure shows that the 60/40 portfolio and the risk targeting strategies successfully reduce the portfolio's drawdown, where the best drawdown reduction is found for the switching strategy. The bad performance of the HSD managed strategy during the corona crisis mainly occurs since this strategy fails to capture the upside potential of the DAX. In contrast, the switching strategy successfully captures the upside potential in the calm period and significantly reduces the drawdown in the crash period. This again shows that a conditional risk model, that success-

are superior to the results shown here. However, in order to be in line with the results shown before, we only show results for the EWMA model and the TSMOM indicator.

fully adapts to changing market environments, is needed when portfolio risk is managed.

In summary, results in this section show that our risk targeting strategies also perform well during the recent corona crisis and successfully reduce the DAX's crash risk. This holds especially for dynamically managed strategies. Unconditionally managed strategies can be used as an adequate tail risk mitigation tool, but these strategies fail to capture the portfolio's upside potential in calm markets. The switching strategy also performs well during this crisis but is, as expected, outperformed by the CVaR targeting strategies. This is in line with the motivation of the switching strategy that CVaR targeting is advantageous in crises but too conservative in the long run. However, switching between volatility and CVaR targeting outperforms the strategy that manages volatility on every day. Thus, the switching strategy is optimal for long-term investors who want to capture an asset's upside potential while simultaneously downside risk is limited in crash periods.

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